

Modelling of Temperature Evolution and Crystal Growth in Additive Manufacturing of Polylactic Acid (PLA)

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Nomenclature

Δt	Time step (s)
ρ	Density (kg / m^3)
v	Nozzle velocity (m / s)
C_p, c	Specific heat capacity ($J / kg \cdot ^\circ C$)
G_0	Pre-exponential factor
k_{plat}	Thermal contact conductance with the platform ($W / m^2 \cdot ^\circ C$)
K_g	Surface nucleation constant
k	Thermal conductivity ($W / m \cdot ^\circ C$)
U^*	Activation energy
h	Heat transfer coefficient ($W / m^2 \cdot ^\circ C$)
T_∞	Surrounding temperature ($^\circ C$)
$l \ w \ h$	Filament length, width and height (m)
H	Distance between nozzle to platform (m)
T_{plat}	Platform temperature ($^\circ C$)
ε	Emissivity factor
D	Nozzle diameter (m)
R	Gas constant ($J / mol \cdot K$)
T_{nozzle}	Nozzle temperature ($^\circ C$)

$\Delta x \Delta y \Delta z$	Length, width and height of single controlled volume (m)
σ	Stefan-Boltzmann constant ($W / m^2 \cdot K^4$)
N_0	Reference number of nucleation
L	100% latent heat of crystallisation per unit mass (J / kg)
n_L, n_W	The number of the printed elements in the x and y direction
$\delta x, \delta y$	The distance of neighbour nodes in vertical and horizon direction (m)
k_{mat}	The thermal contact conductance with filaments ($W / m^2 \cdot ^\circ C$)
T_g	Glass transition temperature ($^\circ C$)

List of Acronyms

ABS: acrylonitrile butadiene styrene;

AM: Additive Manufacturing;

BJ: Binder Jetting;

CAD: Computer Aided Design;

DED: Directed Energy Deposition

DLS: Dynamic Light Scattering;

DMA: Dynamic Mechanical Analyser;

DTA: Differential Thermal Analysis;

DSC: Differential Scanning Calorimeter;

FDM: Fused Deposition Modelling;

IP: Inkjet Printing;

iPP: isotactic polypropylene

IR: Infra-Red;

LOM: Laminated Object Manufacturing;

ME: Material Extrusion;

MJ: Material Jetting;

MW: Microwave;

PA: polyamides

PBF: Powder Bed Fusion

PC: Polycarbonate;

PLA: Poly Lactic Acid;

PM: Polarization Microscopy;

SL: Sheet Lamination

SLA: Stereolithography apparatus;

SLM: Selective Laser Melting;

SLS: Selective laser sintering;

VP: Vat Photopolymerization.

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Abstract

Additive manufacturing (AM), also known as 3D printing, is becoming widely popular in the manufacturing world due to many advantages that it can offer compared to traditional methods: single components with no joints, easiness of printing from a CAD model without a physical mould, possibility of achieving complex shapes with a high specific strength and so forth. However, the evolution of material's temperature during printing and the degree of crystallinity in Fused Deposition Modelling (FDM), which plays an essential role in product quality, has not yet well understood and currently there is no any comprehensive model to explain these phenomena. Hence, this work is aimed to develop a generic model to simulate the temperature evolution via a two-dimensional domain discretization method incorporating the Hoffman-Lauritzen theory to model the crystal growth during the FDM process. Here, PLA was used as the filament material which was extruded using a standard 0.4 mm diameter nozzle. Both single-line and 2×2 multiline printing works are simulated by the proposed model. The predicted temperature evolution shows 4.4 °C of maximum deviations that is a percentage mismatch of 2.2 % with experiment data. Hence, the proposed model is in excellent agreement with the previously published data and experimental measurements, and this verifies its accuracy. In the proposed 2×2 multi-line prediction of temperature evolution, the decrease of cooling rate is clearly expressed and the time required for cooling down to the air temperature is doubled compared with the single line. In addition, the influence of different parameters (such as printing speed, thermal convection coefficient, thermal contact conductance with platform, thermal contact conductance with filament, nozzle diameter and crystallization latent heat) to the temperature and crystallization behaviours has also been discussed in the final part of this work. As was confirmed by the result, the thermal convection coefficient, the thermal contact conductance with the platform and the size of the nozzle diameter have a significant impact on the temperature evolution and degree of crystallization. On the contrary, the printing speed and latent heat of crystallization have little effect on temperature evolution. Consequently, based on the simulation results, the process of printing works could be designed and optimized and the final products could be customized according to the requirement.

Declaration

No portion of the work referred to in the thesis has been submitted in support of an application for another degree or qualification of this or any other university or other institutes of learning.

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Chapter 1 Introduction

1.1 Research background

With the appearance of synthetic polymeric materials, people's life and production mode have been influenced to a great extent. No matter what product/area it is, food packaging, electrical and electronic shells, medical catheters and structure materials, or aeronautics and astronautics, bio-medical and marine, polymeric materials are widely used in all of these areas. Most polymer materials are resistant to corrosion while brings severe pollution because it is hard to decompress in the natural environment.

Poly (lactic acid) (PLA) is a kind of biodegradable semi-crystallisation thermoplastic materials which could be a replacement for traditional petroleum-based polymeric materials. It is mainly produced by agriculture products, which contains lots of starch. After the original products fermenting to the lactic acid, PLA can be made via polymerisation reaction. Besides the biocompatibility and biological degradability, PLA offer severs other benefits such as excellent mechanical properties, high transparency and impressionability^[1-6].

Additive manufacturing (AM), widely known as 3D printing, is a rapidly developing technology, which is leading the design of production, way of products and revolution of manufacturing. Usually, AM products are created by the following four steps:

- (1) Digital Model: Creating a digital model of the product using a relevant software whole sine can be from a micro-electro-mechanical device to an aeroplane wing;
- (2) Forming Materials: Materials used in AM are shaped progressively by wire, powders, filaments and sheets to develop the desired object;
- (3) Tools for extruding the materials;
- (4) Automatic control system to command the tools layout materials in the precise position layer by layer, showing extraordinary dimensional flexibility.

The AM is not a technology revolution to replace conventional technology, but the AM could produce, if required, quantities of independent part at a low cost compared with traditional

manufacturing. Also, the AM could make functional shapes and structures which are not economical for conventional mass manufacturing. That is the reason why AM be followed with enthusiastically interest by the world's leading manufacture companies, such as Airbus, Boeing, GE and Siemens^[7] planning to a leap forward Industry 4.0. The AM has a series of potential benefits, and some of them are list below^[8-10]:

- (1) AM could directly translate a soft model to a physical product;
- (2) AM could use customizable material properties to make complex geometrical artefacts without additional tooling or manufacturing costs;
- (3) AM could directly manufacture the final or near-final shape with no additional processes;
- (4) AM could have the potential in optimizing the use of materials with very little waste;
- (5) AM could significantly reduce the manufacturing time which leads to transfer new concepts to the market quite quickly;
- (6) AM could operate smaller foot-print to manufacture a large variety of components;
- (7) AM could move away from a manufacturing based on projection. It could depend on actual demand to process;
- (8) AM could be remarkably scalable. The size of the products could be from 10 nm to meters depending on the way used;
- (9) AM could offer joint-free products (i.e. single body product) and hence could decrease assembly time and associated costs.

Therefore, it appeals to much attention during the last three decades. By applying AM freedom of materials usage and geometry design and redesign, the resulting products could face the world of a much chance. However, not all parts could benefit from this possibility or sufficient produce via the AM process due to some limitations still should be sorted. This technology should be more comprehensively understood when, why and how to achieve and overcome the related limitation while enabling more opportunities^[11].

As an example, process parameters should be optimised to achieve good deposition quality

in the Fused Deposition Modelling (FDM). However, experimental trials are normally time-consuming and costly. Also, equipment requirement is high for conducting such research under different process directions. Numerical modelling can be a both time-saving and cost-effective approach to obtaining a full understanding of how different parameters will affect the deposition processes.

A considerable amount of research has been found on the temperature evolution of extruded filaments^[12-22] and crystal kinetics of polymers^[4,23-31]. However, up to now, previous proposed model had only considered limited factors and parameters, which is reasonable for the deviation between the experimental and numerical results and at the same time takes crystal kinetics into consideration when the process is influenced by different critical parameters. Therefore, a generalised model for these two aspects needed to be developed in order to get a better understanding of included phenomena.

1.2 Aims and objectives

Traditional manufacturing mostly depends on dimensional metrology to measure length, width and height such as contact and optical coordinate measurements, which are accurate for ordinary shape. However, these traditional ways are limited to sufficiently precisely measure AM products with neoteric structure and frame. The inside of the AM products is also complex to observe without destructive way. Therefore, according to existing experimental results and data, building numerical modelling is a satisfying way to predict and analyse the quality of the final products. The final products quality will be influenced by several ambient factors. In these factors, temperature evolution after the materials extruded from the nozzle is a crucial one, which decides the fusion, solidification of materials and crystallisation and bonding process. Hence, this work aims to numerically model the temperature evolution and degree of crystallinity during 3D printing. In order to accomplish this aim, the following objectives are selected:

1. To model the temperature evolution during 3D printing, and a generalised model should be proposed for temperature evolution as a function of all the related key parameters;
2. To understand the kinetics of polymer crystallisation and model this process with temperature evolution;
3. To model the latent heat of crystallisation and combine this energy as feedback to the temperature evolution;
4. To verify the computational/numerical results with the previously reported experimental and numerical works.

1.3 Thesis structure

This thesis covers various parameters of the fused deposition modelling process with experimental and theoretical analysis. The thesis consists of seven chapters which can be described as **Fig. 1**.

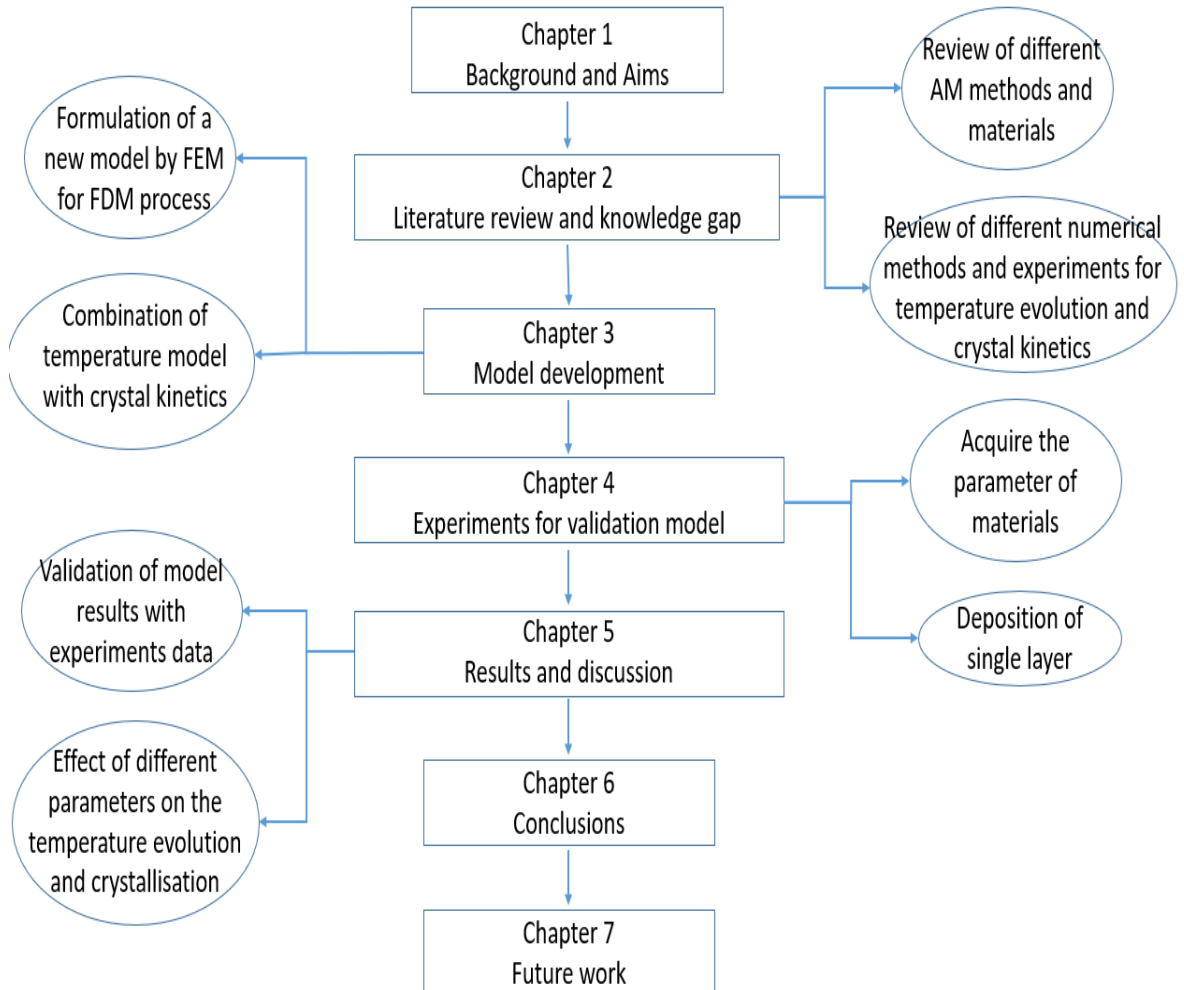


Fig. 1 Diagram of thesis structure

Chapter 2 Literature review

2.1 Classification of Additive Manufacturing

AM is rightly positioned in bringing forward the Fourth Industrial Revolution^[32]. Over the past three decades, the AM process had experienced a big revolution. In early periods, AM was mainly applied on mould and prototypes^[33,34]. With the technology being more and more matured, AM has been playing a significant role in producing soft and rapid tooling^[35]. Today, AM has been implemented for turning out final use parts and products.

AM processes are mainly categorized into seven classifications by the International Organization for Standardization (ISO)/ American Society for Testing and Materials (ASTM) 52900:2015 standard classify standard: (1) binder jetting (BJ); (2) directed energy deposition (DED); (3) material extrusion (ME); (4) material jetting (MJ); (5) powder bed fusion (PBF); (6) sheet lamination (SL) and (7) vat photopolymerization (VP). **Table 1** shows the basic principles, examples of suitable materials, advantages and disadvantages of each of these seven systems.

Listed below is the information on most commonly used 3D printing methods, principles, and how they work. Additionally, **Fig. 2** provides a schematic representation of these methods:

1) Stereolithography apparatus (SLA)

The first concept and patent for photopolymerization were reported in the late 1960s^[36]. SLA uses a UV light to scan photopolymer resin to make the feedstocks robust, and with the platform lowering, the whole layer forming process could repeat again and again. This older AM methods own the highest surface quality and resolution.

Table 1 Basic principles, materials, advantages, disadvantages, typical build volumes and tool manufacturers of seven ASTM categories of AM^[10,37-39]

ASTM category	Basic principle	Example technology	Advantages	Disadvantages	Materials	Build volume (mm × mm × mm)	Tool manufacturer/country
BJ	Liquid binder/s jet printed onto thin layers of powder. The part is built up layer by layer By glueing the partides together	• 3D inkjet technology	• Free of support/substrate • Design freedom • Large build volume • High print speed • Relatively low cost	• Fragile parts with limited mechanical properties • May require post processing	• Polymers • Ceramics • Composites • Metals • Hybrid	Versatile (small to large) X = <4000 Y = <2000 Z = <1000	ExOne, USA PolyPico, Ireland
DED	Focused thermal energy melts materials <i>during</i> deposition	• Laser deposition (LD) • Laser Engineered NetShaping (LENS) • Electron beam • Plasma arc melting	• High degree control of grain structure • High quality parts • Excellent for repair applications	• Surface quality and speed requires a balance • Limited to metals/metal based hybrids	• Metals • Hybrid	Versatile X = 600–3000 Y = 500–3500 Z = 350–5000	Optomec, USA InssTek, USA Sciaky, USA Irepa Laser, France Trumpf, Germany
ME	Material is selectively pushed out through a nozzle or orifice	• Fused Deposition Modelling (FDM)/Fused Filament Fabrication (FFF), Fused Layer Modelling (FLM)	• Widespread use • Inexpensive • Scalable • Can build fully functional parts	• Vertical anisotropy • Step-structured surface • Not amenable to fine details	• Polymers • Composites	Small to medium X = <900 Y = <600 Z = <900	Stratasys, USA
MJ	Droplets of build materials are deposited	• 3D inkjet technology • Direct Ink writing	• High accuracy of droplet deposition • Low waste • Multiple material parts • Multicolour	• Support material is often required • Mainly photopolymers and thermoset resins can be used	• Polymers • Ceramics • Composites • Hybrid • Biologicals	Small X = <300 Y = <200 Z = <200	Stratasys, USA 3D Systems, USA PolyPico, Ireland 3Dinks, USA WASP, Italy
PBF	Thermal energy fuses a small region of the powder bed of the build material	• Electron beam melting (EBM) • Direct Metal Laser Sintering (DMLS) • Selective Laser Sintering/Melting (SLS/SLM)	• Relatively inexpensive • Small footprint • Powder bed acts as an integrated support structure • Large range of material options	• Relatively slow • Lack of structural integrity • Size limitations • High power required • Finish depends on precursor powder size	• Metals • Ceramics • Polymers • Composites • Hybrid	Small X = 200–300 Y = 200–300 Z = 200–350	ARCAM, Sweden; EOS, Germany; Concept Laser Cusing, Germany; MTT, Germany; Phoenix System Group, France; Renishaw, UK;Realizer, Germany; Matsuura, Japan, Voxeljet, 3Dsystems, USA
SL	Sheets/foils of materials are bonded	• Laminated Object Manufacturing (LOM) • Ultrasound consolidation/Ultrasound Additive Manufacturing (UC/UAM)	• High speed, • Low cost, • Ease of material handling	• Strength and integrity of parts depend on adhesive used • Finishes may require post processing • Limited material use	• Polymers • Metals • Ceramics • Hybrids	Small X = 150–250 Y = 200 Z = 100–150	3D systems, USA MCor, Ireland
VP	Liquid polymer in a vat is light-cured	• Stereo Lithography (SLA) • Digital Light Processing (DLP)	• Large parts • Excellent accuracy • Excellent surface finish and details	• Limited to photopolymers only • Low shelf life, poor mechanical properties of photopolymers • Expensive precursors/Slow build process	• Polymers • Ceramics	Medium X < 2100 Y < 700 Z < 800	Lithoz, Austria 3D Ceram, France

2) **Inkjet Printing (IP)**

IP has a similar principle to SLA. However, IP uses a movable nozzle to eject feedstocks. Then a UV light is applied to activate the materials. The platform moves up in steps followed by the layer of materials deposited in the vertical direction. IP offers a high resolution with moderate surface quality. This method is also referred to as poly-jet.

3) **Selective laser sintering (SLS)**

SLS was invented in the early 1990s and successfully commercialised in 1992^[40]. The feedstocks of this method are mainly metals, but before the final products are being made, post-processing is necessary to achieve infiltration, sintering and finishing. In this method, a levelling roller would roll the metal particle across the build chamber, and then a laser or binder would directly act on the powder to form a one-layer pattern. Repeating this process with platform lowering would lead to complete the final products. The left-off powder will remain to support this process completion and could be recycled for the next use. SLS is a low solution method compared with IP and SLA, limited by metal powder sizes.

4) **Fused deposition modelling (FDM)**

By heating the filament in a nozzle, transferred from materials spool, the molten-state filaments extruded and reached the platform to form the patterns. FDM is a low cost but reliable, user-friendly and almost no need for post-processing method. However, warping, low surface quality and resolution limit further boom in the market.

5) **Laminated object manufacturing (LOM)**

Laminated object manufacturing initially appeared in 1991^[36]. The next layer was rolled on to the former layer and combined via a binder. The laser would be used for cutting the pattern out in one-layer thickness, and the rest of the part will remain for supporting. The resolution of this method was limited by the layer thickness, which is usually larger than other AM methods. Besides, taking out the final products is also a time-consuming process compared with other conventional AM methods.

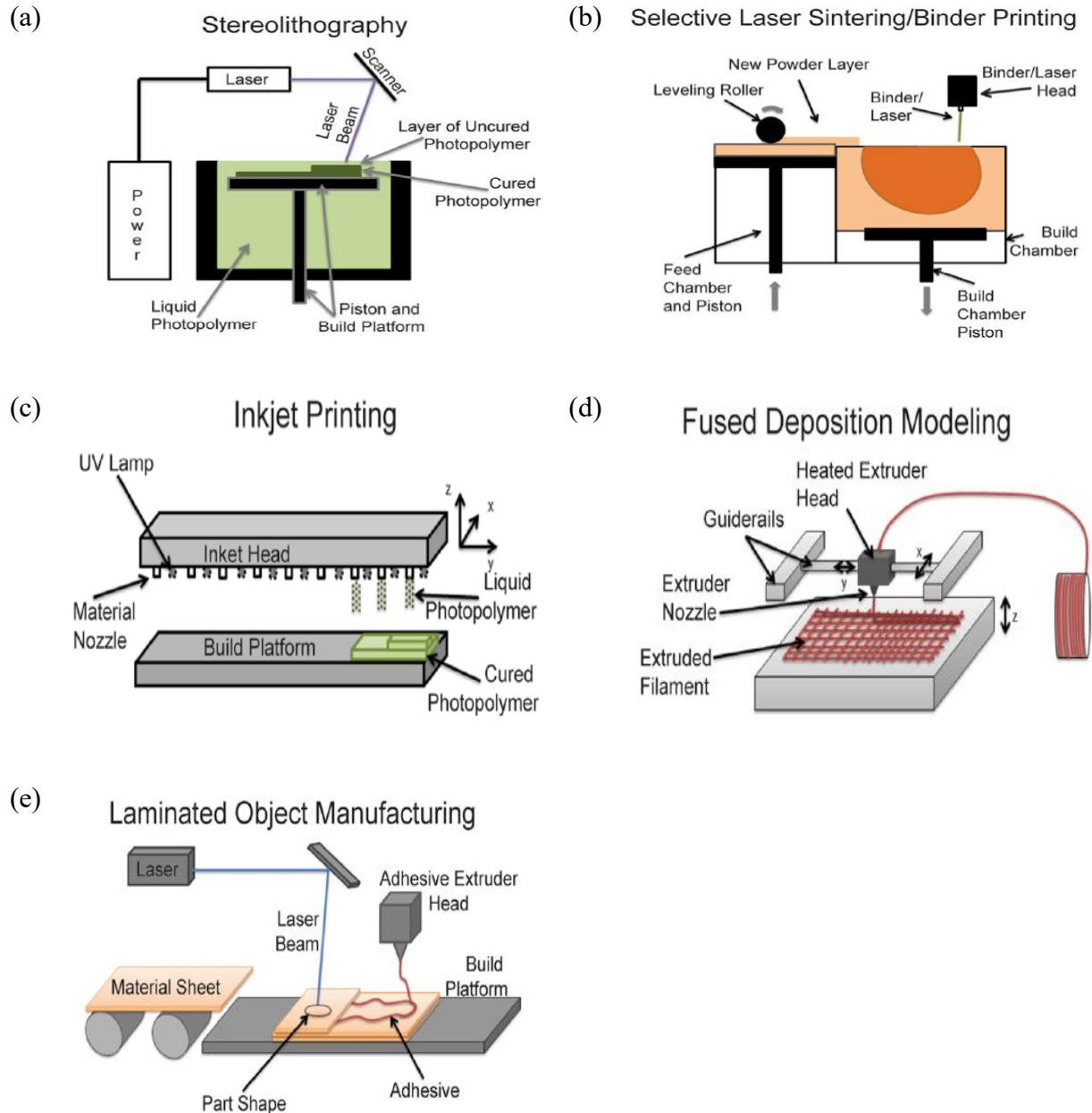


Fig. 2 Schematic of the five most commonly used AM methods^[32]

2.2 The Fused Deposition Modelling Process

Compared to conventional manufacturing, AM methods have their superiorities. It is possible to use Computer-Aided Design (CAD) software to create the file input into the AM system directly and obtain the products without building a mould before producing. Every part of the products could use appointed materials to combine. Besides, by using this way,

the waste of materials could be maximum reduce, and the size of the final products could be easily changed from micrometres to tens of meters.

Fused deposition Modelling (FDM), which was originally developed by Stratasys, is one of the most popular AM methods due to its easy operation, inexpensiveness, utilisation of environmentally safe materials and reproducibility^[41]. However, some of the drawbacks of FDM are also obvious: vertical anisotropy, step-structured surface and lack of strength as the fully functional and load-bearing parts. Several researchers used reinforced materials like carbon fibre with conventional AM materials to strengthen the loading ability to overcome the above-mentioned limitations^[42-48].

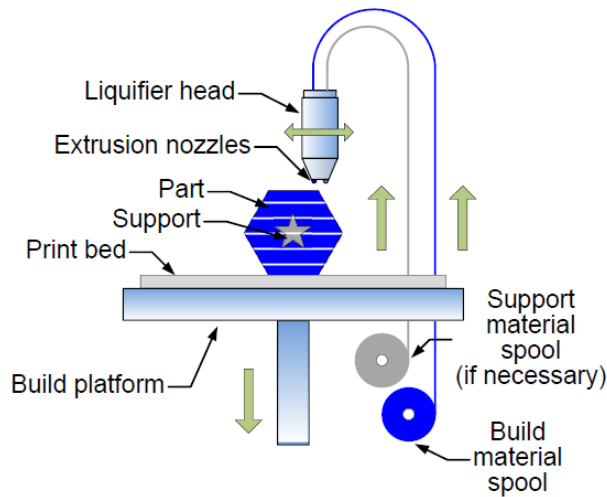


Fig. 3 Schematic of the FDM process with its main components^[42]

As shown in **Fig. 3**, in the FDM process, the feedstocks are divided into two parts: support material and build material. The support material is used for building a hollow structure for adding the build material. After the product is being made, the support material would be removed easily without causing any damage to the product^[49]. Build materials are mainly thermoplastics and their composites such as acrylonitrile butadiene styrene (ABS)^[39,43,45,50,51], Poly (Lactic Acid) PLA^[39,51,52], polycarbonate (PC)^[39,45,51], polyamides (PA)^[45,51] and so on. Both support and built materials are made in filaments and rowed up onto a spool. The spool rotates slowly and transports materials into a liquefier head which heats the feedstock to a temperature slightly higher than the melting temperature and

transfers to the semi-liquid phase. Then, the material in the semi-liquid phase will be extruded from the nozzle, which applies to move in a horizontal plane to print a two-dimensional pattern on the platform. While the first layer is finished, the platform could vertically move one-layer thickness, so that the products could be produced via layer by layer process in FDM.

2.3 The materials used in Additive Manufacturing

With the fast development of Additive manufacturing, 3D printing could apply with a variety of materials from chocolate to advanced functional materials. The form of feedstocks could be filaments, wire, powder, sheets and inks; different forms of materials could be applied to different classification of AM machines.

Polymers are considered as the most common type of materials used in the AM industry, mainly due to their diversity and easiness of using different kinds of AM process. The main form of polymers in AM would be filaments, powder, resin and reactive monomers. In the past few years, 3D printing polymers are widely applied in aerospace, construction, toy and medical fields^[39,51]. The benefits that AM polymers could bring are high accuracy customised products, cost-effectiveness than conventional methods like moulding or extrusion. Meanwhile, pure polymer products built by 3D printing, which are lack of strength and functionality, are often used as a conceptional prototype. Many types of research have used different methods to develop better performance and advanced composite materials to enhance the mechanical properties of 3D printed polymers^[9,42,45,47,50,52,53].

Compared with traditional methods, AM could reduce the waste of expensive metals such as titanium. Besides, it also eliminates the installation time, decreases the risk of localised stresses while assembling and increases the freedom of design. Aerospace, defence and automobile industry are using metals and alloys with 3D printing technology to fabricate complicated parts. The form of materials used in the AM process mainly in powder or wires. Selective Laser Sintering (SLS), Selective laser melting (SLM) and Directed Energy

Deposition (DED) are the main methods used with laser or electron-beam melting and sintering powder to complete the products. The defects of metal AM process including porosity, accumulations of beads and the change of materials property and shapes, limits the future application.

3D printed ceramics can be used to fabricate parts with a high strength-weight ratio and have explored many intricate ceramic lattices for application. One of the main applications is high strength and complexed shaped ceramic scaffolds in tissue engineering. The main forms of ceramics materials used in AM process are powder and ink, sintered by laser and high temperature, respectively. Although only a small number type of ceramic materials could be applied into AM process and the limited accuracy as well as quality affect the comprehensive utility, microstructure and ingredient of ceramic products could be better-controlled by 3D printing. Therefore, further research on new ceramic materials and product optimization will be promising for 3D printing of ceramics in the future^[39,54].

Concrete is the most widely used human-made materials^[54]. The primary method used in concrete AM process is extrusion. However, the development of 3D printed concrete can be seen in modern architecture, processes slowly. Besides, the layer by layer surface, anisotropic mechanical properties and weak inter-layer bonding are the main challenges that should be overcome with 3D printed concretes. Despite these drawbacks, mass-customisation, no mould, lightweight structure and automation still attract much anticipation for a bright future^[10].

Although there are some current limitations for materials applied in different categories of AM methods, multi-materials apply and AM technology revolution will step by step sort these problems. Also, the development of numerical modelling and optimisation techniques of materials processing will assist the improvements of future AM process.

2.4 Introduction to PLA

The first AM machine used for manufacturing with pure plastic materials was rapid prototypes^[42]. Currently, thermoplastics and their composites are widely used in FDM. Build materials are mainly include: acrylonitrile butadiene styrene (ABS)^[39,43,45,50,51], Poly (Lactic Acid) PLA^[39,51,52], polycarbonate (PC)^[39,45,51], polyamides (PA)^[45,51] and so on. **Table 2** shows the main properties of these materials and their common applications in the industry^[39].

Table 2 Material properties of FDM thermoplastics and their application in various industries ^[39]

Material	Properties	Applications/industries
ABS ASA	Tough and strong Mechanical Strength and UV stability	Automotive, aerospace, medical-device Functional prototyping from brackets and electrical housings to automotive prototypes and practical production parts for outdoor use under the sun
Nylon 12	Good chemical resistance, high fatigue resistance and high impact strength	Ideal material for applications that demand impact-protective components and high fatigue endurance, including antenna covers, custom production tooling, friction-fit inserts and snap fits in automotive and aerospace industries
PC	High tensile and flexural strength	Functional prototypes, tooling and fixtures, blow-molding master in automotive and aerospace industries
PPSF/PPSU	Excellent chemical and heat resistance and mechanical strength	PPSF/PPSU can withstand various sterilization methods including ethylene oxide, autoclaving, and radiation. Sterilizable medical devices, automotive prototypes, and tooling for demanding applications in a variety of industries
PEI or ULTEM	Biocompatible, excellent mechanical, chemical and thermal stability	Due to its high strength-to-weight ratio and existing certification, ULTEM is ideal for rapid prototyping and advanced tooling applications in aerospace, automotive, medical and food-production industries
PLA	Good tensile strength and surface quality	Ideal for model and prototypes that require aesthetic detail and environmentally-friendly for both home and office
TPU	Excellent tear and wear resistance, high impact strength and hardness	Exceptional flexibility (i.e. elongation at break) and corrosion resistance to many common industrial chemicals and oils. Highly versatile material with the both rubber and plastics properties for a variety of industrial application

PLA is a biodegradable polymer material, which could be produced by renewable resources^[55]. The first report for independent lactic acid from milk could be found in 1780^[56] and then first commercially production was made by Swedish chemist Scheele in 1881^[57]. Due to PLA possesses high modulus, high strength and excellent purity, it arose vast attraction as a potential material to represent petroleum-based polymer.

Before PLA is applied into packaging^[55] and commercial materials^[3,58] in the market, it has been developed for biomedical applications^[1]. Excellent biocompatibility and absorbing ability make PLA to be an ideal material for applications in medicine transfer systems, structures and artificial blood vessels^[1]. As shown in **Fig. 4**, PLA could be made in two ways^[2,59]: 1) Lactic acid polycondensate to low-molecular-weight PLA, and then by adding chain coupling agents to create high weight average molecular weight PLA; 2) Low

molecular weight polymer could be obtained after condensation, then a depolymerization method makes low molecular polymer chain become Poly(L/D Lactic acid) ring. By using a ring-opening polymerisation process, high molecular weight PLA could be acquired. Nowadays, the majority of commercial grades of PLA are produced by bacterial fermentation of carbohydrates. The fermentation process can be classified based on the type of bacteria used.

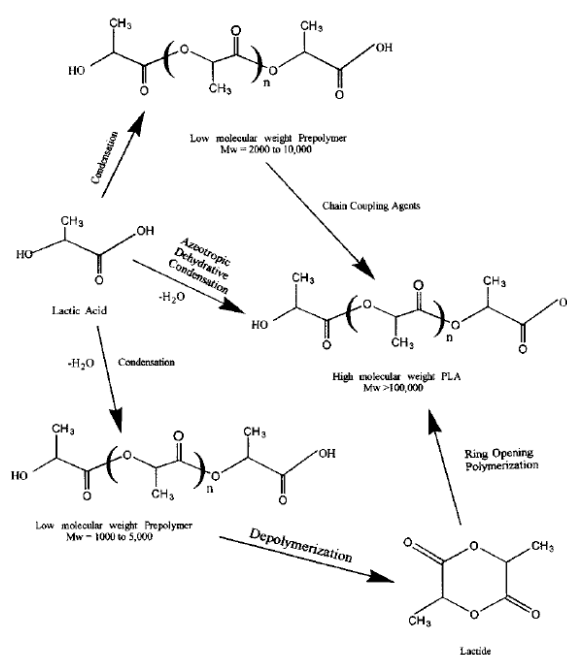


Fig. 4 Synthesis methods for high-molecular-weight PLA^[2]

One disadvantage of PLA family materials is that, compared to other polyesters, PLA has lower glass transition temperature (T_g), approximately of around 60 °C, which makes PLA has weak heat resistance unless having a high crystallisation rate^[59].

Pyde et al^[60] investigated that the heat capacity of semi-crystalline PLA can vary from temperature of 190 to 470 K by combining differential thermal analysis (DSC) method. At low temperature (5 to 200 K), they assumed that only vibration motion occurred. Moreover, at high temperature (332.5 to 600 K) when PLA in the liquid phase, they divided the capacity into two parts by their functional group and linearly added the heat capacity of each value to get the final liquid heat capacity of PLA. The results showed that a good match between

experimental and calculated heat capacity when the temperature below the glass transition temperature. When PLA turned into liquid phase with temperature increasing, a larger deviation appeared between experimental and calculated values. The calculated value is lower than the experimental one due to samples have a different length along the chain, and other experiments also approved this phenomenon^[61].

Compared with traditional materials like ABS, PLA is more difficult to print, which is stronger than ABS but more brittle. Some studies have researched the mechanical properties of PLA ^[62-65], and some these information is presented in **Table 3**, and it showed that PLA had better performance in tensile strength than other investigated 3D printing thermoplastic materials such as ABS, PC and PA, etc^[65-69].

Letcher et al^[65]. studied the tensile strength, flexural strength and fatigue along tested orientation for printed PLA. All the samples were printed independently in the centre of the platform to make sure those samples are in the same situation as much as possible. The nozzle and platform surface temperature were set as 230 °C and 65 °C, respectively, with a nozzle velocity of 100 mm/s. The orientation of printed filaments was from 0° to 90°. The stretching speed was 5 mm/min and data was collected with frequency of 100 Hz in tensile testing; the grips moving speed was set at 10 mm/min in the 3-point flexure test; and 3 periods of cycle loading were added to the specimen: first 2 Hz up to 1000 cycles, then 5 Hz up to 10000 cycles and final 20 Hz until failure. For every single test, five same specimens were prepared to proceed. The results showed that: for 0°, 45° and 90° printing orientation specimens, the average ultimate stress were 58.45, 64.03 and 54.01 MPa, the average extension length were 2.02 %, 2.50 % and 4.14 % and the average modulus of elasticity were 3.33, 3.60 and 3.49 GPa in tensile testing; the average ultimate stress were 102.203, 90.649 and 86.136 MPa in flexural test. The 45° printing orientation specimen performed the most robust behaviour in the tensile testing while 0° printing orientation specimen could resist stronger bending stresses. In the fatigue test, the 0° and 45° printing orientation specimens showed a similar performance, which is much better than 90° item.

Rezayat et al^[66]. used both experimental and numerical modelling method to test the mechanical properties of printed ABS. The samples used in the experiment were 0.2540 mm in thickness and 0.3806 mm in width, respectively. And the results relating to the different air gap and raster angle conditions showed in **Fig. 5**. In the simulation part, they applied the ANSYS FEM program to initialise the FDM process with various parameters. The model has assumed it occurred at the ideal situation on feedstocks, bonding and loading. By observing the strain energy distribution in the materials, they could mathematically work out the external loading conditions. Tymrak with his co-workers^[63] also did similar experimental works on printed the PLA and ABS. Their results approved Rezayat et al work^[66].

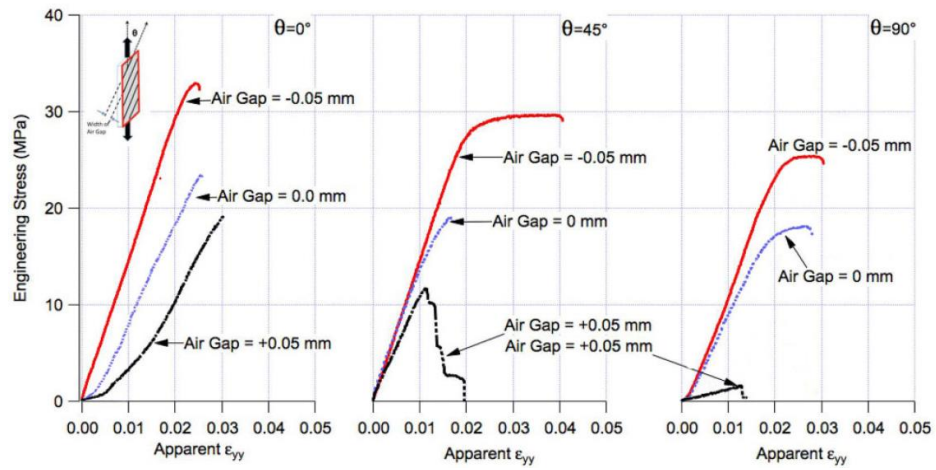


Fig. 5 Stress-strain curves for the samples in loading direction with different air gap and raster angle^[66]

Carneiro et al^[68]. compared conventional and 3D printed products made by polypropylene to study the effect of different printing conditions such as printing orientation, layer thickness, infill degree and raw/reinforced materials.

Table 3 Ultimate tensile strength (MPa) of different thermoplastic 3D-printed by FDM ^[65-69]

Raster Angle	ABS	Polypropylene	Polycarbonate	PLA	PEI
90°	26	32	19	54	40
0°	34	36	59.7	58	59
Authors	Rezayat et al.	Carneiro et al.	Hill et al.	Letcher et al.	Bagsik et al.

Torres et al^[62]. had researched on the shear stress-strain response based on changing layer thickness, infill density and postprocessing heat-treatment time at 100 °C for 3D printed PLA. Average ultimate shear strength, shear modulus, 0.2% yield strength, proportional limit, ductility and fracture strain were also calculated. The thickness (0.1-0.3 mm), infill density (20-100%) and postprocessing heat-treatment time (0-20 min) ranges are divided into three levels: low, middle and high, respectively. For each condition, three experimental trails carried out, and hence 27 trails in total. Heat treatment would be in a Barnstead Thermolyne F48055 resistance furnace to process, and then the samples would be secured in an MTS Bionix Electromechanical Torsion tester with a 50 N·m capacity to test mechanical property. According to the results shown in **Table 4** that the average ultimate shear strength, shear modulus, 0.2% yield strength, proportional limit, ductility and fracture strain, would be primarily influenced by infill density. The second effect ranking was thickness except fracture strain, which plays a more responsive to heat treatment than thickness. Tsouknidas et al^[64]. Followed similar conditions for investigating absorbed energy/volume fraction response based on three variables: thickness, infill density and filling pattern, respectively. They found that the optimal choice for force attenuation and energy absorption are based on 25% infill density, while the thickness and filling pattern had only a slight effect on absorbed energy/volume fraction. The ideal result from the test would be 0.1mm in thickness and the rectilinear filling pattern. Besides, they tested the relationship between printing speed and tensile strength and found that. The tensile strength decreases with the increasing printing velocity.

Table 4 Parameter ranking influential on the average ultimate shear strength, shear modulus, 0.2% yield strength, proportional limit, ductility and fracture strain ^[62]

Material property	1st	2nd	3rd
0.2% shear yield strength (τ_y)	Infill ³	Thickness ¹	Heat treatment ³
Ultimate shear strength (S_{us})	Infill ³	Thickness ¹	Heat treatment ³
Fracture strain (ϵ_f)	Infill ³	Heat treatment ^{1, 2}	Thickness ¹
Shear modulus (G)	Infill ³	Thickness ¹	Heat treatment ³
Proportional limit (τ_{pl})	Infill ³	Thickness ¹	Heat treatment ³

Within the table, 1 = low setting, 2 = middle setting, and 3 = high setting.

Song et al^[6] carried out further research on the mechanical properties of 3D printed PLA via tensile, compression and fracture test. The conventional injection-moulded method was seen

as a control group for comparison. Their results showed that the specimens performed asymmetry in tension/compression test in each direction and displayed tougher in the printing direction than that in the transverse printing direction in the fracture test. Also, compared to the traditional injection-moulded method, 3D printed PLA owned more robust character and performed better behaviour in mechanical property.

In order to enhance mechanical properties, fibre-reinforced polymers could strengthen physical performance, reduce warpage and improve dimensional stability during printing process^[70]. Hinchcliffe et al^[5]. combined 3D printing and fibre-reinforced method and tested six classes of specimens with two different reinforcement fibres and post-tension force for tension test; 4 class specimens with non- or reinforcement and post-tension force for the flexural strength test. The printing parameters were set as 215 °C, 0.2 mm and 60 mm/s for nozzle temperature, resolution and printing speed, respectively. Moreover, all the printed specimens were let to cool down under room temperature. The results showed that the tensile strength and modulus of fibre-reinforced PLA increased by 116% and 62%, respectively. The flexural strength and modulus increased by 14% and 10%, respectively.

2.5 Filament deformation and bonding

Several papers have envisioned polymer sintering during the FDM process. Bellehumeur et al.^[71] studied filaments coalescence in the rotation modelling process. They used high and low-density polyethylene resins in powder form to perform the experiments. The powder particle size distribution was from 250 to 600 µm and the pictures taken were analysed by LECO 2001 (42) software. After that, they found for the high-density ones, the bigger the size of the powder particles, the higher the rate of the sintering. However, for the low-density ones, the trends did not show distinctly as high-density ones, except for the initial duration of the whole sintering process. Besides, they have explored the influence of particle geometry, whose results showed that the sintering rate of irregular powder particles is slightly slower than the cylindrical particles during the first 200s, which may be due to shear and thermal profile according to the authors. Therefore, they concluded that the critical the

factors influencing the rate of sintering are the viscosity of polymers and particle size. Compared the experiments result with previously developed models, Hopper's model^[72,73] is the most satisfying fitting the curve than others.

Pokluda et al.^[74] proposed the modifications to the Frenkel's model. They made assumption to reduce the variables in the model with similarly simplification as Frenkel. Additionally, the particle radius of the polymer was considered in this model. The authors compared their results with different models and a similar trend was achieved at the initial stage with the predictions of the Frenkel-Eshelby model, Hooper's model, Jagota and Dawson model and Modified Frenkel model. When compared with the exist experiment data, the present model showed some deviation in the late stage. However, the main satisfying things for this model are the simplification of the numerical Simulink with considering viscoelasticity, and it could add more possible variables in the industrial process.

Sun et al.^[22] investigated the acrylonitrile butadiene styrene (ABS) bonding formation during an FDM process through experimental work. ABS was first heated to 270 °C and then extruded to the platform in 70°C. 0.003" K type thermocouples monitored the temperature of the printed part via a DAQ assisted by LabVIEW. Then the specimens were sectioned by Leica RM2165 microtome. Pictures of cross-section were taken with a CCD camera installed on an Olympus BX60 microscope and features were analysed using Image-Pro software. Then, the flexural strength of the specimens was tested, and the results exhibited that the strength of the original specimens is stronger than the ones cut from the originals. They claimed that this might be due to two possible reasons: (1) inner fault of the printed products escaped from being discarded, leading to the more significant standard deviation. (2) The temperature history is non-uniform in different position resulting in varying flexural strength. Furthermore, the temperature evolution also influences the flexural strength in the vertical orientation, due to the times of the reheating process. In the end, the authors predicted the bond formation by building a numerical model, which mainly includes two phases: First, when the filaments extruded from the nozzle, the temperature of the filaments is higher than 200°C. This stage is the contact and wetting section with the adjacent filament. After that, the filament is well-contact and enter into diffusion section, whose temperature is below than

200°C even with the reheating process.

In 2004, Bellehumeur^[75] and workers launched a study on filaments bonding in the FDM process. They hypothesised ideally that the geometry of the printed filaments' cross-section is circular. After that, they combined the sintering experimental data, which was under a controlled condition, with the heat transfer model to predict the neck growth of adjacent filaments. The sintering experiments in this operation were setting in a Mettler FP82 hot stage, while monitored by a CCD camera with an optical microscope. The diameter and thickness of the extruded specimens from an FDM machine, are 0.47 mm and 0.3 mm, respectively. And the coalescence experiments were under two circumstances that are constant temperature and ramped temperature. The images were analysed by ImagePro software. Combining a Newtonian sintering model of polymer and Lumped capacity heat transfer model, a prediction model was built for the neck growth and this can be used study the effect of different variables. Analytical results produced with commercial software MINITAB showed that the extruded temperature had a less vital influence on the envelope temperature than the cooling process, and the increase of extruded temperature had promoted the neck growth. Another critical parameter was the convective heat transfer coefficient, which has strong repercussion on both the temperature evolution and bonding process. With the increase of the convective heat transfer coefficient, the temperature of the extruded filaments drops faster and the dimensionless neck radius (neck radius/ particle radius) is also decreased.

Sun et al.^[76] used ABS P400 made by Stratasys FDM 2000 as the specimens to carry out their experiments. The nozzle temperature and envelope temperature were set as 270°C and 70°C, respectively. 0.0118mm K-type thermocouples were used to monitor the temperature history of the extruded filaments and the measured temperature of the tip was between 258°C to 260°C. A CCD camera, mounted on an Olympus BX60 optical microscope, was used to take pictures of the cross-section of the sectioned filaments. The data was recorded and analysed by a high-frequency analogue-digital converter, data acquisition card 6024E and LabVIEW 6/7 software. The consequence of the experiments indicated that not only because of the temperature above the sintering temperature and the influence of the gravity resulting

in the creep deformation^[75]; but also due to the time when the temperature is higher than the T_g and lower than the critical sintering temperature, the bottom voids are smaller than the top. Two geometries had been applied in studying the thermal properties and mesostructured: (1) 200×12×2.8 mm, in which filaments were deposited in the longitudinal direction; (2) 130×28×2.8 mm, in which the filaments were deposited in the lateral direction. The average temperature of the lateral part (112.2°C) is higher than the longitudinal part (99.9°C). Hence, the final results showed that the dimensionless radius of the laterally part (0.45) is also higher than the longitudinal part (0.35). Besides, the envelope temperature has a significant influence on the quality of the bond growth, and the repercussions of the variations in the convective conditions should also be considered.

Abinesh et al. ^[77] reported an in-process laser local pre-deposition heating method in order to strengthen the bonding quality during the FDM process. This method demonstrated in their paper could help local ABS precisely where the next bonding process happened, and the temperature was just above the T_g without bringing geometry deformation. For this experiment used 2.89 mm diameter ABS filaments and 0.5 mm nozzle. The initial temperature of the nozzle and platform were set at 230 °C and 110 °C, respectively. The laser heating system was mounted with a lens system around the 3D printing platform as showing in **Fig. 6**. In the 3-point bending tests, the samples were made by laser pre-deposition heating method showing more than 60% increase in strength, near 100% growth in elasticity and more than 300% toughness than the original extruding process with the same condition. Moreover, they investigated the influence of the laser intensity, and found that there existed a peak point of laser intensity that made the flexural strength increase about 50%, other laser power points could not reach the maximum strength, which was due to the lack of enough energy to heat the previous layer in the beginning several points or over-heating led the evaporation defect in the later points.

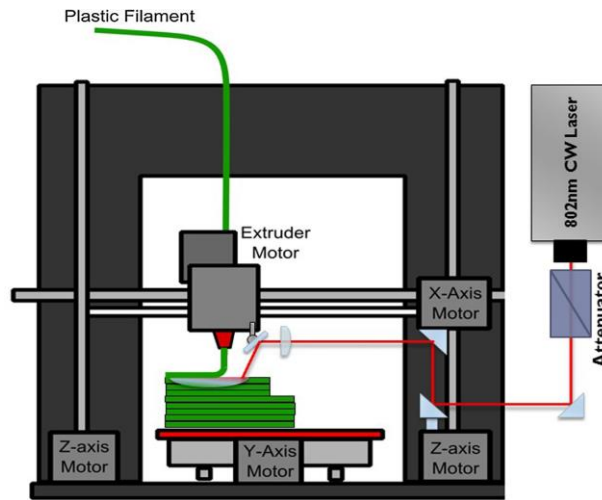


Fig. 6 In-process laser local pre-deposition heating illustration^[77]

During the cooling process in 3D printing, the buckling and residual stresses will appear, which will lead to printed part warping and delamination. The temperature evolution plays an indispensable role in bonding quality and mechanical properties of final products. Therefore, it is significant to know how the temperature evolution of the printed part is affected by process variables.

2.6 Crystallization kinetics and spherulite growth of polymers

The pioneers of crystallisation kinetics theoretically researches were Turnbull and Fisher in 1949^[78]. The aims of crystallisation models are two aspects: First, to explain growth kinetics and morphology of crystal. Second, to provide analytical or numerical methods by which the measured kinetics and structure development of real process could be predicted. The predictable behaviour of crystallisation in a numerical model is useful, which could be applied to the processing and manufacturing process to control and develop the industrial process.

Considering molecules in a liquid at a temperature below the melting temperature point. The free energy of the solid is lower than that of the liquid. Therefore, there is a thermodynamic driving force to push the liquid transfer to the solid. However, the first reliable aggregate to form must be a tiny one that has a large surface-to-volume ratio. It is required that

considerable energy has been stored in the surfaces, described it as σ . If ΔG_v represents the energy difference per unit volume between the solid and the liquid, then the free energy of an embryo is the sum of bulk term and surface term as Eq. (1):

$$\Delta G = 4\pi r^2 \sigma + \frac{4}{3}\pi r^3 \Delta G_v \quad (1)$$

Where, r is the radius of the embryo. And a relationship between r and ΔG would be as sketched as in **Fig. 7**. Where y_1 and y_2 is the surface energy and energy difference between the solid and liquid of the embryo at radius r .

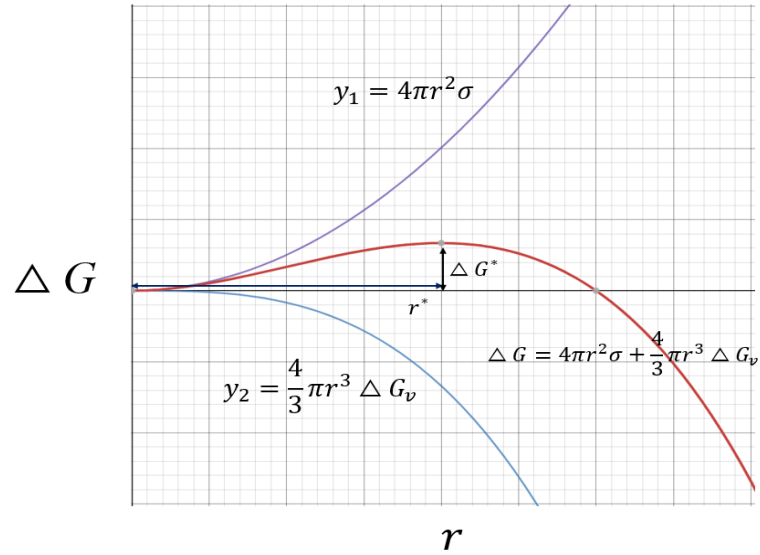


Fig. 7 Sketch of the two terms and their sum as a function of radius

The curve also exhibits the critical radius r^* , where the ΔG gets the maximum value ΔG^* . Once the radius forms more massive than the critical radius, the embryo would become unstable and grow spontaneously. An embryonic crystal would be seen as a nucleus when it reaches that size. The critical radius size could be calculated by the first derivative in Eq. (2):

$$r^* = \frac{-2G}{\Delta G_v}, \Delta G^* = \frac{16\pi\sigma^3}{3\Delta G_v^2} \quad (2)$$

Moreover, from the experimental stand point, this theory grants tuning of the critical radius through the dependence of $\Delta G = -H_f \frac{T_m - T}{T_m}$ on temperature. The critical radius could be expressed as a function of undercooling in Eq. (3):

$$r^* = -\frac{2\sigma T_m^0}{H_f \Delta T} \quad (3)$$

H_f is the entropy of fusion per unit volume, $\Delta T = T_m - T$ is the undercooling for free growth. Thus, when the temperature increases; in other words, the undercooling decreases, the critical radius increases.

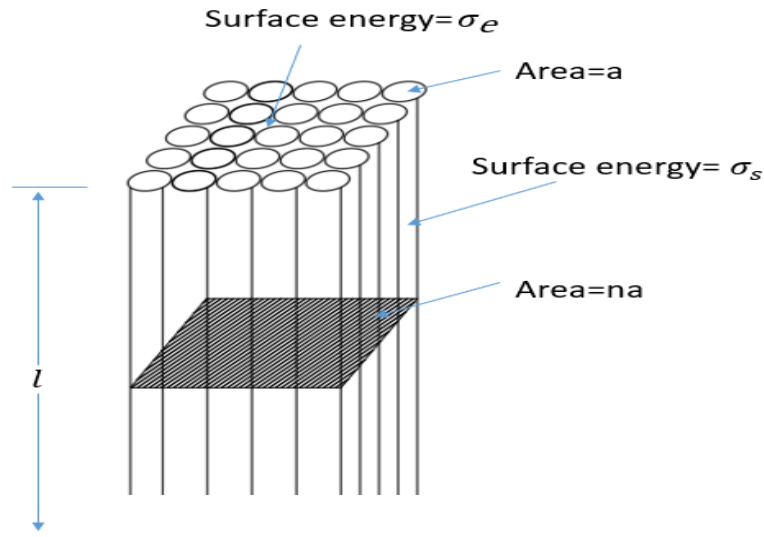


Fig. 8 n chain strands in the process of attachment to a polymer crystal embryo

The time until the appearance of the first crystal is also called the primary nucleation time, which distinguishes it from secondary nucleation time. The primary nucleation describes the first nucleus form process. For polymer crystallisation, we could assume that an embryo consists of n strands, each of length l and cross-section, a as shown in **Fig. 8**. The surface free energies for the ends and the sides of this cylindrical embryo are given by σ_e and σ_s , respectively. Therefore, the free energy difference between the embryo and the liquid is in Eq. (4):

$$\Delta G(n, l) = 2na\sigma_e + 2l\sqrt{\pi na}\sigma_s + nal\Delta G_v \quad (4)$$

For a given length of the strand l , the critical number of strands n^* , at which makes embryo has sufficient volume-to-surface ratio that driving force outweighs the surface term and the embryo could spontaneously grow, is expressed as Eq. (5):

$$n^* = \frac{\pi l^2 \sigma_s^2}{a(l\Delta G_v + 2\sigma_e)^2} \quad (5)$$

If we further search the available minimum energy barrier, then the l could get from the first derivative of $\Delta G(n^*, l)$ versus l in Eq. (6):

$$l = -\frac{4\sigma_e}{\Delta G_v} \quad (6)$$

Then, the critical n^* and ΔG^* would be in Eq. (7) and based on the relationship between ΔG and temperature, we could obtain the Eq. (8) mathematically:

$$n^* = \frac{\pi \left(-\frac{4\sigma_e}{\Delta G_v}\right)^2 \sigma_s^2}{a \left(\left(-\frac{4\sigma_e}{\Delta G_v}\right) \Delta G_v + 2\sigma_e\right)^2} = \frac{(C\sigma_s)^2}{a\Delta G_v^2} \quad (7)$$

$$\begin{aligned} \Delta G^* &= 2 \frac{(C\sigma_s)^2}{a\Delta G_v^2} a\sigma_e + 2 \left(-\frac{4\sigma_e}{\Delta G_v}\right) \sqrt{\pi \frac{(C\sigma_s)^2}{a\Delta G_v^2}} a\sigma_s + \frac{(C\sigma_s)^2}{a\Delta G_v^2} a \left(-\frac{4\sigma_e}{\Delta G_v}\right) \Delta G_v \\ \Delta G^* &= \frac{2C^2 \sigma_s^2 \sigma_e (T_m^0)^2}{\Delta H_f^2 \Delta T^2} \end{aligned} \quad (8)$$

Although the primary and secondary nucleation both contribute to the increase in the number of crystals in the system. Primary nucleation describes the transition to a new phase that does not depends on the new phase already being present. Secondary nucleation depends on crystals which are already present. Hoffman and co-workers^[27,29,31,79] developed the linear growth rate theory, in which they studied linear polyethylene ranging in molecular from 3600 to 80700. And their model for the crystallisation growth rate is given by Eq. (9):

$$G = G_0 \exp\left[\frac{-U^*}{R(T_c - T_\infty)}\right] \exp\left[\frac{-K_g}{T_c \Delta T f}\right] \quad (9)$$

Where, G_0 is the pre-exponential factor containing terms which are essentially temperature-independent, K_g is the surface nucleation constant which is given by

$$K_g = \frac{nb_0\sigma\sigma_e T_m^0}{\Delta h_f k}, \text{ in which } n \text{ is a constant that depends on the different crystallization}$$

regime (2 for regime II, 4 for regime I and III), σ and σ_e are the lateral and end-surface free energies, b_0 is the molecular thickness, Δh_f is enthalpy fusion and k is the

Boltzmann constant, U^* is the activation energy for transport of crystallizable segments to the crystallization front, T_∞ is the temperature below which such motions cease, T_c is the crystallization temperature, $\Delta T = T_m^0 - T_c$ is the degree of undercooling, T_m^0 is the

equilibrium melting point and could calculate by $T_m = T_m^0 \left(1 - \frac{1}{\gamma}\right) + \frac{T_c}{\gamma}$, where T_m is the

observed melting point and γ is the ratio of the initial to the final lamellar thicknesses. f ,

which is given by $f = \frac{2T_c}{T_m^0 + T_c}$, is a factor which accounts for the variation in the enthalpy

fusion Δh_f with temperature.

Vasanthakumari et al.^[80] first carried out experiments for poly(L-lactic acid) to investigate the nucleation constants using in Hoffman-Lauritzen theory. In the pre-processing, they prepared different molecular weight PLLA supplied by Boehringer and Sons, Ingelheim, Germany. The feedstocks were purified by recrystallisation from ethyl acetate before using and dissolved in methylene chloride (distilled over CaCl_2), precipitated in methanol, dried in vacuum at 55 °C, and then stored in phosphorous pentoxide to avoid water causing degradation. In crystallisation growth experiments, the samples were first heated above the melting temperature of around 10-15 °C for 10 minutes, then cooled down rapidly to the set isothermal crystallisation temperature. Then, the radius growth rates were calculated by the

radius growth difference divided by time. Radius growth of samples was measured by a hot stage microscope. The results showed that the growth rate was first increasing to the maximum plot and then decreased with the temperature increased. And the growth rate of samples was dependent upon the molecular weight, the heavier the samples were, the slower the growth rate at the same temperature. However, all the samples reached the peak at the same temperature.

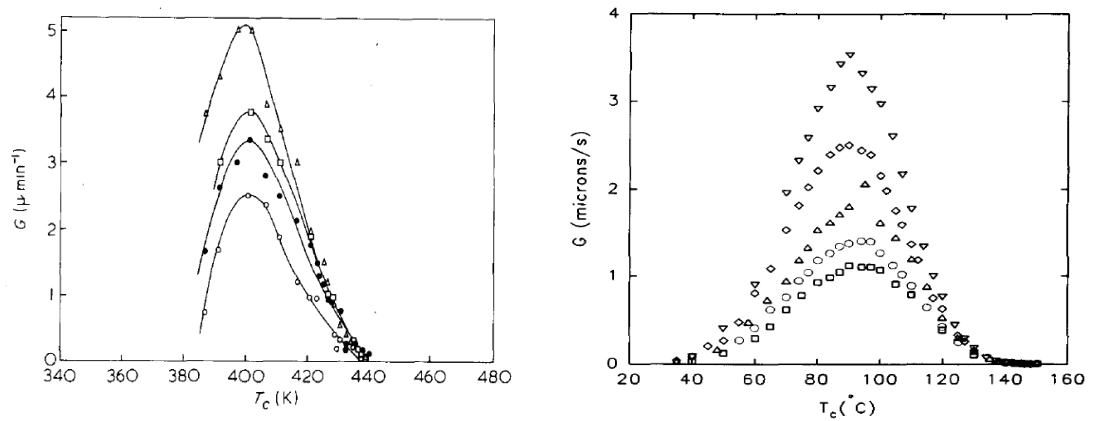


Fig. 9 Radius growth rate, G as a function of crystallisation temperature^[80,81]

Pearce et al.^[81] studied the spherulitic growth rate of isotactic and atactic(0-50 wt.%) bacterial poly(β -hydroxybutyrate)(PHB). The bacterial and atactic samples were procured from Marlborough Biopolymers (Billingham, UK) and Polysar Rubber Corporation (Sarnia, Ontario, Canada), and were dissolved into N, N-dimethylformamide (DMF) for purifying and chloroform separately. Then, the purified samples were reprecipitated into a significant excess of diethyl ether. By using standard polystyrene samples for calibration, the molecular weight was measured by gel permeation chromatography in chloroform. For isothermal crystallization experiments, specimens were heated to 15 $^{\circ}\text{C}$ above the melting temperature holding for 1 minute, then quenched to the pre-set temperature. The spherulitic growth rate was measured by optical polarising microscopy. The results illustrated that the higher the purity of the specimens, the larger the maximum growth rate at the same crystallisation temperature. And with the purity of samples decreasing, the temperature of maximum growth rate was right-shift. They also indicated that the ratio of the regime three to regime two is 2.4, which is higher than predicted 2.0, and the best-fit activation energy and motions

cease temperature for Hoffman-Lauritzen equation was 2700 Cal/mol and 223 K rather than 1500 Cal/mol and 226 K, respectively. Meanwhile, they suggested that the regime transition temperature was the same when the purity of the specimen changed.

Jiang et al.^[82] used the poly(l-lactide-co-meso-lactide) by mixing l-lactide and meso-lactide under nitrogen in a flask as samples to study. The mixture was heated at 180 °C for 4 hours, then dissolved in dichloromethane, filtered and precipitated from methanol. The copolymer was dried in a vacuum chamber for the whole night until the weight was constant. Next, the samples were further dried at 80 °C for another 48 hours in a vacuum environment and erased previous thermal history by melting at 30 °C above observed melting temperature for 3 minutes. In the end, the samples were rapidly cooled to the crystallisation temperature for 15 minutes to observe and record the growth rate via an Olympus intervalometer with the microscope setup (Olympus BHSP-300). According to the results of their experiments, they indicated that bulk degrees of crystallinity increase with decreasing the degree of supercooling, except for sample D, where it was constant with a varied supercooling degree. Besides, they did not observe any regime transition from growth rate or L-H plots. Therefore, they claimed that the crystallisation process in their isothermal experiments happened in the same regime.

Hideki et al.^[83] studied morphology with the crystal regime transition periods. They prepared the samples by chloroform solutions and spanned on a glass substrate surface to create a thin PLA film. Then the samples were heated to 220 °C and maintained 30 s. After that, the temperature was rapidly cooling down to the pre-set isothermal crystal temperature with the range of 100-160 °C. Crystalline morphology and growth rate were observed by an optical microscope (Nikon OPTIPHOTO-2) assembled with a phase-contrast lens, and the melting temperature was determined by differential scanning calorimetry (DSC) measurements. When isothermal crystal temperature was higher than 150 °C, the shape of crystals would be hexagonal until contact with other growing crystals, and when the temperature is lower than 150 °C, the shape of that tended to be spherulitic. According to their isothermal growth rate results, two peaks were presented, and the first peak happened at around 130 °C, and when temperature decreased to 120 °C, a sudden increase happened which indicated the regime

transition of the crystallisation.

Hideto et al.^[84] first reported the spherulite growth rate of multiarm PLLA. In their report, they declared that linear and 3-arm PLLA was used in their experiments and the specimen was purified by reprecipitation, which was used by methylene chloride and methanol as solvent and nonsolvent, respectively. Then, the samples were dried in a vacuum environment for one week, melted at 200 °C for 3 min under nitrogen gas, and crystallised at desired crystallization temperature between 100 and 160 °C for 600 min. The spherulite growth was observed via Olympus polarisation optical microscope (BX50) installed with a controlled heating stage. The results indicated that the growth rate for linear and 3-arm PLLA was in the range of 1-70 $\mu\text{m/s}$ and 1-8 $\mu\text{m/s}$, respectively. When absolute number average molecular weight of linear and 3-arm PLLA was similar, the former growth rate is much higher than the 3-arm sample. Besides, with the decrease of the absolute number average molecular weight, the linear PLLA growth rate increased. Nevertheless, the 3-arm sample showed independent results between absolute number average molecular weight with growth rate, which was due to branching effect. The regime transition temperature was 120 °C both in linear and 3-arm PLLA samples, which also proves previous results. ^[80,81,83]

Tsuji et al.^[23] investigated the different effect of molecular weight, tacticity and comonomers for PLLA on the spherulite growth behaviour. They have prepared four types copolymers: PLLA, P(LLA-DLA), P(LLA-GA) and P(LLA-CL). The results are shown in **Fig. 10**. The growth rate decreases with the comonomer content increases and the larger the molecular weight, the slower the growth rate. Besides, not all regime transition presented. **Fig. 11** illustrated that with increasing molecular weight, less remarkable regime transition happened even all periods remained at regime II without transition in **Fig. 11** (b), (c) and (d).

Takahiko et al.^[85] used wide-angle (WAXS) and small-angle (SAXS) X-ray scattering, optical microscopy and differential scanning calorimetry (DSC) to investigate the melting and crystallisation behaviour. The specimens supplied by Toyota Motor Corporation were melted at 200 °C to eliminate temperature profile, then cooled down to the isothermal crystal temperature in the range of 80-160 °C. The regime transition temperature for regime III to

regime II and regime II to regime I happened at about 120 °C and 150 °C, respectively^[23]. They also found that when crystall temperature below 90 °C, the shape of the formed crystal would be a hexagonal phase, which supplied the previous work PLLA in regime III morphology^[83].

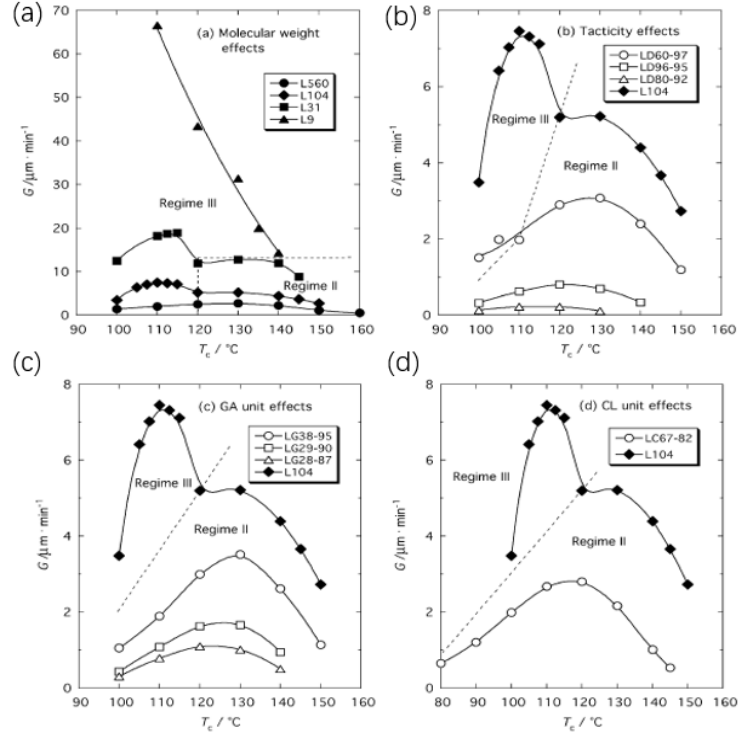


Fig. 10 Spherulites radius growth rate (a) PLLA, (b)P(LLA-DLA) (c) P(LLA-GA), and (d)P(LLA-CL) films in different crystallisation temperature^[23]

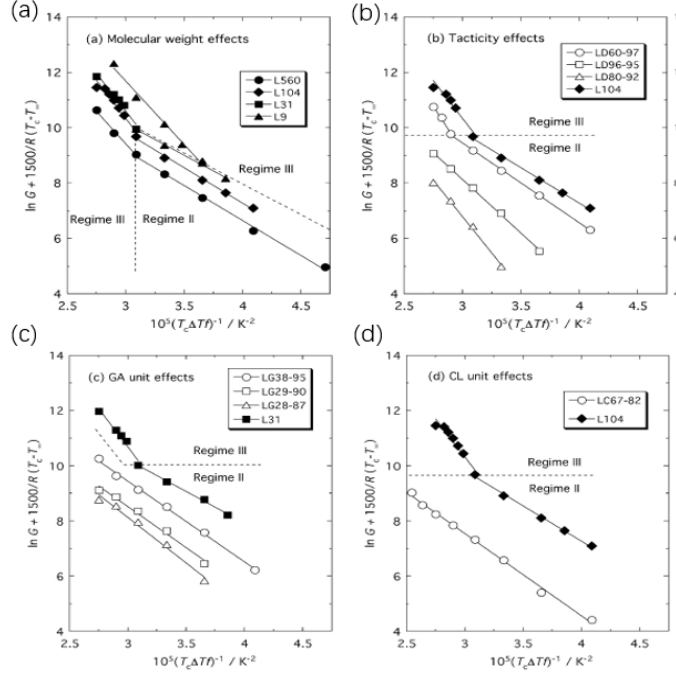


Fig. 11 $\ln G + 1500 / R(T_c - T_\infty)$ of (a) PLLA (b) P(LLA-DLA) (c) P(LLA-GA) and (d) P(LLA-CL) as a function of $1 / (T_c \Delta T f)$ [23]

Pan et al.^[86] researched the crystalline structure of PLLA in the range of 100-120 °C, which is the ordinary temperature for the industry to process PLLA. The feedstock was first purified in ethanol from chloroform solution and dried in a vacuum environment at 40 °C for one week. Then the samples were melted to 200 °C for 2 minutes to erase temperature history, then cooled down to isothermal crystalline temperature. In their experiments, they found that the growth rate showed bimodal with a minimum value at 110 °C between two peaks. The different crystallisation temperature led to different final morphology, and they found that 110 °C was a critical temperature. When temperature below 110 °C, α' crystals would be formed and when the temperature is higher than the critical temperature, α crystals would be produced. Besides, the α crystals unit cell is little smaller than α' crystals under the same growth time and other conditions.

Santis et al.^[24] systemically investigated the crystallisation of PLA by both experimental and numerical methods. They carried out isothermal crystallisation with PLA samples whose glass temperature and melting temperatures are 58 and 153 °C, respectively. And they used

DSC to detect the transition behaviour of PLA. The isothermal crystallisation temperature they used ranged from 95 to 125 °C changing with 5 °C per step. For the theoretical calculation, they combined Avrami and Hoffman-Lauritzen theory to calculate crystal rate and nucleation and growth rate behaviour. As a consequence, good agreement was between experimental data and calculation result.

Apart from the previous isothermal way for studying crystallisation growth rate of PLLA, Chung and his co-workers^[87] first came up with non-isothermal method via a differential equation to overcome the difficulty that peak growth rate was too high to measure by conventional method. In their experiments, the powder samples were first melted at 400 °C for 25 min for compression moulding, then cooled down into an ice-water bath. Their isothermal experiment was similar with previous work that was erased the temperature profile and cooled down to different crystallisation temperature via polarized microscopy to measure radius increase with time. When they did the dynamic work, the sample was heated to 400 °C for 200 min to erase temperature history, quickly cooled down to 302 °C for 50 minutes in order to get sufficient spherulites, then observed the crystallisation process from 302 °C to room temperature with a 10°C/min cooling rate. The differential equation is given by Eq. (10) and (11):

$$R = f(T), T = g(t) \quad (10)$$

$$\frac{dR}{dT} = f'(T) = \frac{dR}{dt} \frac{dt}{dT} \quad (11)$$

Where, R is the radius of spherulite, temperature T is a function of time t. Therefore, the temperature-radius relation is shown in Eq. (11). **Fig. 12** showed the results were in well matching between two methods.

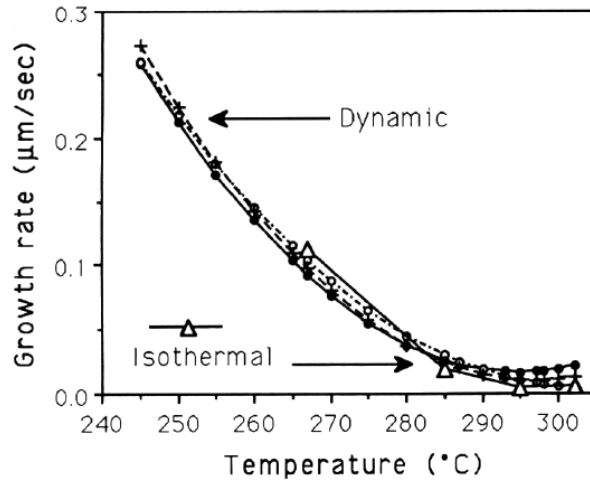


Fig. 12 Comparison of dynamically calculated growth rate with isothermal growth rate ^[87]

Maria et al.^[88] combined isothermal and non-isothermal methods to measure PLLA spherulite growth rates. The specimens were first heated from 30 °C to 200 °C at the rate of 20 °C/min and kept for 2 min for melted sufficiently. Then the samples cooled down to 145 °C with the rate of 1 °C/min and kept at that temperature until the first spherulite was observed. After that, the sample was reheated to 175 °C, cooled down at the rate of 1°C /min. The whole process was under observed via a Zeiss polarizing microscope assembled a Linkam TMHS 600 heat stage. As a comparison, the conventional isothermal crystallization process was under the cooling rate of 30 °C/min to reach the desired temperature. The results showed a good match between non-isothermal and isothermal data, and by using Hoffman and Lauritzen theory, they also detected regime II to III transition temperature at 120 °C.

Later, a series of similar studies on PLLA via non-isothermal methods were applied: Yong et al.^[89] used zinc catalysed PLLA to acquire melt crystallisation behaviour, morphology and spherulite growth rate. Tsuji et al.^[90] investigated the effects of different amount of PDLA content to PLLA crystallisation via non-isothermal methods and the results showed that it is beneficial to accelerate overall PLLA crystallisation with the addition of a small amount of PDLA. Leevameng, Tsuji and co-workers^[91] used non-isothermal methods to investigate the PLLA/PDLA blends crystallisation and structural change with different content.

Not only experimental works, modelling research are also investigated by cooperating with temperature evolution, and further temperature evolution work will be introduced on Section 2.6. Christian and Sriram^[19] developed a one-dimensional model to study temperature evolution of polyethylene via using the finite difference method (FDM), which ignored the latent heat. Then, the temperature profile was added to the growth rate equation to simulate the solidification. It was the first to present the details of the orientation of fibrils in different spherulites in simulation.

Piorkowaska^[92] modelled isotactic polypropylene(iPP) film spherulitic crystallisation with uniaxial linear temperature evolution. Besides, they also included the conversion of the melt into spherulites. They simulated the spherulite growth rate of iPP with Hoffman equation, where they set parameters in universal value: $U^*=1500\text{cal/mol}$, $T_\infty=231.2\text{K}$, $K_{gII}=1.47\times10^5$, $K_{gIII}=3.30\times10^5$, $T_m^0=458.2\text{K}$ and the regime transition temperature is at 409.2K. By computer simulation, they also carried out visualise spherulitic evolution patterns to predict the spherulitic structure.

2.7 Temperature Evolution of Printed Filaments

Successful and widespread utilisation of FDM/FFF requires the capacity to generate products with satisfactory dimensional accuracy, surface quality and mechanical performance. These properties are determined by the filament deformation and solidification during deposition and the quality of the bonding contents between contiguous filaments^[75,93,94]. When the filament of the bottom layer is deposited onto the platform, heat exchanges develop with the surroundings by convection and radiation and with the platform or previous layer by interface convection.

Amount of experiments have been investigated in order to explore the cooling rate of the extruded filaments. As is mentioned above, Sun et al.^[22] used FDM 2000 from Stratasys Inc for studying with commercial material ABS 400. The ABS 400 is first heated to 270 °C, then extruded to the 70 °C platforms. The temperature of the printed part was monitored by 0.003”

K type thermocouples whose measurements were saved and analysed through a high-frequency analogue-digital converter, amplifier, data acquisition card 6024E and LabVIEW 6.0 software on a PC. The specimen was collected sectioned by Leica RM2165 microtome. Pictures of cross-section were taken with a CCD camera installed on an Olympus BX60 microscope and features were analysed by Image-Pro software. After collecting plentiful data of experiments, they developed a sintering model to predict the neck growing and use a non-isothermal diffusion model to predict the interface between filaments bonding evolution.

Seppala et al.^[95] purchased ABS copolymer filaments and 3D printing from MakerBot Industries. Then, they used Autodesk Inventor to design high-resolution stereolithography mould and transferred it to G-code by the software the MakerBot Desktop V3.7. The Infrared imaging was captured by CEDIP Jade III mid-wave IR camera whose precision is 1 °C. Besides, when their experiments are in the process, they consider many possible factors which may lead to the deviation, like indirect heating from the high-temperature nozzle, IR thermography emissivity, solid angle and detector efficiency. They applied Plank's law with parameters to correct the curve. From the results of experiments, the temperature of printed filaments decreases quickly, which is less than 2s. Meanwhile, the sublayer gets small part heat transfer from the printed parts, and the sub-sublayer will not reach the temperature T_g .

Recently, Srinivas et al.^[96] also detected the temperature profile of FDM by series experiments. The commercial materials, including different purity and molar mass of PLA, were bought from Corbion Purac. The initial nozzle and platform temperature were set as 200 °C and 70 °C, respectively. The temperature of the platform is just above the T_g . The print speed was set as 20 and 50 mm/s. The sputter-process was used Scanning electron microscopy (SEM) on the Philips XL30 system with conductive carbon tape and thin gold coating to complete. Micro-computed X-ray tomography was used to do the observation for the internal characteristic of the extruded filaments. The acquired pictures were processed by NRecon reconstruction software and noise reduction. Recording printed layer temperature was used by 0.15mm K type thermocouple, which was embedded in the base

plate of the nozzle. The DataQ DI718 data catching system made sure the high accuracy in data with 1 kHz sampling frequency. After detecting the temperature profile by the ultrathin thermocouple, they concluded that a steep peak was shown after each new filament printed on the previous one, the time for temperature increase and decrease is transitory. Additionally, when the speed is 50mm/s, due to the high speed with non-sufficient heat, the maximum temperature is limited by this variable. Another critical factor is that the high motion speed nozzle limited the previous layer cooling entirely, which had a vital effect on crystallization. The previous layer would maintain at relatively high temperature with a large crystal growth rate, which will be beneficial for the rate of final crystallisation in the printed filament. And due to the PLA is water-soluble, it is convenient to not break the temperature history to separate each layer for observation.

After abundant experiments work has been done, lots of research begin to carry on the Simulink and modelling work of the temperature profile and history. Yardimci et al.^[12] published one of the first papers to focus on the indispensability of the heat transfer analysis for the cooling down process. They mainly focus on four aspects, including liquefier thermal design, melt front location analysis in the liquefier, the degree of cooling in the tip and build a detailed deposition model. For the Modelling, they have made some assumptions during the deposition progress: First, previously deposited materials have enough time to cool down until the temperature not change; then reduce the temperature dimension by averaging across the thickness; finally, using sink terms to model effectively the heat transfer to environments and bottom layers. They identified Peclet and Biot number as important parameters from dimensional analysing and the effects of these parameters are showed in the numerical experiments.

Thomas et al.^[97] developed an analytical heat transfer model assuming rectangular cross-section in shape of the extruded filaments to predict fracture strength based on reptation theory for the interdiffusion of long-chain polymer molecular, neglecting the thermal resistance. The most influential factor in the cooling process they found is filament cross-section area, and the time to cool down to T_g takes from 1.74 to 15.9 s for the half and

double-sized Fused Deposition nozzle, respectively. However, there still exists some limitation and approximations, such as: the nozzle speed is an essential factor, and their model did not include the heat transfer along the printing direction. Consequently, the bond strength increased when the solidification times of filaments are prolonged. Besides, post-fabrication annealing also increase bond strength.

Bellehumeur et al.^[75] model the bonding quality among polymer filaments in the fused deposition process, investigate the bond formation among extruded filaments and analyse the thermal condition during the cooling period by lumped-capacity analysis. they established this model using lumped-capacity model with three assumptions: (1) consistent temperature distributed the cross-sectional area of the extruded parts; (2) semi-infinite length of filament; (3) constant heat conduction and convection coefficients. The first two hypotheses were used to simplify the geometry of the filament. After the numerical model, they found that with the increasing of the initial temperature, the quality of bonding between two adjacent filaments grows.

Due to the thermal contact resistance, r_c ($m^2 \cdot ^\circ C / W$), a temperature difference arises at the interface^[98]. The value of r_c is governed by surface roughness, the hardness of the materials, applied pressure and temperature^[99]. Sun et al.^[76] compared the result from both experiments data and two model methods, and they found the presented heat transfer model describing the FDM process is not adequate, because underestimating the heat transferred through conduction, or missing variations in the convective conditions and during fabricating process. The final mesostructured and strength of extruded filaments sincerely rely on the temperature history repercussions.

Zhou et al.^[13] used the interfacial conductance approach to model the heat conduction from the bottom surface. For the convection coefficient function, they adopted machine learning algorithms such as Artificial Neural Networks to avoid using a traditional solution like time-consuming and iterative solution. Moreover, Artificial Neural Networks aims to build implicit regression with several hidden layers, which include linear or non-linear computing elements and blocks. These elements and blocks interconnect together to activate function

as neurons, which works like a linear combiner with multiple weighted inputs in the hidden layer. Based on the above method, the coefficient of heat transfer process was trained by the machine learning program, and this developed model could offer essential advantages when referring to the heat transfer coefficient. However, the accuracy of the model needs to be increased, the response by the infrared camera should be more sensible, and more complexed boundary situation also should be included.

Zhang et al.^[14] use finite difference method to describe a new approach for thermal simulation, this new approach includes almost all relevant thermal factors, such as surface convection and radiation of filament to the surrounding areas, thermal contact conductance with the substrate and adjacent layers. In this way, they achieved linear time complexity by exploiting spatial and temporal locality based on the experimental data.

Xia et al.^[100] combined the N-S equation for both the air and the polymer in the computational domain on FDM numerical methods. In their numerical model, it not only includes the traditional variables, but the significant changes of this model are also adding the nozzle modelling, the implicit treatment of the diffusion terms in the momentum, the energy equation, temperature and shear-dependent viscosity. The effect of changing the nozzle speed is intuitive than changing the injection temperature only. However, the initial nozzle temperature has a strong influence on the previous layer reheating process.

Moreover, Zhou et al.^[20] contribute to developing both experimentally and numerically process of FDM thermal profile, and the influence of some related things had done, such as platform temperature, nozzle temperature, the speed of nozzle movement and layer thickness. After several experiments, they built the model to predict the FDM process, which had less than 13% in bias, and this bias was lower than the existing simulation methodology and thermal prediction.

Costa et al.^[15-17] developed a series of heat transfer methodology to predict thermal histories at first. Based on the deposition time, the temperature was allowed to change in the printing axis of the filaments. They initially built an analytical solution to research the temperature evolution of the FDM process, including the basement condition and boundary condition

with adjacent filaments. The numerical result showed that contact characteristics have a high impact on temperature history and bonding. After that, they examine the contribution to the overall transfer of various thermal phenomena developing during the manufacturing sequence (convection and radiation with the environment, thermal contact conductance with the support). Consequently, heat transfer during the whole printing process is mainly controlled by convecting with air, thermal contact conductance with the platform and adjacent filaments. The significant factor that radiation with the environment could be neglected when the heat transfer coefficient is larger than $60 \text{ W} / \text{m}^2 \cdot ^\circ\text{C}$. In the end, they have built an analytical solution for the transient heat transfer during filament deposition and cooling profile, taking into consideration of additional various physical contacts and parameters between filaments during the progressive built-up of a three-dimensional structure, which could predict the time evolution of filament temperature as well as the bond quality.

Zhou et al.^[101] modelled a three-dimensional bead on a support platform for thermal analysis in ANSYS, which included the non-linear behaviour of thermal conductivity and specific heat varying with temperature. Latent heat of phase change and the variable distribution by finite element methods with a Chernoff strategy was based on the continuous media theory. Meanwhile, they built this model with the following assumption: (1) the same temperature distributes across the cross-section area; (2) semi-infinite filament length of the printed filaments; (3) the conduction transfer with the platform could be seen as convection cause the mass of the platform is much higher than filaments. The producing program of this model using 'dead' and 'birth' technology method in ANSYS makes the extruded and non-extruded filaments easily identified. The result shows almost fitting to the experiments data curve.

D'Amico et al.^[102] presented an adaptable 3D finite element methods model capable of simulating heat transfer in 3D printing and at sufficiently small time scales to capture the rapid cooling in temperature history. Plenty of details were investigated, which may affect the results, such as: nozzle movement and material deposition, internal heat transfer, material properties and geometry, external heat transfer and boundary conditions. Consequently, they

demonstrated a view of the internal temperature heat transfer of the cooling process, which is not readily achievable with surface temperature measurement experimental methods. And their model method could be satisfied with SLS modelling by changing the boundary conditions.

Holzinger et al.^[18] simplified the calculation of temperature evolution and the results difference is small between the experimental data and theoretical data. They simplified the heat flow only included the convection with air and conduction with the platform. The radiation^[16] and adjacent filaments heat transfer are neglected cause the contact area is small compared with the whole filament surface. The heat conduction between the bead and substrate is seen by a gap conductance h_{gap} , which is used to fit the theoretical data to the experiments. Moreover, they assumed that the cross-section area temperature was uniformed when the Biot number is less than 0.2. In the end, they compared several experiments data with numerical simulation, where the difference is small to predict the cooling process of the temperature evolution.

Poorland et al.^[21] predict temperature evolution and heat flow calculated by a transient enthalpy formulation with moving boundary condition, and the results have been compared with experiments measurements, which would be expanded to forecast the interaction of heat flow and temperature profiles. For the heat transfer, it is complex to include properly modelling conduction in adjacent parts contacting and substrate-filaments heat flow, radiation, natural or forced convection and boundary condition. They began with a 1D thermal problem to solve the 3D cylinder model; records thermograms in-situ while a part was fabricated with the specific parameter; ran the numerical calculation program and compared the theoretical and experiments data; adjusted the heat transfer coefficient until met the convergence criterion. Forward finite differences are used for both time and space domain discretisation to solve the 1D problem.

Zhang et al.^[103] proposed and applied the three-dimensional transient numerical model to calculate the heat transfer during and after the FDM process with boundary condition by the finite difference method. They assumed some hypothesis to build this model: first, the

influence of polymer crystallization latent heat in the heat transfer process is insignificant; then, they neglected the radiation on temperature evolution and thermal expansion and friction of the filaments; in the end, the interior of the constructed component is pore-free, and there is no contact heat resistance between filaments to filaments or filaments to the platform. Based on the above assumptions, they researched several factors which may affect the final products of the FDM: Reheating is one common factor that will influence the previous filaments temperature profile. Cooling rate and temperature gradient are main parameters for temperature evolution. The influence of cooling rate will affect the internal part bonding strength and crystallisation rate while the temperature gradient will lead to the distortion and wrapping situation of stack filaments. The temperature of initial printing, substrate and surrounding are considered as constant temperature. Moreover, another critical factor dimensional effect influence also should be considered. When the dimensional is small that means the resolution of the printed part is high, the heat transfer of printed component is rapid, and the cooling rate is inverse proportional to the square of the resolution which will lead less bonding and crystallization in the internal. Thus, the control of the substrate and air temperature is indispensable. Finally, the effect of the layer thickness and printing speed is compensable for control the temperature of filaments, which allows the main temperature of the filaments stay longer at desired crystal temperature.

2.8 Latent heat of solidification

Brent et al.^[104] numerically investigated the melting of pure gallium phase transfer by the enthalpy-porosity two-dimensional approach, and natural-convection effect was also included. Since the reference melting temperature of gallium is 29.78 °C, they built the boundary condition with initial temperature as 28.3 °C, heated to the final temperature as 38 °C, and the dimension of the samples were 8.89 cm long and 6.35 cm width. All other physical property values were measured at 32 °C: the thermal conductivity was 88.5 W / m · K along the width axis and 16 W / m · K along the height axis. The latent heat part is defined as a function of temperature, as Eq. (12) and (13):

$$S_h = \frac{\partial(\rho \Delta H)}{\partial t}, \Delta H = f(T) \quad (12)$$

$$f(T) = \begin{cases} L & T > T_m \\ 0 & T \leq T_m \end{cases} \quad (13)$$

Where, S_h is the energy-equation source term, T_m is phase transfer temperature and L is heat latent of fusion. By combining with natural convection and conservative momentum and mass part, the numerical results have got sound validation with experimental results.

Brown and Hunt^[105] built a model to simulate alloy solidification numerically. For the diffusion during phase change, they used the energy equation to model the solidification process in Eq. (14) and (15):

$$\rho C_p \frac{\partial T}{\partial t} = k \nabla^2 T + E, E = E_a + E_t \quad (14)$$

$$E_a = \frac{(d - d^0)g_s^0 L}{\Delta t \Delta x \Delta y}, E_t = \frac{d^0(g_s - g_s^0)L}{\Delta t \Delta x \Delta y} \quad (15)$$

Where, E is the latent heat source which includes two parts: 1) E_a the advance of the solidification front; 2) E_t subsequent thickening of the mushy zone. d is the volume of a controlled volume covered by semi-solid at the end of the time step; d^0 is covered at the initial of the time step; L is the latent heat of fusion per unit volume; g_s and g_s^0 are current and initial time step of solid fraction, respectively. And for the g_s , they simplified the solid fraction evolution with linear variation in Eq. (16):

$$\begin{cases} g_s = 0, T > T_L \\ g_s = \frac{T_L - T}{T_L - T_s}, T_s \leq T \leq T_L \\ g_s = 1, T < T_s \end{cases} \quad (16)$$

When the temperature is higher than T_L or lower than T_s , the single-phase heat equation

will be used again.

Wajira et al.^[106] have built a numerical 2D model combined thermal convection into solidification of equiaxed grain. The energy equation they used in this work is Eq. (17):

$$\frac{\partial T}{\partial t} + \frac{\partial(uT)}{\partial x} + \frac{\partial(vT)}{\partial y} = \frac{k}{\rho C_p} \left[\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} \right] + E \quad (17)$$

Where, E is the source term account for latent heat, and E is expressed as Eq. (18):

$$E = \frac{L}{C_p} \left[f_s \frac{\partial V}{\partial t} + V \frac{\partial f_s}{\partial t} \right] \quad (18)$$

L is latent heat of fusion, V is the volume of controlled volume, f_s is solid volume fraction of an arbitrary controlled volume, and is given by Eq.(19):

$$f_s = g_s \left(\frac{V_{ij}}{\Delta x \Delta y} \right) \quad (19)$$

g_s is the solid fraction and calculated linearly with temperature. Non-linear solid fraction could be applied from Scheil equation^[107], which only brought small change for the simulation results. Therefore, in order to simplify the calculation, they used linear solid fraction. The final results showed that four-fold symmetry would not appear in the square mould when presented the convective effects. Duggan et al.^[108] also used a similar source term to account for latent heat for modelling melting and solidification in laser welding.

Bayani et al.^[109] built a model to simulate solidification of A356 aluminium alloy. In their model, the latent heat term is written as Eq. (20):

$$E = \rho \Delta H_f \frac{\partial f_s}{\partial t} \text{ and } f_s = 1 - \left[\frac{T - T_{sol}}{T_{liq} - T_{sol}} \right]^{-\frac{1}{1-k_0}} \quad (20)$$

Where, ρ is the density, f_s is the solid fraction, k_0 is the partition coefficient and T_{sol} and T_{liq} are solidus and liquidus temperature, respectively. Then, the equation could be expressed as a function of temperature in Eq. (21):

$$E = \rho_{\Delta} H_f \frac{\partial f_s}{\partial T} \frac{\partial T}{\partial t} = \rho_{\Delta} H_f \left[\frac{-1}{(T_{liq} - T_{sol})(k_0 - 1)} \right] \times \left[\frac{T - T_{sol}}{T_{liq} - T_{sol}} \right]^{\frac{2-k_0}{k_0-1}} \frac{\partial T}{\partial t} \quad (21)$$

And when the temperature is higher than T_{liq} or lower than T_{sol} , the heat latent will be constant.

2.9 Summary of the literature review

Nowadays, Additive Manufacturing could use Computer-Aided Design method to create a file which can directly input into the AM software system without building a mould before producing. Compared to traditional manufacturing, the AM could bring lots of advantages such as maximise the use of the materials with little waste and customise the final products complex geometries without additional tooling or manufacturing processes^[8-10].

With the fast development of AM technology, more and more kinds of materials could be applied to the AM process. The form of these materials could be filaments, wires, powders and inks depending on different categories of the AM machines. Polymers, considered the most common materials in AM, have been applied in aerospace, structure toy and medical areas^[39,51]. According to the Wohlers Report in 2019^[110], metallic materials became highly popular in AM market, which increased 900% over the last decade,. And the research of metal additive manufacturing should have more focused to sort-out the defects founded in processing periods to acquire better performance in mechanical property. Concrete and ceramics also benefit a lot from using the AM technology, which associate to better-control structure and ingredient of the products. Shape complexed and lightweight structures could also be accomplished for ceramics and concretes, respectively.

Fused Deposition Modelling is one of the most popular AM techniques, first invented by Stratasys, owns lots of advantages, including smooth operation, using environment-friendly materials and economical products and so forth^[41]. Whereas, some of the drawback of this method should also overcome by using reinforced materials to strength mechanical properties and also to increase the accuracy and quality of final products^[42-48].

Poly Lactic Acid is a kind of biodegradable materials, which could be produced from renewable resources^[55]. Excellent biocompatibility and absorbing ability make PLA to become an ideal choice for applying in biomedical applications. Many researchers performed a number of experiments to explore different kinds of properties of PLA^[55,61-65]. Compared with ABS and other 3D printing materials, PLA is more challenging to print as they are stronger and more brittle than other materials. The tensile strength could up to 54 and 58 MPa in 90° and 0° raster angle, respectively. In order to further increase the mechanical properties of PLA, fibre-reinforced methods such as graphene, carbon nanotube and nano clay etc. are applied to make better performance^[5,70].

Since 1949, Turnbull and Fisher have started research on the crystallisation kinetics and spherulite growth of polymers^[78]. Then, Hoffman and his co-workers^[27-30,79] make a significant contribution in this area and proposed a well-known equation used for crystallisation growth rate. Later, lots of isothermal^[23,24,80-86] and non-isothermal^[87-91] experiments were reported for different kinds of materials and the results were in good agreement with the previously presented equation by Hoffman.

FDM has the ability to produce the desired products in good accuracy, surface quality and mechanical properties. These properties are determined by the filaments extruded from the nozzle and the deformation and solidification affected by the surroundings. The key parameter for solidification of extruded materials is the temperature profile during the whole process, which will be influenced by the nozzle speed, surrounding and platform temperature, materials properties and boundary condition. When the temperature evolution could be found real-time during the whole process, the crystallization and nuclear forming could be predicted, which will lead to the final mechanical property of the products. Both experiments^[22,95,96] and modelling works are studied in recent years. Several scientists built one-dimensional model to simulate the temperature changes^[12-19]. However, the temperature distribution in the vertical direction could not be performed, and the results have an obvious deviation with the experiments data. The three-dimensional model applied in Zhang's work suffers by limitations and the time to calculate the whole process is too complex to be accepted, which should be optimized in the future^[101,103]. Therefore, two-dimensional model

would be desirable now and could be applied in widespread^[20-22]. However, previous two-dimensional model had only considered limited parameters, which is reasonable for the deviation between the experimental and numerical results. And there were no previous works which simultaneously considers the crystal kinetics into consideration when the process is influenced by different critical process parameters such as printing speed, thermal convection coefficient, thermal contact conductance with platform, thermal contact conductance with filament, nozzle diameter and crystallization latent heat. Moreover, there were no previous works investigating the temperature distribution during printing. Therefore, a generalised model for these aspects needed to be developed. To fill this gap, the current study contributes a novel model to simulate the whole process in order to get a better understanding of included phenomena and optimise the process parameters. This would be the first reported model of its kind which could help to predict the crystal structure and the quality of the final printed products.

Chapter 3 Model development

This chapter mainly presents the modelling methods, basic heat transfer and crystallization theory. In the first section, the derivation of the two-dimensional domain discretisation methods is presented and concerning implementation simulation in the single filament extruded from the FDM printer. Moreover, a moulded product printed by FDM is made layer by layer. Therefore, as an example of the layer by layer process, a 2×2 multi-line simulation work has been also presented in this section, which include all the boundary situation in the whole manufacturing process of FDM. Then, the theoretical polymer crystallisation work by Hoffman-Lauritzen theory has been outlined to model the crystallisation process as a function of temperature. Section 3.3 provides the flow chart of the general procedure to simulate the whole process.

3.1 Model of temperature evolution

3.1.1 Two-dimensional domain discretisation in single line works

In the transient or unsteady state engineering problem, the field variables such as displacement, stress, temperature and velocity are not only as a function of the spatial coordinates but also changing with time. In order to analyse these parameters varying with time, the first step is to discretise the body into smaller and regular subdomains, known as finite elements. This discretisation process is equivalent to replacing the domain which has an infinite number of freedom degrees, by a system having a finite number of freedom degrees.

Fig. 13 shows the schematic of the two-dimensional domain discretisation in an arbitrary body. In domain discretisation, there are four essential geometry factors: (1) Gridline. The gridline partitions the domain into the subdomain;(2) Nodes. The plots where gridline intersects with another gridline or boundary line are called the nodes;(3) Element. Element is also called controlled volume. Every node represents a change in a small area around it,

and the size of the controlled area depends on the distance of two gridlines;(4) Interface. The boundary of two elements doubled is called the interface.

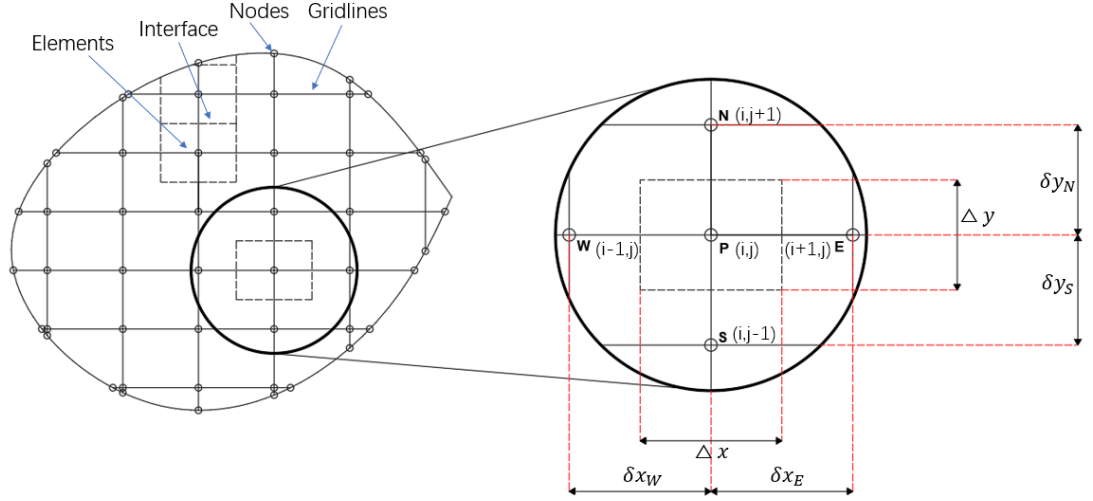


Fig. 13 Schematic of two-dimensional domain discretisation

For a random plot $P(i, j)$ in this domain, the upper, down, left and right node could be expressed as $N(i, j+1)$, $S(i, j-1)$, $W(i-1, j)$ and $E(i+1, j)$, respectively. The distance between P and the rest of the nodes are δy_N , δy_S , δx_W and δx_E , respectively. Moreover, the area of the plot P controlled volume is equal to $\Delta x \times \Delta y$. Therefore, based on this theory, the extruded filament has been meshed and simplified, as shown in **Fig. 14**.

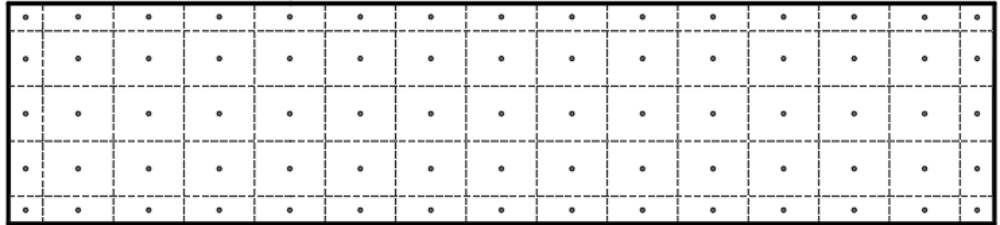


Fig. 14 the meshed filament extruded from the FDM nozzle

In the proposed two-dimensional model, it was assumed that the heat flux only happens in x and y direction, and there will be no heat transfer in the z -direction, which means the

temperature will be uniform in the z-direction. Besides, the filaments' cross-section shape after extruded from circle nozzle was simplified as rectangle shape. The cross-section would keep this shape all the time, neglecting the gravity effect and the density change with temperature. However, the area of the cross-section would be used as same as the nozzle cross-section. When the controlled volume is close to the boundary, the size of it will be more sensitive, which will benefit the simulation results more accurate. The boundary and the corner elements area will be 1/2 and 1/4 of the standard elements.

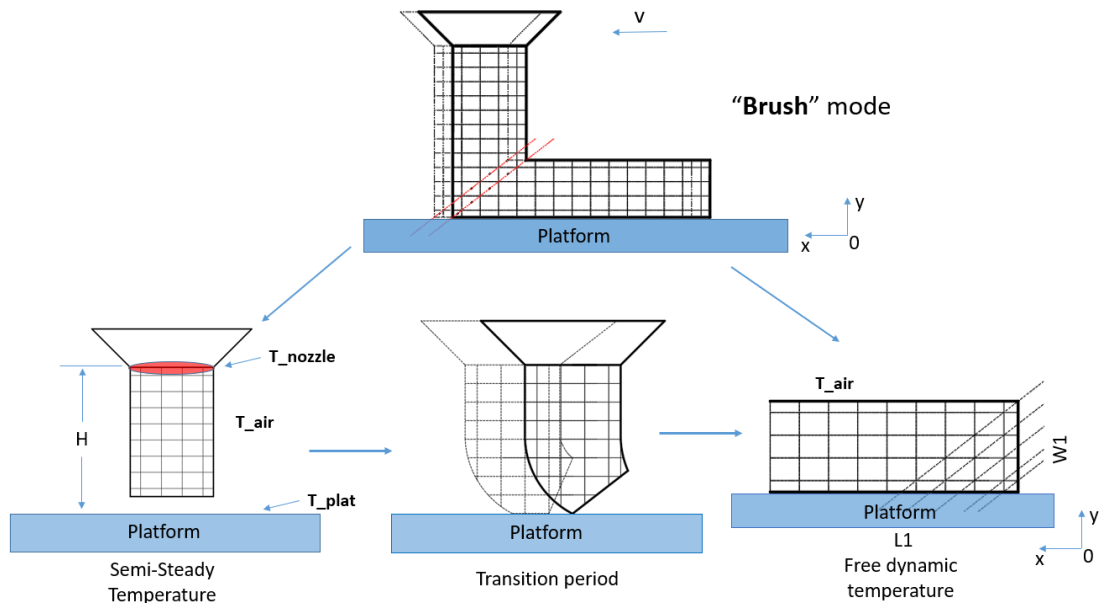


Fig. 15 Three periods of extruded filaments cooling down with two-dimensional mesh methods

Then, the filament extruded from the nozzle, reaching the platform until completely solidification, could be divided into three periods as shown in **Fig. 15**. The first period is the semi-steady temperature evolution periods. When the filament is heated over the melting temperature, the filaments would be melted and ready to be extruded from the nozzle. Then, the nozzle would move to the exact position and release the filament above the platform with a constant moving speed v , and distance H between the nozzle tip and the platform upper surface would also be constant value during the nozzle moving. At the time of the first filament layer nodes contacting the platform, all cross-section layers of following unextruded filament will be in the same boundary condition with the already printed filament layer nodes in the same position. Therefore, it is assumed that from the next cross-section

extruded from the nozzle, all later cross-section nodes temperature would follow and as same as the presented cross-section nodes in the same height. Later cross-section nodes have slightly different situation compared with the initial several cross-section nodes due to their bottom cross-section surface will not happen convection with air as same as the first cross-section. In that case, the later filament's cross-section will have relatively low cooling rate compared with the initial several cross-sections. However, due to the gap height between the nozzle tip to the platform surface is about 1.2-1.5 times filament diameter, it will take about 0.012s under a standard nozzle diameter in 0.4 mm and the nozzle velocity of 50 mm / s . As a consequence, the temperature evolution difference of later cross-section is smaller enough to be neglected between the real situation and developed model.

Next, when the filament length is extruded enough, the first nodes will contact on the platform, as shown in **Fig. 15 (c)**. The filament shape would have transformed with the high-speed velocity of the nozzle, which means the left bottom node will contact the platform first and not all the same horizontal cross-section nodes will contact on the platform at the same time. After that, the left side nodes will one by one reach the platform and begin the next periods of the free cooling process.

Finally, when the followed nodes contact the platform, the temperature profile of these nodes would have rules to follow. Two red slashes showed the time t and time $t + \Delta t$ moving of the whole system, as shown in **Fig. 15 (a)**. And in this system, the temperature profile of nodes in the second slash at time $t + \Delta t$ will copy from the first slash at time t on the same horizon line. Hence, the temperature profile of the following nodes would like a paintbrush draw in parallel, and this way would be called as the “Brush Mode”. When all filaments have entirely on the platform, the free dynamic temperature evolution begins for the whole system, which shows internal thermal conduction, surface convection with platform and air and radiation with the surroundings, until it cools down to the environmental temperature and keeps stable.

In requiring the discretisation functions, the primary methods are using Taylor’s Formula, Polynomial fitting method, balanced control volume method and control volume integration.

In this model, the balanced control volume method will be used as the prime method to build the discretisation function with all variables. For the steady heat transfer problem, the balanced control volume method is that the sum of all nodes heat flux flows into the plot P is zero, and for the unsteady problem, the sum of heat flux is equal to the internal energy change of node P controlled volume in **Fig. 13**.

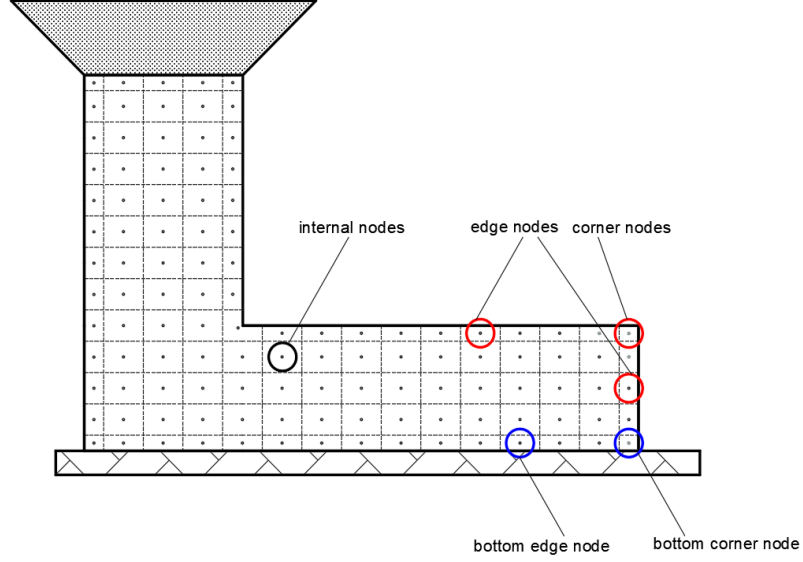


Fig. 16 Different boundary condition nodes in meshed filament

In order to analyse every node's heat transfer with its neighbor, the different boundary situations of the plots should be defined. There are 5 different boundary situations in the single line model: internal nodes, edge nodes, corner nodes, bottom edge nodes and bottom corner nodes, respectively, as shown in **Fig. 16**. Different boundary nodes have different heat flux flow from their neighbour nodes and different areas convect and radiate with the surroundings. For the internal nodes, the node control volume will get heat flux from the upper, below, left and right nearest nodes and the front and rear surface will convect and radiate with the air. Therefore, the discretisation function could be obtained as Eq. (22):

$$\rho C \Delta x \Delta y \Delta z \frac{T_{i,j}^{n+1} - T_{i,j}^n}{\Delta t} = k \frac{T_{i+1,j}^n + T_{i-1,j}^n - 2T_{i,j}^n}{\delta x} \Delta y \Delta z + k \frac{T_{i,j+1}^n + T_{i,j-1}^n - 2T_{i,j}^n}{\delta y} \Delta x \Delta z \quad (22)$$

$$+ 2\Delta x \Delta y h (T_{\infty} - T_{i,j}^n) + 2\Delta x \Delta y \varepsilon \sigma (T_{\infty}^4 - T_{i,j}^{n4}) + E$$

According to **Fig. 13**, if any two adjacent parallel grid line is in the same distance, then:

$$\Delta x = \delta x = \delta_1 \quad \Delta y = \delta y = \delta_2 \quad (23)$$

The Eq. (22) would be transferred into Eq. (24) and (25):

$$\begin{aligned} \frac{T_{i,j}^{n+1} - T_{i,j}^n}{\Delta t} = & \frac{k}{\rho c} \left(\frac{T_{i+1,j}^n + T_{i-1,j}^n - 2T_{i,j}^n}{\delta x^2} + \frac{T_{i,j+1}^n + T_{i,j-1}^n - 2T_{i,j}^n}{\delta y^2} \right) \\ & + \frac{2h}{\rho c} \left(\frac{T_\infty - T_{i,j}^n}{\Delta z} \right) + \frac{2\varepsilon\sigma}{\rho c} \frac{(T_\infty^4 - T_{i,j}^{n4})}{\Delta z} + E \end{aligned} \quad (24)$$

$$\begin{aligned} T_{i,j}^{n+1} = & T_{i,j}^n + \Delta t \frac{k}{\rho c} \left(\frac{T_{i+1,j}^n + T_{i-1,j}^n - 2T_{i,j}^n}{\delta x^2} + \frac{T_{i,j+1}^n + T_{i,j-1}^n - 2T_{i,j}^n}{\delta y^2} \right) \\ & + \frac{2h}{\rho c} \Delta t \left(\frac{T_\infty - T_{i,j}^n}{\Delta z} \right) + \frac{2\varepsilon\sigma}{\rho c} \Delta t \frac{(T_\infty^4 - T_{i,j}^{n4})}{\Delta z} + \frac{L}{c} \frac{\Delta V}{V_{\text{element}}} \end{aligned} \quad (25)$$

Where, Δx , Δy and Δz are the length, height and width of the node controlled volume, respectively. δx is the distance between two nodes in the x-direction and δy is the distance between two nodes in the y-direction. Δt relies on the nozzle velocity and mesh size, which is equal to the ratio of the horizontal two neighbour nodes with the nozzle velocity. $T_{i,j}^{n+1}$ and $T_{i,j}^n$ are the filaments nodes temperature at (n+1) th and n th Δt time, respectively. T_∞ is the air temperature. $T_{i+1,j}^n$, $T_{i-1,j}^n$, $T_{i,j+1}^n$ and $T_{i,j-1}^n$ are the left, the right, the upper and the below nodes temperature for plot (i, j) at the n th Δt time, respectively. ρ , c and k are the density, specific heat capacity and thermal conductivity of the feedstocks, respectively. h is the thermal convection coefficient, σ is the Stefan-Boltzmann constant and ε is the emissivity factor of the feedstocks. E is the latent heat source of crystallization. L is the value of 100% crystallization heat release per gram. $\Delta V = V^{n+1} - V^n$ is the growth volume of crystallization during Δt time and $V_{\text{element}} = \Delta x \Delta y \Delta z$ is the volume of plot-controlled volume.

Similarly, the discretisation function for other types of boundary nodes could be built. For the edge nodes in Eq. (26) and (27):

$$\rho c \frac{\Delta x}{2} \Delta y \Delta z \frac{T_{i,j}^{n+1} - T_{i,j}^n}{\Delta t} = k \frac{T_{i+1,j}^n - T_{i,j}^n}{\delta x} \Delta y \Delta z + k \frac{T_{i,j+1}^n - T_{i,j}^n}{\delta y} \frac{\Delta x}{2} \Delta z + k \frac{T_{i,j-1}^n - T_{i,j}^n}{\delta y} \frac{\Delta x}{2} \Delta z$$

$$+ (\Delta y \Delta z + 2 \frac{\Delta x}{2} \Delta z) h(T_{\infty} - T_{i,j}^n) + (\Delta y \Delta z + 2 \frac{\Delta x}{2} \Delta z) \varepsilon \sigma (T_{\infty}^4 - T_{i,j}^{n4}) + E$$
(26)

$$T_{i,j}^{n+1} = T_{i,j}^n + \Delta t \frac{k}{\rho c} \left(\frac{2(T_{i+1,j}^n - T_{i,j}^n)}{\delta x^2} + \frac{T_{i,j+1}^n + T_{i,j-1}^n - 2T_{i,j}^n}{\delta y^2} \right) + \Delta t \frac{2h}{\rho c} \frac{(T_{\infty} - T_{i,j}^n)}{\delta x} +$$

$$\Delta t \frac{2\varepsilon \sigma}{\rho c} \frac{(T_{\infty}^4 - T_{i,j}^{n4})}{\delta x} + \frac{2h}{\rho c} \Delta t \left(\frac{T_{\infty} - T_{i,j}^n}{\Delta z} \right) + \frac{2\varepsilon \sigma}{\rho c} \Delta t \frac{(T_{\infty}^4 - T_{i,j}^{n4})}{\Delta z} + \frac{L}{c} \frac{\Delta V}{V_{element}}$$
(27)

For the corner nodes in Eq. (28) and (29):

$$\rho c \frac{\Delta x}{2} \frac{\Delta y}{2} \Delta z \frac{T_{i,j}^{n+1} - T_{i,j}^n}{\Delta t} = k \frac{T_{i+1,j}^n - T_{i,j}^n}{\delta x} \frac{\Delta y}{2} \Delta z + k \frac{T_{i,j+1}^n - T_{i,j}^n}{\delta y} \frac{\Delta x}{2} \Delta z$$

$$+ \left(\frac{\Delta x + \Delta y}{2} \Delta z + 2 \frac{\Delta x \Delta y}{4} \right) h(T_{\infty} - T_{i,j}^n) + \left(\frac{\Delta x + \Delta y}{2} \Delta z + 2 \frac{\Delta x \Delta y}{4} \right) \varepsilon \sigma (T_{\infty}^4 - T_{i,j}^{n4}) + E$$
(28)

$$T_{i,j}^{n+1} = T_{i,j}^n + \Delta t \frac{k}{\rho c} \left(\frac{2(T_{i+1,j}^n - T_{i,j}^n)}{\delta x^2} + \frac{2(T_{i,j+1}^n - T_{i,j}^n)}{\delta y^2} \right) + \Delta t \frac{4h}{\rho c} \frac{(T_{\infty} - T_{i,j}^n)}{\delta x}$$

$$+ \Delta t \frac{4\varepsilon \sigma}{\rho c} \frac{(T_{\infty}^4 - T_{i,j}^{n4})}{\delta x} + \frac{2h}{\rho c} \Delta t \left(\frac{T_{\infty} - T_{i,j}^n}{\Delta z} \right) + \frac{2\varepsilon \sigma}{\rho c} \Delta t \frac{(T_{\infty}^4 - T_{i,j}^{n4})}{\Delta z} + \frac{L}{c} \frac{\Delta V}{V_{element}}$$
(29)

For the bottom corner nodes in Eq. (30) and (31):

$$\rho c \frac{\Delta x}{2} \frac{\Delta y}{2} \Delta z \frac{T_{i,j}^{n+1} - T_{i,j}^n}{\Delta t} = k \frac{T_{i+1,j}^n - T_{i,j}^n}{\delta x} \frac{\Delta y}{2} \Delta z + k \frac{T_{i,j+1}^n - T_{i,j}^n}{\delta y} \frac{\Delta x}{2} \Delta z + k_{plat} (T_{plat} - T_{i,j}^n) \frac{\Delta x}{2} \Delta z$$

$$+ \left(\frac{\Delta y}{2} \Delta z + 2 \frac{\Delta x \Delta y}{4} \right) h(T_{\infty} - T_{i,j}^n) + \left(\frac{\Delta y}{2} \Delta z + 2 \frac{\Delta x \Delta y}{4} \right) \varepsilon \sigma (T_{\infty}^4 - T_{i,j}^{n4}) + E$$
(30)

$$T_{i,j}^{n+1} = T_{i,j}^n + \Delta t \frac{k}{\rho c} \left(\frac{2(T_{i+1,j}^n - T_{i,j}^n)}{\delta x^2} + \frac{2(T_{i,j+1}^n - T_{i,j}^n)}{\delta y^2} \right) + \Delta t \frac{2h}{\rho c} \frac{(T_{\infty} - T_{i,j}^n)}{\delta x} + \Delta t \frac{2\varepsilon \sigma}{\rho c} \frac{(T_{\infty}^4 - T_{i,j}^{n4})}{\delta x}$$

$$+ \frac{2h}{\rho c} \Delta t \left(\frac{T_{\infty} - T_{i,j}^n}{\Delta z} \right) + \frac{2\varepsilon \sigma}{\rho c} \Delta t \frac{(T_{\infty}^4 - T_{i,j}^{n4})}{\Delta z} + \Delta t \frac{k_{plat}}{\rho c} \left(\frac{2(T_{i-1,j}^n - T_{i,j}^n)}{\delta y} \right) + \frac{L}{c} \frac{\Delta V}{V_{element}}$$
(31)

For the bottom edge nodes in Eq. (32) and (33):

$$\begin{aligned} \rho c \Delta x \frac{\Delta y}{2} \Delta z \frac{T_{i,j}^{n+1} - T_{i,j}^n}{\Delta t} = & k \frac{T_{i+1,j}^n + T_{i-1,j}^n - 2T_{i,j}^n}{\delta x} \frac{\Delta y}{2} \Delta z + k \frac{T_{i,j+1}^n - T_{i,j}^n}{\delta y} \Delta x \Delta z \\ & + k_{plat} (T_{plat} - T_{i,j}^n) \Delta x \Delta z + 2 \Delta x \frac{\Delta y}{2} h (T_{\infty} - T_{i,j}^n) + 2 \Delta x \frac{\Delta y}{2} \varepsilon \sigma (T_{\infty}^4 - T_{i,j}^n) + E \end{aligned} \quad (32)$$

$$\begin{aligned} T_{i,j}^{n+1} = & T_{i,j}^n + \Delta t \frac{k}{\rho c} \left(\frac{T_{i+1,j}^n - T_{i,j}^n}{\delta x^2} + \frac{T_{i-1,j}^n - T_{i,j}^n}{\delta x^2} + \frac{2(T_{i,j+1}^n - T_{i,j}^n)}{\delta y^2} \right) + \Delta t \frac{2k_{plat}}{\rho c} \frac{(T_{plat} - T_{i,j}^n)}{\delta y} \\ & + \frac{2h}{\rho c} \Delta t \left(\frac{T_{\infty} - T_{i,j}^n}{\Delta z} \right) + \frac{2\varepsilon \sigma}{\rho c} \Delta t \frac{(T_{\infty}^4 - T_{i,j}^n)}{\Delta z} + \frac{L}{c} \frac{\Delta V}{V_{element}} \end{aligned} \quad (33)$$

Where, k_{plat} is thermal contact conductance with the platform and T_{plat} is the temperature of the platform.

3.1.2 Simulation of multi-line works

By heating the filament in a nozzle, transferred from materials spool, the molten-state filaments extruded and reached the platform to form the patterns layer by layer in the FDM process. The simulation work with multi-line is indispensable and is considered in this section. With the high-speed moving of the nozzle, the previous layer would be re-melted by the newly deposited filament before the temperature below the glass transition temperature. Therefore, the crystallizing time for extruded filament has been prolonged. The re-melting process would significantly influence the crystallization behaviour of printed polymer which will enhance the quality of final products. As shown in **Fig. 17**, an example of a 2×2 filaments printing strategy has been presented. When the first line has been successfully printed into the exact position, the nozzle would be controlled by motor moving to the starting point of the next line. During this transient time, the first line would be free cooling with the surroundings. Then, the second line was printed on the platform and the left lateral surface contact with the first line right lateral surface. The surface heat transfer would occur between two adjacent filaments and the first line will be heated by the second line. Therefore,

the mathematic temperature relationship for the first line changes to:

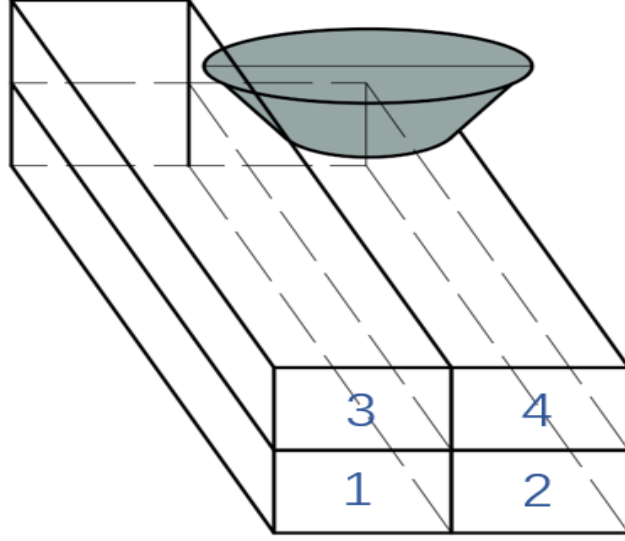


Fig. 17 Schematic of multi-line printing process

For the internal nodes in Eq. (34):

$$T_{i,j}^{n+1} = T_{i,j}^n + \Delta t \frac{k}{\rho c} \left(\frac{T_{i+1,j}^n + T_{i-1,j}^n - 2T_{i,j}^n}{\delta x^2} + \frac{T_{i,j+1}^n + T_{i,j-1}^n - 2T_{i,j}^n}{\delta y^2} \right) + \frac{h}{\rho c} \Delta t \left(\frac{T_\infty - T_{i,j}^n}{\Delta z} \right) + \frac{\varepsilon \sigma}{\rho c} \Delta t \frac{(T_\infty^4 - T_{i,j}^{n4})}{\Delta z} + \frac{k_{mat}}{\rho c} \Delta t \left(\frac{T_{i+n-L,j}^n - T_{i,j}^n}{\Delta z} \right) + \frac{L}{c} \frac{\Delta V}{V_{element}} \quad (34)$$

For the edge nodes in Eq. (35):

$$T_{i,j}^{n+1} = T_{i,j}^n + \Delta t \frac{k}{\rho c} \left(\frac{2(T_{i+1,j}^n - T_{i,j}^n)}{\delta x^2} + \frac{T_{i,j+1}^n + T_{i,j-1}^n - 2T_{i,j}^n}{\delta y^2} \right) + \Delta t \frac{2h}{\rho c} \frac{(T_\infty - T_{i,j}^n)}{\delta x} + \Delta t \frac{2\varepsilon \sigma}{\rho c} \frac{(T_\infty^4 - T_{i,j}^{n4})}{\delta x} + \frac{h}{\rho c} \Delta t \left(\frac{T_\infty - T_{i,j}^n}{\Delta z} \right) + \frac{\varepsilon \sigma}{\rho c} \Delta t \frac{(T_\infty^4 - T_{i,j}^{n4})}{\Delta z} + \frac{k_{mat}}{\rho c} \Delta t \left(\frac{T_{i+n-L,j}^n - T_{i,j}^n}{\Delta z} \right) + \frac{L}{c} \frac{\Delta V}{V_{element}} \quad (35)$$

For the corner nodes in Eq. (36):

$$T_{i,j}^{n+1} = T_{i,j}^n + \Delta t \frac{k}{\rho c} \left(\frac{2(T_{i+1,j}^n - T_{i,j}^n)}{\delta x^2} + \frac{2(T_{i,j+1}^n - T_{i,j}^n)}{\delta y^2} \right) + \Delta t \frac{4h}{\rho c} \frac{(T_\infty - T_{i,j}^n)}{\delta x} + \Delta t \frac{4\varepsilon \sigma}{\rho c} \frac{(T_\infty^4 - T_{i,j}^{n4})}{\delta x} + \frac{h}{\rho c} \Delta t \left(\frac{T_\infty - T_{i,j}^n}{\Delta z} \right) + \frac{\varepsilon \sigma}{\rho c} \Delta t \frac{(T_\infty^4 - T_{i,j}^{n4})}{\Delta z} + \frac{k_{mat}}{\rho c} \Delta t \left(\frac{T_{i+n-L,j}^n - T_{i,j}^n}{\Delta z} \right) + \frac{L}{c} \frac{\Delta V}{V_{element}} \quad (36)$$

For the bottom corner nodes in Eq. (37):

$$T_{i,j}^{n+1} = T_{i,j}^n + \Delta t \frac{k}{\rho c} \left(\frac{2(T_{i+1,j}^n - T_{i,j}^n)}{\delta x^2} + \frac{2(T_{i,j+1}^n - T_{i,j}^n)}{\delta y^2} \right) + \Delta t \frac{2h}{\rho c} \frac{(T_\infty - T_{i,j}^n)}{\delta x} + \Delta t \frac{2\varepsilon\sigma}{\rho c} \frac{(T_\infty^4 - T_{i,j}^{n4})}{\delta x} \quad (37)$$

$$+ \frac{h}{\rho c} \Delta t \left(\frac{T_\infty - T_{i,j}^n}{\Delta z} \right) + \frac{\varepsilon\sigma}{\rho c} \Delta t \frac{(T_\infty^4 - T_{i,j}^{n4})}{\Delta z} + \Delta t \frac{k_{plat}}{\rho c} \left(\frac{(T_{i-1,j}^n - T_{i,j}^n)}{\delta y} \right) + \frac{k_{mat}}{\rho c} \Delta t \left(\frac{(T_{i+n-L,j}^n - T_{i,j}^n)}{\Delta z} \right) + \frac{L}{c} \frac{\Delta V}{V_{element}}$$

For the bottom edge nodes in Eq. (38):

$$T_{i,j}^{n+1} = T_{i,j}^n + \Delta t \frac{k}{\rho c} \left(\frac{T_{i+1,j}^n - T_{i,j}^n}{\delta x^2} + \frac{T_{i-1,j}^n - T_{i,j}^n}{\delta x^2} + \frac{2(T_{i,j+1}^n - T_{i,j}^n)}{\delta y^2} \right) + \Delta t \frac{2k_{plat}}{\rho c} \frac{(T_{plat} - T_{i,j}^n)}{\delta y} \quad (38)$$

$$+ \frac{h}{\rho c} \Delta t \left(\frac{T_\infty - T_{i,j}^n}{\Delta z} \right) + \frac{\varepsilon\sigma}{\rho c} \Delta t \frac{(T_\infty^4 - T_{i,j}^{n4})}{\Delta z} + \frac{k_{mat}}{\rho c} \Delta t \left(\frac{(T_{i+n-L,j}^n - T_{i,j}^n)}{\Delta z} \right) + \frac{L}{c} \frac{\Delta V}{V_{element}}$$

Where, k_{mat} is the thermal contact conductance with filaments, $T_{i+n-L,j}^n$ is the temperature of the adjacent element, $n-L$ is the number of the printed elements in the x-direction. Similarly, the third line deposit on the upper surface of first line, the heat transfer occurs between the first and the third lines on the contact area. The boundary condition of the fourth line is regrouping of the previous situation.

Fig. 18 shows the whole printing procedure from the first line to the fourth line until finished. All other multi-line printing processes are based on the 2×2 model, and the boundary condition could be found the same one in this 2×2 model. Hence, the two-dimensional model has been built and the temperature profile would be as a function of time and other parameters. The next step is to combine the temperature evolution with the crystallisation growth situation, and the latent heat of crystallisation will give feedback heat to the whole heat transfer system.

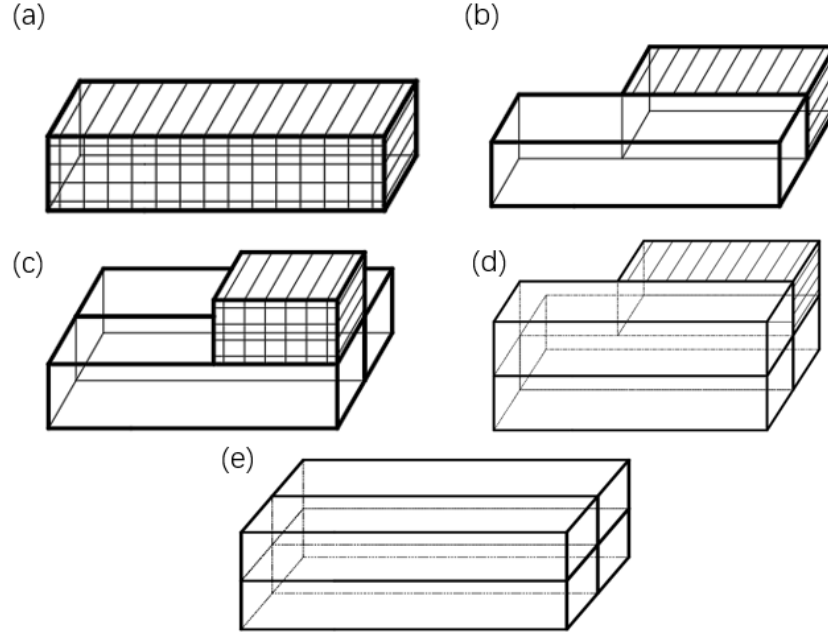


Fig. 18 Filaments in calculation with different periods and boundaries. The schematic of meshed printing order: (a) the first line; (b) the second line; (c) the third line; (d) the fourth line and (e) the final free cooling period.

3.2 Model of crystallisation

Although the primary and secondary nucleation both contribute to the increase in the number of crystals in the system. Primary nucleation describes the transition to a new phase that does not depends on the new phase already being present. Secondary nucleation depends on crystals which are already present. In this work, the Hoffman theory was used to compute the polymer crystallization in the FDM process. Due to different growth speeds of front laterally g and secondary nucleation rate i at different temperature situation, three regimes of crystallisation kinetics have been presented in **Fig. 19**^[29,30,79].

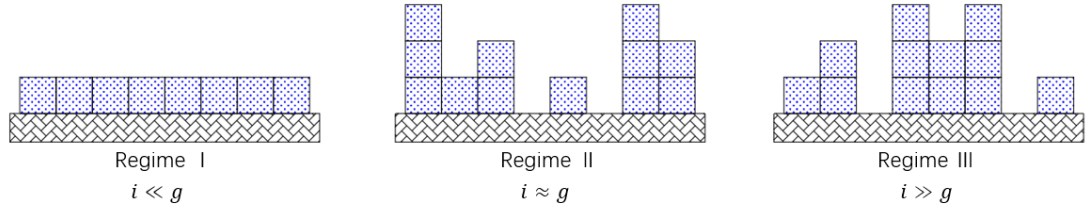


Fig. 19 Three regimes of crystallisation kinetics

In the regime I , each layer has completed before the next nucleate; in the regime II , each growth layer faces multiple nucleations, and in the regime III , molecular step is always much forward than growth speed of front laterally and ready for the next strand. The surface is molecularly rough. Also, Hoffman expressed the growth rates for different regimes via combining the surface energy, driving force, energy barrier and temperature with physical and mathematical approximation in Eq. (39).

$$\begin{cases} G_I = G_0 \times \exp\left[\frac{-U^*}{R(T_c - T_\infty)}\right] \times \exp\left[\frac{-K_{gI}}{T_c \Delta T f}\right], i \ll g \\ G_{II} = G_0 \times \exp\left[\frac{-U^*}{R(T_c - T_\infty)}\right] \times \exp\left[\frac{-K_{gII}}{T_c \Delta T f}\right], i \approx g \\ G_{III} = G_0 \times \exp\left[\frac{-U^*}{R(T_c - T_\infty)}\right] \times \exp\left[\frac{-K_{gIII}}{T_c \Delta T f}\right], i \gg g \end{cases} \quad (39)$$

Where, G_0 is the pre-exponential factor containing terms which are essentially temperature-independent, K_g is the surface nucleation constant which is given by

$$K_g = \frac{nb_0\sigma_s\sigma_e T_m^0}{\Delta H_f k}, \text{ where } \sigma \text{ and } \sigma_e \text{ are the lateral and end-surface free energies, } b_0 \text{ is}$$

molecular thickness, ΔH_f is enthalpy fusion, k is Boltzmann constant and n is different in different regimes where $n=4$ in the regime I and III and $n=2$ in the regime II , respectively. Therefore, $K_{gI} = K_{gIII} = 2K_{gII}$. U^* is the activation energy for transport of crystallizable segments to the crystallization front, T_∞ is the temperature below which such motions cease, T_c is the crystallization temperature, $\Delta T = T_m^0 - T_c$ is the

degree of undercooling, T_m^0 is the equilibrium melting point and could calculate by

$T_m = T_m^0(1 - \frac{1}{\gamma}) + \frac{T_c}{\gamma}$, where T_m is the observed melting point and γ is the ratio of the

initial to the final lamellar thicknesses. f , which is given by $f = \frac{2T_c}{T_m^0 + T_c}$, is a factor

which accounts for the variation in the enthalpy fusion ΔH_f with temperature.

Through the derivation of the formula, the slope K_g and intercept $\ln G_0$ values of different materials and regimes could be acquired from isothermal experiments observation for crystallisation growth at different temperatures in Eq. (40) [23].

$$\ln G + \frac{U^*}{R(T_c - T_\infty)} = -\frac{K_g}{T_c \Delta T f} + \ln G_0 \quad (40)$$

The radius growth rate of the crystal could be expressed as Eq. (41):

$$\Delta r = \Delta t \times G(T_c) \quad (41)$$

And the accumulative radius at the time t is in Eq. (42):

$$r_i = \sum_0^t (\Delta t \times G_i(T_c)) \quad (42)$$

The volume of crystal at the time t is in Eq. (43):

$$V = \frac{4\pi}{3} (\sum_0^t r_i)^3 = \frac{4\pi}{3} (\sum_0^t (\Delta t \times G_i(T_c)))^3 \quad (43)$$

The formed maximum number of nucleation N under specific crystal temperature T could be expressed as Eq. (44) and (45) [4,24]:

$$N = N_0, T_c < T_g \quad (44)$$

$$N = N_0 \exp(-\frac{1/k}{T_c \Delta T}), T_c > T_g \quad (45)$$

Where, N_0 is the final reference number of nucleation per unit volume at specific temperature. The time t_{nuc} for PLA nucleation formed to the maximum number is obtained from Zhou et al.^[111] The relationship between t_{nuc} and crystal temperature T could be simplified polynomial fitted as Eq. (46):

$$t_{nuc} = AT^2 + BT + C \quad (46)$$

The nucleation rate in every second n_s could be expressed from Eq. (45) and (46):

$$n_s = \frac{N}{t_{nuc}} = \frac{N_0 \exp(-\frac{1/k}{T_c \Delta T})}{AT^2 + BT + C} \quad (47)$$

The total number of nucleation from time 0 to t could be calculated by Eq. (47):

$$n_{total} = \int_0^t n_s dt \quad (48)$$

Therefore, the accumulative volume of crystallisation could be in Eq. (49):

$$V_{total} = \sum_0^{n_{total}} V_i = \sum_0^{n_{total}} [\frac{4\pi}{3} (\sum_0^t (\Delta t \times G_i(T_i)))^3] \quad (49)$$

3.3 General procedure of model simulation process

Based on a series of theories mentioned above, a general model has been proposed to simulate the temperature evolution and crystallisation situation of filaments extruded from the FDM machine. The flow chart in **Fig. 20** shows the general procedure to run the whole evolution program. First, input all related parameters into the program, as shown in Nomenclature. Then, according to the accuracy of the results demands, the mesh sensibility could be set and be applied to the whole filament.

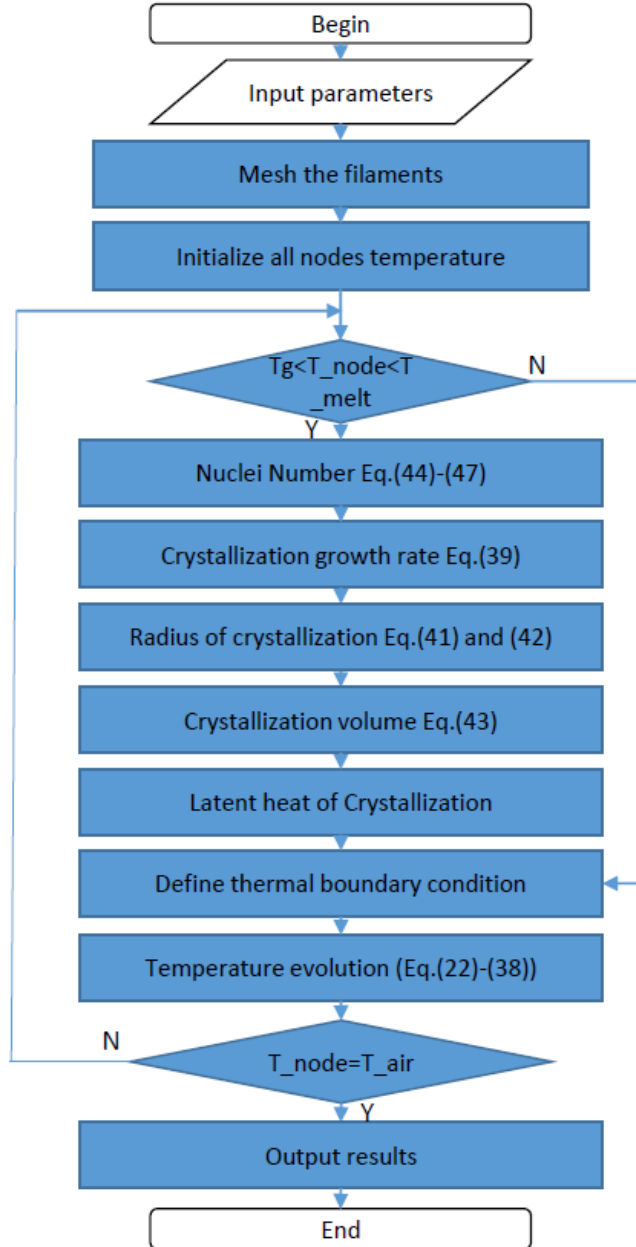


Fig. 20 Procedure to compute heat transfer evolution and crystallisation kinetics

All filament elements are extruded from the same temperature. Therefore, the initial temperature of all nodes could be pre-set. When the nozzle temperature is higher than the melting temperature, the crystallisation process will not present in that situation. Hence, the filament part will directly run into the temperature evolution part to process heat transfer with neighbour elements, platform and surroundings. After the temperature of the element cools down below the melting temperature, the first spherulite begins to form and grow. The velocity of nucleation forming rate and growth speed are as a function of the temperature of

the element. On the base of the above process, the crystallisation of the filaments could be got step by step, and the crystallisation forming process will release heat, which will as a feedback act on the element heat energy. Next, the position and boundary situation of each element would be analysed. Different boundary condition will lead to different area exposed to the air and contacting area to the platform. During the temperature evolution and crystallisation process, all data would be output with time information. It is a complete loop for the whole model simulation work before the temperature of all nodes keeps the same with the time. Finally, the temperature evolution is as a function of time and crystallisation process is as functions of time and temperature.

3.4 Summery

This chapter presented a two-dimensional temperature evolution model on both single line and multi-line works, which covered different boundary conditions, latent heat source and related parameters, such as printing speed, thermal convection coefficient, thermal contact conductance with platform, thermal contact conductance with filament, nozzle diameter and crystallization latent heat. Also, the Hoffman theory was applied to simulated the nucleation forming and crystal growth. Some experimental works will be presented in the next chapter to collect the parameter value of materials and temperature evolution data to validate this model.

Chapter 4 Experiment part

In this chapter, the type and fundamental of machine for measuring the temperature evolution and basic parameters of materials have been introduced. Firstly, the essential parameters of a common standard FDM printer and silver PLA filaments has been presented. Then, an IR thermal camera basic information was introduced. Next, section 4.3 introduced the test principle of a normal DSC machine. In the end, section 4.4 showed the machine operating principle for the thermal conductivity test.

4.1 Fused Deposition Modelling machine and Material

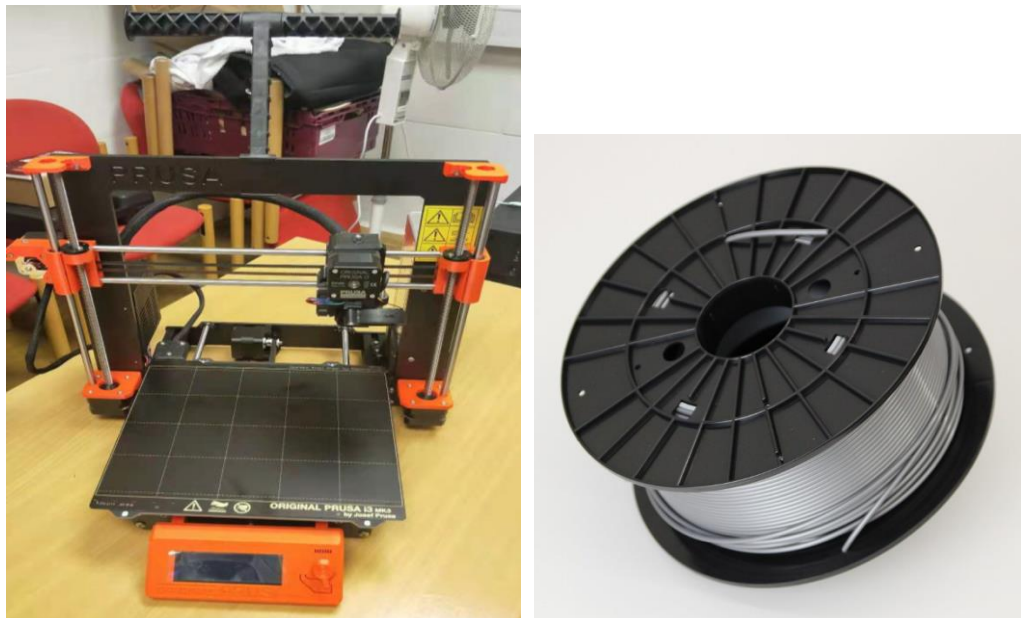


Fig. 21 the original PRUSA I3 MK3S FDM printer and PLA filaments

In order to build the experiments to detect the temperature evolution of printed filaments. The original PRUSA I3 MK3S FDM printer and poly (lactic acid) filaments without further purification nor thermal treatment were used in this work, both of which were bought from Prusa Research (Praha Czech Republic). Some properties of PLA filaments and 3D printer are listed in **Table 5**^[112].

Table 5 Some properties of PLA filaments and 3D printer^[112]

	Description	Value
FDM printer:	Build volume	25×21×21cm
	Layer height	0.05 – 0.35mm
	Speed	200 + mm / s
	Max nozzle temperature	300 °C
	Max platform temperature	120 °C
Filaments:	Diameter	1.75mm
	Molecular Weight	15.000g / mol
	Density	1.24g / cm ³
	Peak Melt Temperature	145 – 160 °C
	Glass Transition Temperature	55 – 60 °C
	Moisture Absorption 4 weeks	0.3
	Tensile Yield Strength	57.4 ± 0.4MPa
	Filament	

4.2 Thermal camera for recording temperature data

In order to measure and record the filaments temperature evolution after extruded from the nozzle, the thermoIMAGER TIM 160 infrared camera from Micro-Epsilon (German) was used to record the data. This camera could be able to withstand ambient temperature up to 180 °C, and the camera measurement temperature range from -20 °C to 900°C.

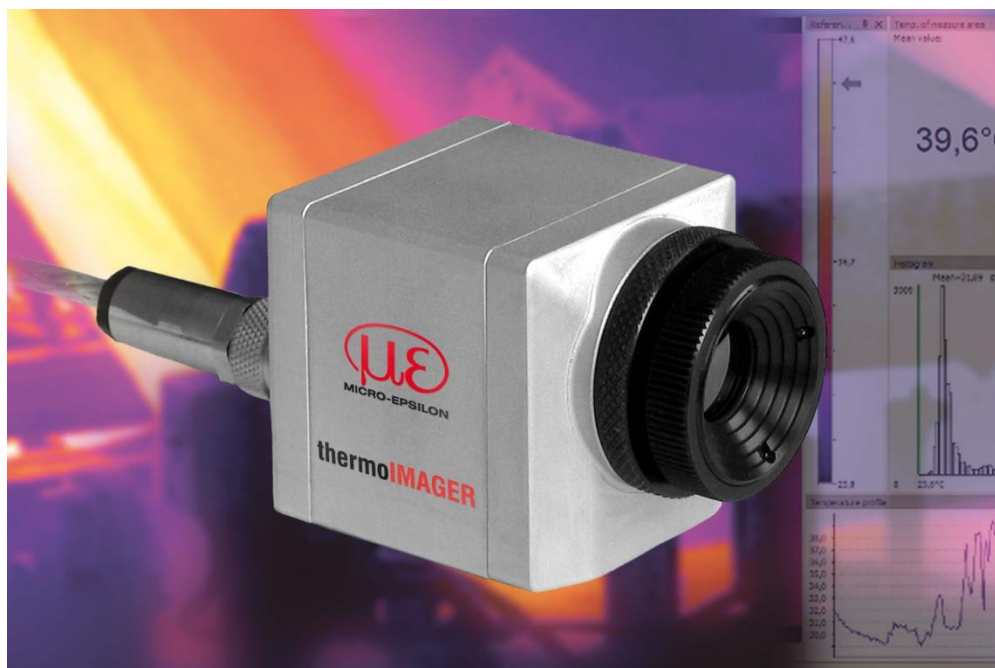


Fig. 22 thermoIMAGER TIM 160 infrared camera

4.3 Differential Scanning Calorimeter (DSC)



Fig. 23 TA Instruments Q100 DSC machine

According to the handbook and introduction of the PLA filaments from the purchased company, some critical parameters are given by a range. In order to get the specific value of

these parameters applying into the program, a series of experiments have been done to test these key values. Thermal analysis measurements were used TA Instruments Q100 DSC machine (German). This machine is capable of temperatures ranges from $-90\text{ }^{\circ}\text{C}$ to $550\text{ }^{\circ}\text{C}$, and sapphire was used as reference material. In the DSC, the sample's weight was required between 2 mg to 20 mg . Then the sample would be sent under the same heat flux as the sapphire. When the phase transition occurs of the samples, the temperature of the sample is different from that of sapphire. In order to keep the sample as same temperature as the sapphire, the heat flux should be increased or decreased and the sensor will record the differential heating power between sample and sapphire. According to the known mass of the sample and the sapphire, the differential power and the known specific heat of sapphire, then the phase transition temperature and specific heat of sample could be acquired.

4.4 Dr Colin and FOX 50 for thermal conductivity test



Fig. 24 Dr Colin P300 P/M from Colin Lab & Pilot Solution (German)

Thermal conductivity is another key factor of the feedstocks to simulate the temperature evolution in the proposed model. The samples for testing the value of the thermal

conductivity were heated above the melting temperature. Then, the PLA filaments were compressed in the mould by Dr Colin P300 P/M from Colin Lab & Pilot Solution (German) to transfer into the discs. The hot-pressed samples will be used for the thermal conductivity test by FOX 50 TA Instruments (German). **Table 6** presents some specifications of FOX 50 thermal conductivity instruments.

Table 6 Some specifications of FOX 50 thermal conductivity instruments

Property	Value
Sample Diameter	51mm
Temperature Range	$-10^{\circ}\text{C} - 110^{\circ}\text{C}$
Temperature Resolution	$\pm 0.01^{\circ}\text{C}$
Thermal Conductivity Range	0.1 to 10 $\text{W} / (\text{m} \cdot \text{K})$
Thermal Resistance Range	0.003 to 0.05 $\text{m}^2 \cdot \text{K} / \text{W}$
Accuracy, Standard	$\pm 3\%$
Heat Flux Transducers Area	25 × 25mm

As shown in **Fig. 25**, the schematic of FOX 50 thermal conductivity is presented. The fundamental theory used in the FOX heat flow meter instruments is based on one-dimensional Fourier-Biot law in Eq. (50):

$$q = -\lambda \frac{dT}{dx} \quad (50)$$

Where q is heat flux (W / m^2) flowing through the sample, λ is the thermal conductivity $\text{W} / (\text{m} \cdot \text{K})$ of the sample and the $\frac{dT}{dx}$ is the temperature gradient (K / m) between the two isothermal flat surfaces in the sample.

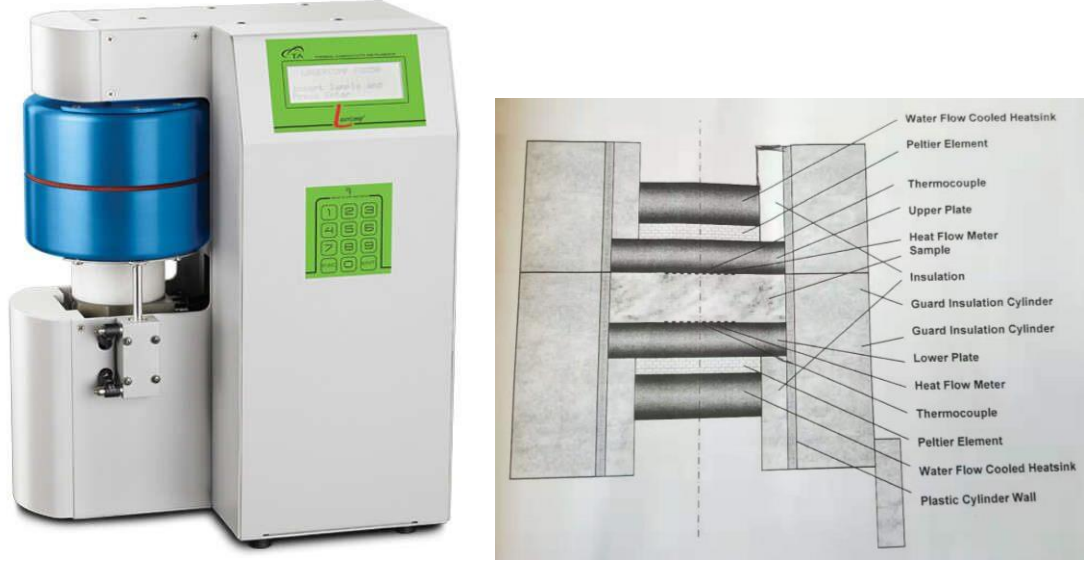


Fig. 25 Schematic of FOX 50 Heat Flow Meter Instrument for Thermal Conductivity Measurements

Due to the thermal contact resistance between the sample surface to the plate is not much smaller than the thermal resistance of the sample. Therefore, temperature distribution could be divided into three parts in Eq. (51):

$$\Delta T_{plates} = \delta T + \Delta T_{sample} + \delta T \quad (51)$$

Temperature jump between plate to the surface of the sample is δT which is proportional to the heat flux q (W / m^2) and thermal contact resistance R ($m^2 \cdot K / W$) as in Eq.(52):

$$\delta T = q \cdot R \quad (52)$$

Then, the electric signal in heat flow meter instruments could be expressed as Eq. (53):

$$Q = \frac{q}{S_{cal}} = \frac{\Delta T}{(2R + \frac{\Delta x}{\lambda})S_{cal}} \quad (53)$$

The S_{cal} is calibration factor which is determined by using a standard material with known thermal conductivity. Thermal contact resistance depends on the types of contacting materials, surface roughness and the interface pressure, etc. Hence, in order to get more accurate results, two samples with different thickness are used to get the value of thermal contact resistance R by Eq. (53).

$$Q_1 = \frac{\Delta T}{(2R + \frac{\Delta x_1}{\lambda})S_{cal}}$$

$$Q_2 = \frac{\Delta T}{(2R + \frac{\Delta x_2}{\lambda})S_{cal}}$$

Mathematically, the thermal conductivity λ and thermal contact resistance R could be obtained as Eq. (54) and (55):

$$\lambda = \frac{S_{cal} Q_1 Q_2 (\Delta x_2 - \Delta x_1)}{\Delta T (Q_1 - Q_2)} \quad (54)$$

$$R = \frac{(\Delta x_2 Q_2 - \Delta x_1 Q_1) \Delta T}{[Q_1 Q_2 S_{cal} (\Delta x_2 - \Delta x_1)]} \quad (55)$$

4.5 Summary

This chapter presented the fundamental of a series of experiments details of applied feedstocks. Different experimental machine has been introduced with the main parameters and experimental principle. The collected experiments data will be discussed specifically and the temperature evolution of printed materials will be used to validate the model under same parameters used in experiments in the next chapter.

Chapter 5 Results and Discussion

There are 5 sections presented in this chapter. At first, section 5.1 presented the experimental testing results of feedstocks' key values used in the modelling work and recorded temperature evolution of filament from an FDM printer. Section 5.2 compared two group results of experimental with the developed model with different process parameters and analysed the temperature distribution in the filaments. Section 5.3 simulated crystal kinetics by combining temperature modelling. Then, a 2×2 filaments as a multi-line computational work example successfully presented in section 5.4, and the effect was discussed of the difference results between single and multi line. The effect of the re-melting process on previously deposited filaments surface was also investigated. Finally, section 5.5 compared different parameters effects on the both simulated temperature and crystallization results including printing speed, thermal convection coefficient, thermal contact conductance with platform and filaments, used nozzle diameter and latent heat of crystallization.

5.1 Experiments results

5.1.1 Thermal conductivity



Fig. 26 Chopped 3D printing PLA filaments (part length: 2 mm)



Fig. 27 Disc-shaped samples prepared for thermal conductivity experiment (diameter: 50 mm and measured thickness: 1.36 ± 0.03 and 3.02 ± 0.04 mm)

In order to melt the feedstocks in the mould properly, commercially purchased PLA filaments were cut into 2 mm pieces as shown in **Fig. 26**. Then, the weight of materials needed for test specimens could be calculated mathematically with the density and volume of the filaments. Due to the melting temperature of the used PLA filaments is around 160 °C. The mould with chopped material placed flat in Dr Colin at 0 bar pressure and then heated up to 200 °C for over 3 minutes letting whole feedstocks be fully melted. According to previous trained experience for letting the pressed polymer surface be flat with high pressure, the pressure was increased to 10 bar and maintained the pressure for 3 minutes. Then, the specimens were let to cool down to the ambient temperature and cut the edges to remove flash materials to obtain the samples for thermal conductivity test, as shown in **Fig. 27**. The thickness of the specimen was measured by a digital calliper whose minimum resolution is 0.01 mm. In each sample, 5 points were chosen to calculate the average thickness for each sample as the measured thickness (4 points on edge randomly with another one in the centre points).

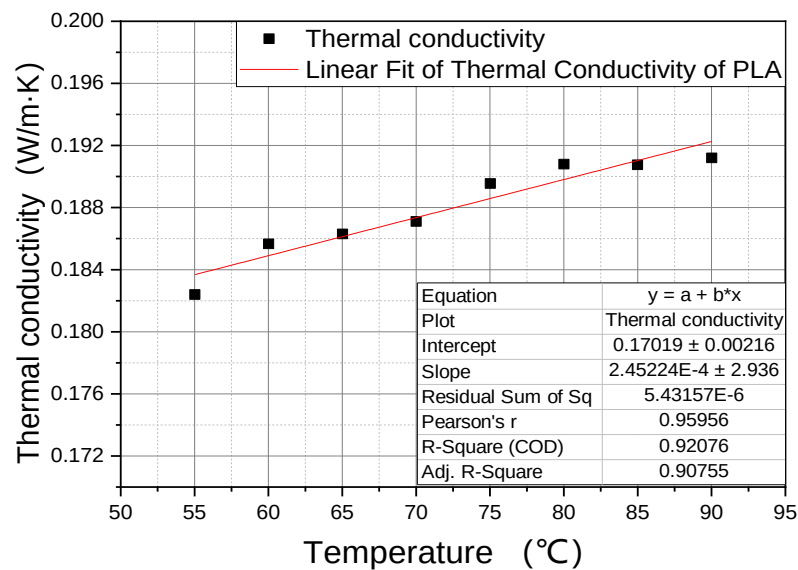


Fig. 28 thermal conductivity of PLA filaments ranges from 55 to 90 °C

The expected recording temperature range of the extruded filaments in the experiments is from the melting temperature cooling down to the glass transition temperature, and the maximum temperature could be reached in the FOX 50 is 100 °C. Therefore, the set temperature for thermal conductivity experiments was ranged from 55 °C to 90 °C, and measurements were taken in every 5 °C step change in FOX 50. From the linear fit line in

Fig. 28, the thermal conductivity value was slightly growing with the temperature increasing. The difference between the maximum value and the minimum value in this range was $0.08 \text{ W / m} \cdot \text{K}$. Hence, the thermal conductivity could be approximately applied as constant value $0.188 \text{ W / m} \cdot \text{K}$ in the program if the detected temperature does not change a lot.

5.1.2 the value of T_m , T_g and C_p for the feedstocks by DSC test

The melting temperature T_m , glass transition temperature T_g and specific heat capacity C_p were measured by the TA Q100 DSC machine. With a constant nitrogen flow rate of 50 mL / min nitrogen flow, all samples were heated from $35 \text{ }^\circ\text{C}$ to $200 \text{ }^\circ\text{C}$ in the DSC. Fresh samples were used for each measurement in order to minimise thermal degradation. Three different heating rates were used, and the resulting curves are shown in **Fig. 29**.

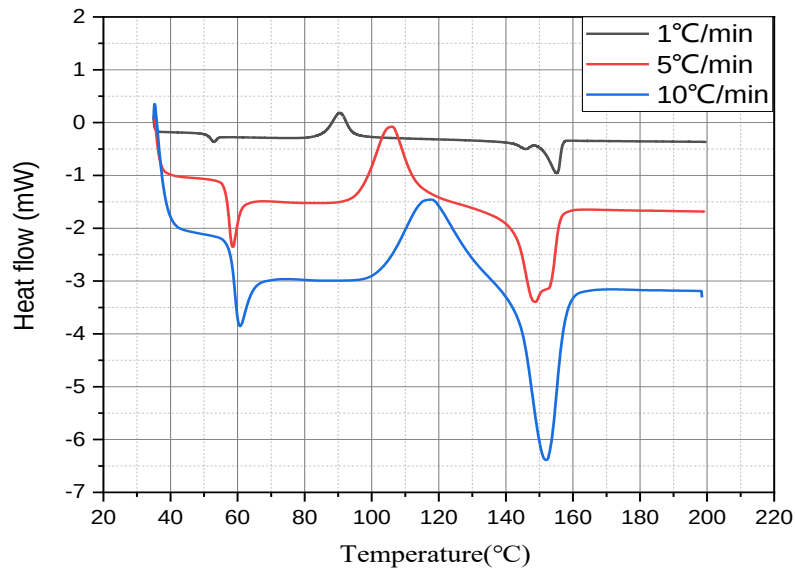


Fig. 29 DSC traces of PLA from $35 \text{ }^\circ\text{C}$ to $200 \text{ }^\circ\text{C}$ with different heating rates

As expected, all peaks and dips showed a shift as the heating rate is increased, and this shift is due to the sensitive of the machine that fast scanning brings the recording delay. When a change of heat flow was observed by the sensor, the recorded temperature of the sample has been heated up. Therefore, the lowest cooling rate was used to get the glass transition temperature and melting temperature which are approximate of 55 and $145 \text{ }^\circ\text{C}$, respectively.

These detected temperature points are in the range of feedstocks given handbook^[112].

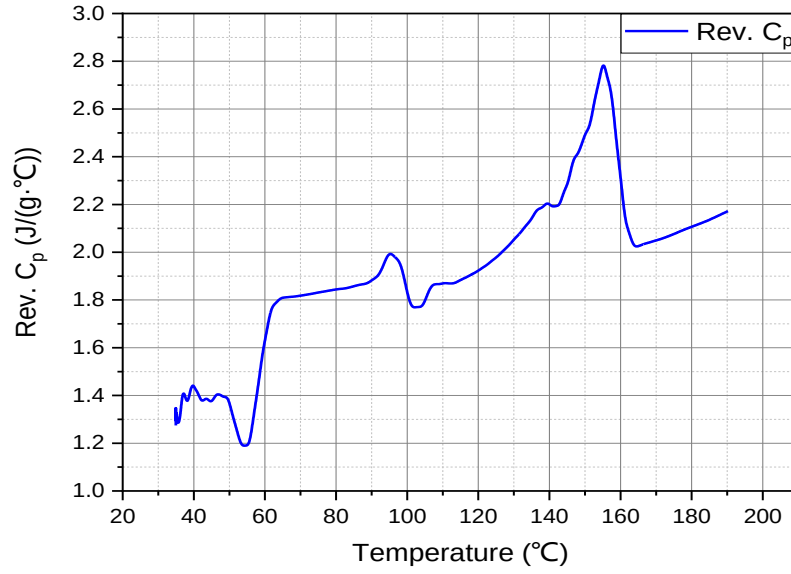


Fig. 30 Specific heat capacity C_p of PLA as a function of time measured by MDSC

Fig. 30 shows the specific heat capacity C_p measured by Modulated DSC. In the beginning, when the temperature is lower than the glass transition temperature T_g , the C_p was measured as a low value at $1.3\text{--}1.4 \text{ J/g} \cdot ^\circ\text{C}$, and when the temperature is close to the melting temperature T_m , the value of C_p increases with temperature and reach a maximum value of $2.78 \text{ J/g} \cdot ^\circ\text{C}$ at the melting point. Between the T_m and T_g , the average specific capacity of the test sample is about $1.8 \text{ J/g} \cdot ^\circ\text{C}$, which agrees with the previously reported values^[60,76,96].

5.1.3 Temperature evolution

In order to observe the temperature evolution of extruded filaments, a single line mould with the dimensions of $0.35 \times 0.35 \times 100 \text{ mm}^3$ was developed using Solidworks software, as shown in **Fig. 31**.

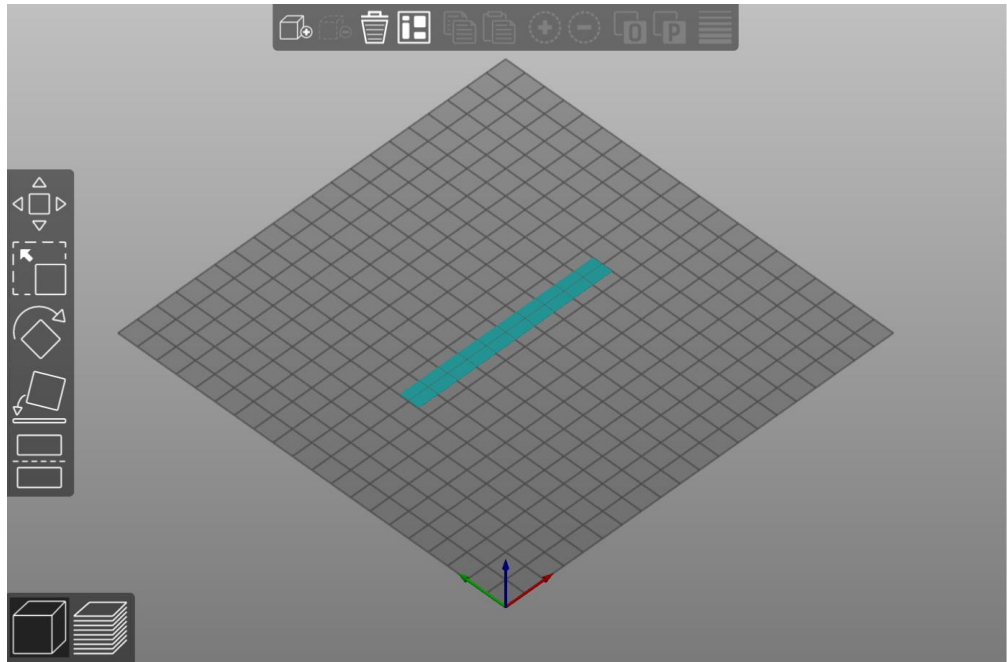


Fig. 31 The single line mould with the dimension of $0.35 \times 0.35 \times 100 \text{ mm}^3$ in the software

The pre-set value of printing PLA was used for the printer parameter setting, as shown in **Fig. 32**, which is the most generalised setting for the printing of PLA in this machine as was recommended by the Prusa company (the printer manufacturer). Moreover, the ambient temperature was kept as 30°C during the test.

Speed for print moves			
Perimeters:	• •	60	mm/s
Small perimeters:	• •	15	mm/s or %
External perimeters:	• •	50%	mm/s or %
Infill:	• •	80	mm/s
Solid infill:	• •	20	mm/s or %
Top solid infill:	• •	15	mm/s or %
Support material:	• •	60	mm/s
Support material interface:	• •	100%	mm/s or %
Bridges:	• •	60	mm/s
Gap fill:	• •	20	mm/s

Speed for non-print moves	
Travel:	• • 130 mm/s

Modifiers	
First layer speed:	• • 30 mm/s or %

Filament			
Color:	• •		
Diameter:	• •	1.75	mm
Extrusion multiplier:	• •	1	
Density:	• •	1.24	g/cm ³
Cost:	• •	0	money/kg

Temperature °C			
Extruder:	First layer: • •	215	Other layers: • • 215
Bed:	First layer: • •	60	Other layers: • • 60

Fig. 32 The pre-set value of printing PLA was used for the printer parameter setting

A thermal camera was used to record the filament temperature when the filament was extruded from the nozzle, and the recorded highest temperature was 124.7°C , as shown in **Fig. 33**.

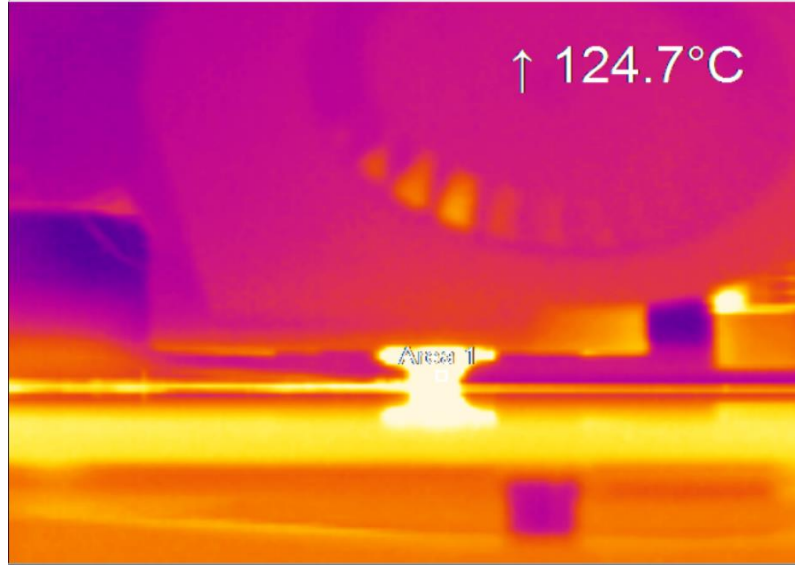


Fig. 33 Thermal image generated by the thermal camera

Fig. 34 shows the recorded temperature evolution of the filament at the midpoint via the thermoIMAGER TIM 160 infrared thermal camera. The maximum temperature was 124.7 °C, and after 5 seconds, the filament surface cooled down to the ambient temperature. Due to the nozzle and platform was moving when the temperature data was recorded, the thermal camera focus changes, and this focus changes caused to create an unsmoothed curve in the second half part of printing between 2.5 to 5.5 s.

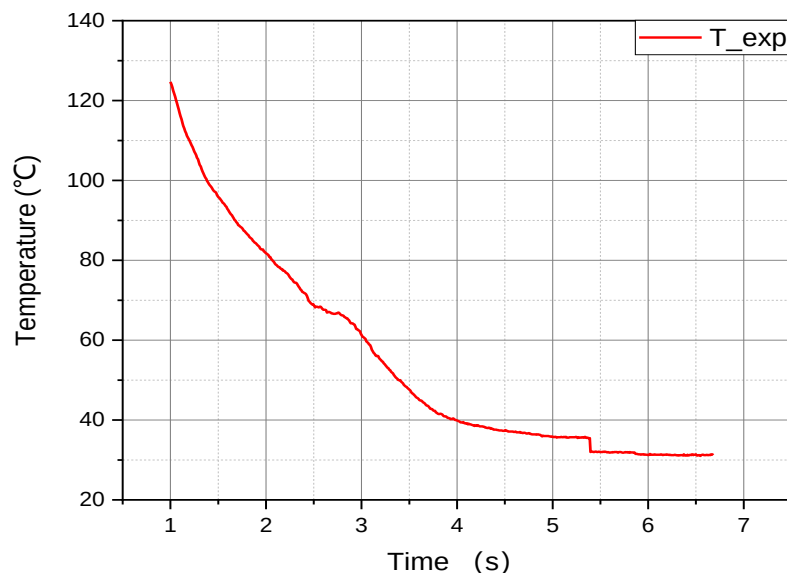


Fig. 34 Temperature evolution as a function of time recorded via thermal camera

After completing and collecting all above experiments, the next step is to use these values

of parameters to validate the proposed model.

5.2 Computational work of temperature evolution

5.2.1 Temperature evolution validated by previously published data

Table 7 the values of FDM variables and material properties in the model to validate the previously published data

Property	Value	Property	Value
Density (kg / m^3)	$\rho=1240$ ^[96]	Time step (s)	$\Delta t=0.00002$
Specific heat capacity ($J / kg \cdot ^\circ C$)	$C_p=1908$ ^[96]	Nozzle velocity (m / s)	$v = 0.5$ ^[96]
Thermal contact conductance with the platform ($W / m^2 \cdot ^\circ C$)	$k_{plat}=30$	Pre-exponential factor	$G_{0I}=3.987 \times 10^7$ $G_{0II}=4.81 \times 10^{11}$ ^[23]
Thermal conductivity ($W / m \cdot ^\circ C$)	$k=0.14$ ^[96]	Surface nucleation constant	$K_{gI}=2.55 \times 10^5$ $K_{gII}=5.51 \times 10^5$ ^[23]
Heat transfer coefficient ($W / m^2 \cdot ^\circ C$)	$h_1=75$ $h_2=50$	Activation energy ($cal / mole$)	$U^*=1500.0$ ^[23]
Surrounding temperature ($^\circ C$)	$T_\infty=70$ ^[96]	Filament length, width and height (m)	$l=0.2$ $w=0.00035$ $h=0.00035$
Platform temperature ($^\circ C$)	$T_{plat}=70$ ^[96]	Distance between nozzle to platform (m)	$H=0.002$
Emissivity factor	$\varepsilon=0.95$	Nozzle diameter (m)	$D=0.0004$

Stefan-Boltzmann constant ($W / m^2 \cdot K^4$)	$\sigma = 5.67 \times 10^{-8}$	Nozzle temperature ($^{\circ}C$)	$T_{nozzle} = 200$ ^[96]
Length, width and height of single controlled volume (m)	$\Delta x = 0.00001$ $\Delta y = 0.00001$ $\Delta z = 0.00035$	Gas constant ($cal / K \cdot mole$)	$R = 1.987$
100% latent heat of crystallisation per unit mass (J / kg)	$L = 90$ ^[96]	Reference number of nucleation	$N_0 = 0.175$
The distance of neighbour nodes in vertical and horizon direction (m)	$dx = 0.00001$ $dy = 0.00001$	the number of the printed elements in the x and y direction	$n_L = 2 \quad n_W = 2$
The thermal contact conductance with filaments ($W / m^2 \cdot ^{\circ}C$)	$k_{mat} = 200$	Glass transition temperature ($^{\circ}C$)	$T_g = 60$ ^[96]

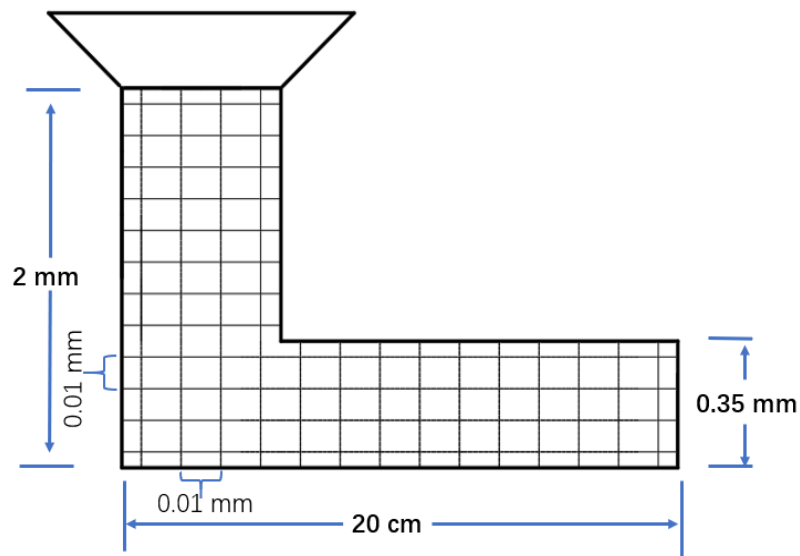


Fig. 35 A 20 cm line printed by a standard 0.4 mm diameter nozzle with two-dimensional mesh

Fig. 35 showed a single line printed by a 0.4 mm nozzle with two-dimensional domain discretisation. The whole filament has been meshed into 20000×35 parts, and the size of each element was in $0.01 \text{ mm} \times 0.01 \text{ mm}$. The gap between the nozzle tip and the platform was set as 2 mm. In this work, PLA was selected as the material to fill in the model, and the related values of PLA's heat transfer parameters were collected by Varun et al.^[96], and the values of FDM variables and material properties in the model to validate the previously published data are shown in **Table 7**.

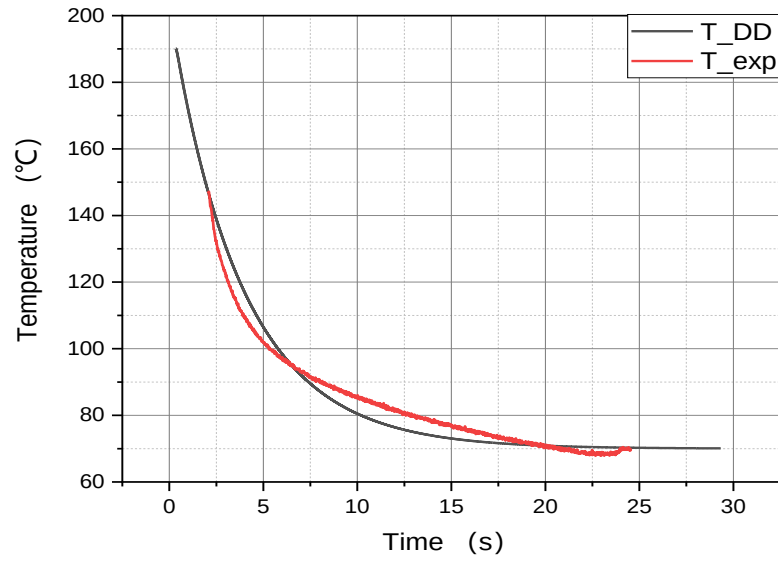
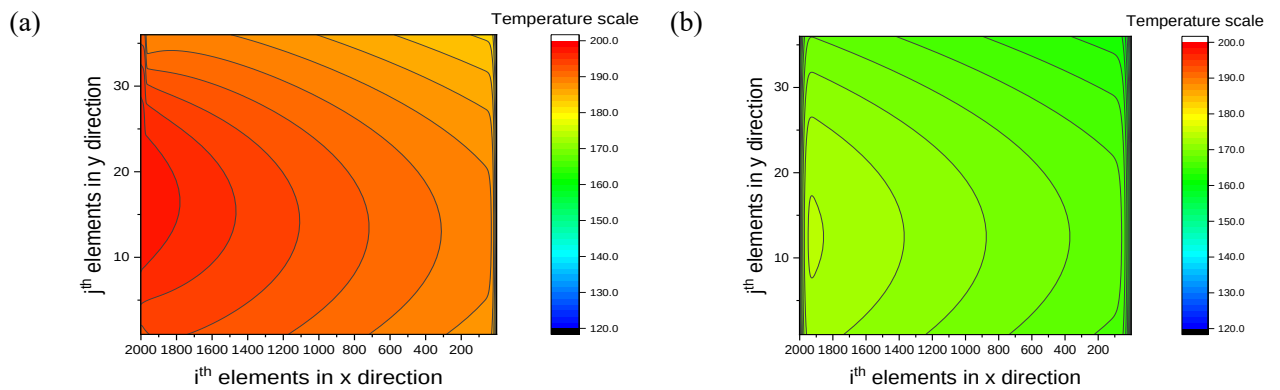


Fig. 36 A comparison of the simulation results by the domain discretisation methods(T_{DD}) with experimental data(T_{exp}) of Varun et al. ^[96]

As shown in **Fig. 36**, the domain discretisation method was applied in the simulation work with the same parameters' value and the same detecting position in experiments. The experimental data were recorded at 147 °C as the maximum point to the 67 °C as the lowest point, detected by a K type thermocouple^[96]. For the modelling work, the nozzle and surrounding temperatures were set as 200 °C and 70 °C, respectively. When the filament was in the semi-dynamic and transition period before contacting to the platform, the heat transfer coefficient would be different between the windward side and lee side. The former value was hypothesised 1.5 times as the latter, and in the free dynamic period, this value was set as $50 \text{ W} / \text{m}^2 \cdot ^\circ\text{C}$ the lee side.

In the proposed modelling, the results were recorded at the mid-bottom point of the whole filament at the same point as experimental work reported by the Varun et al. [96]. All the simulation results presented below including crystallisation and multi-line results used the same position to record the data except the especially instructions. The results of the domain discretisation method present an excellent match with the experiments data. From **Fig. 36**, as time goes on, the cooling rate was decreased on both experiment and simulation results. The predicted temperature evolution showed of around 4.4 °C of maximum deviations compared with the previously published experiments data[96]. This deviation might cause due to two main reasons: 1) In Varun's experiment, the printing object was a rectangular solid rather than only a single line. The temperature recorded midpoint of the central line would be affected by the adjacent material, and this influence will be discussed in Section 5.4 under multi-line works; 2) the range of K type thermocouple temperature measurements is from - 200 °C to 1350 °C, and one of the materials made up this K type thermocouple is nickel, which is magnetic. The deviation will undergo when the temperature reaches the Curie point of the nickel material, which is variable around at 150 °C for the different K type thermocouple. Based on the characteristics and broad range of the K type thermocouple, the numbers of deviation may occur and be recorded. One of the proofs is that the lowest temperature recorded was 67 °C which was below 3 °C than the platform and surrounding temperature (70 °C).



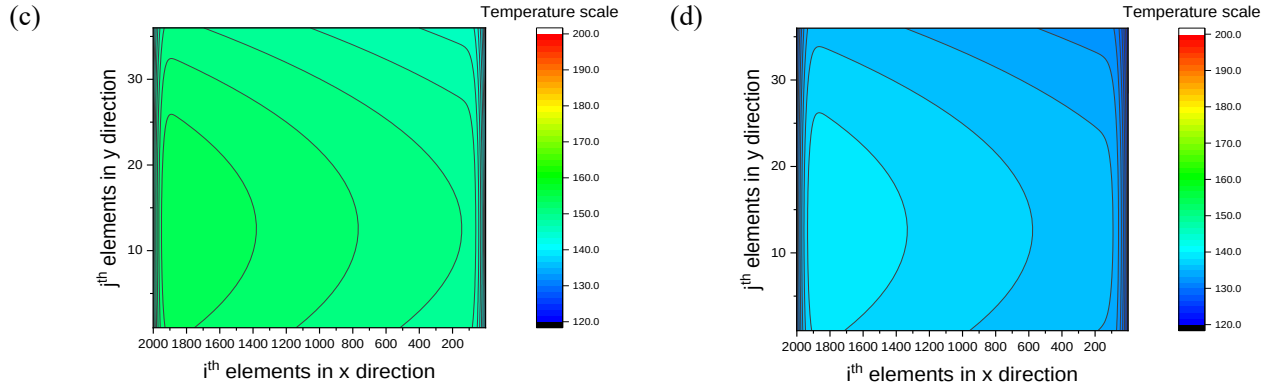


Fig. 37 The contour of the temperature distribution of the internal filament changes with time after the whole single line deposited on the platform: (a) 0s; (b) 1s; (c) 2s; (d) 3s.

In order to explore the whole filaments temperature change with time but not the only point. The whole of filament's nodes temperatures was recorded. When the whole filament deposited on the platform, the temperature distribution contour of the internal filament is shown in **Fig. 37**. At first, when the filament ends part just contacted to the platform, the highest temperature point is at the mid-end of the filament as shown in **Fig. 37 (a)**. Then, the whole filament began to experience natural cooling. Due to heat transfer between the surface of the filament and surroundings, and the cooling rate was different in different boundaries; therefore, the highest temperature gradually right and down shifted, as shown in **Fig. 37 (b)** (c) and (d). However, the highest-temperature area was still close to the end part of the filament because of the cooling time for different cross-section at x-direction is still in priorities.

5.2.2 Temperature evolution validated by the recorded data

In order to further provide validation for the proposed model, model-predicted temperature evolution was compared with experimental measurement, as presented in **Fig. 38**. Moreover, all the related parameters used in program for this comparison are listed in **Table 8**. As it is evident, both predicted and measured temperatures between 125 °C to 70 °C matched quite well, with the maximum deviation of only 4.3 °C. After that range, due to the recorded temperature data via thermal camera fluctuated, the simulated results appeared a big gap

compared with experiments data with a maximum deviation of 7.8 °C. From the above, the developed model is further verified with strong dependence.

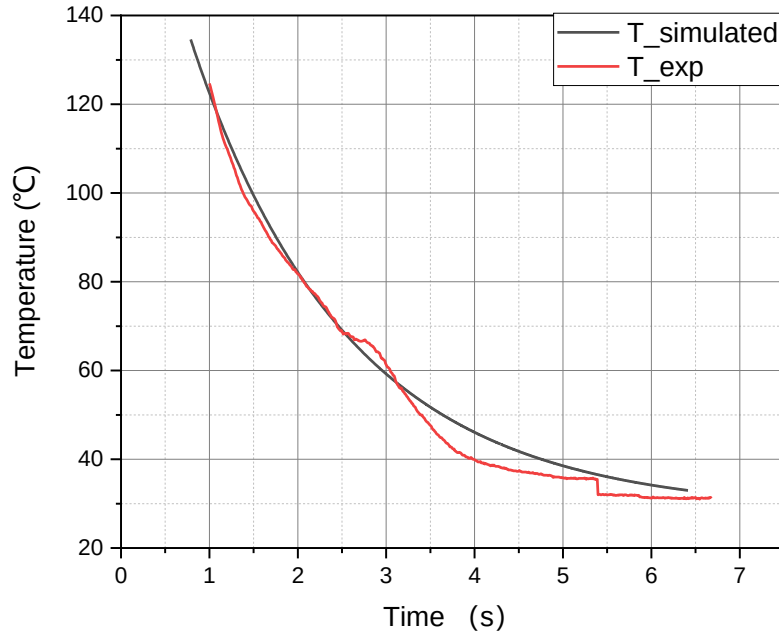


Fig. 38 Comparison of the model predicted results ($T_{\text{simulated}}$) with the recorded experiments data (T_{exp}) by a thermal camera

Table 8 the value of FDM variables and material properties to validate the data of experiments

Property	Value	Property	Value
Density (kg / m^3)	$\rho=1240$	Time step (s)	$\Delta t=0.00002$
Specific heat capacity ($J / kg \cdot ^\circ C$)	$C_p=1800$	Nozzle velocity (m / s)	$v = 0.5$
Thermal contact conductance with the platform ($W / m^2 \cdot ^\circ C$)	$k_{plat}=30$	Pre-exponential factor	$G_{0I}=3.987 \times 10^7$ $G_{0II}=4.81 \times 10^{11} [23]$
Thermal conductivity ($W / m \cdot ^\circ C$)	$k=0.18$	Surface nucleation constant	$K_{gI}=2.55 \times 10^5$ $K_{gII}=5.51 \times 10^5 [23]$

Heat transfer coefficient ($W / m^2 \cdot ^\circ C$)	$h_1=120$ $h_2=80$	Activation energy ($cal / mole$)	$U^*=1500.0^{[23]}$
Surrounding temperature ($^\circ C$)	$T_\infty=30$	Filament length, width and height (m)	$l=0.1$ $w=0.00035$ $h=0.00035$
Platform temperature ($^\circ C$)	$T_{plat}=60$	Distance between nozzle to platform (m)	$H=0.002$
Emissivity factor	$\varepsilon=0.95$	Nozzle diameter (m)	$D=0.0004$
Stefan-Boltzmann constant ($W / m^2 \cdot K^4$)	$\sigma = 5.67 \times 10^{-8}$	Nozzle temperature ($^\circ C$)	$T_{nozzle}=215$
Length, width and height of single controlled volume (m)	$\Delta x=0.00001$ $\Delta y=0.00001$ $\Delta z=0.00035$	Gas constant ($cal / K \cdot mole$)	$R=1.987$
100% latent heat of crystallisation per unit mass (J / kg)	$L=90$	Reference number of nucleation	$N_0=0.175$
The distance of neighbour nodes in vertical and horizon direction (m)	$dx=0.00001$ $dy=0.00001$	Glass transition temperature ($^\circ C$)	$T_g=55$

5.3 Computational modelling of PLA crystallinity

As was clear from the above results, the proposed model for temperature evolution during 3D printing performs well. Hence, it was felt that it would be good to extend this model to predict the materials' crystallisation behaviour as well incorporating the Hoffman theory. Based on the previous study by Tsuji et al.^[23], the critical value for simulating PLA crystallisation has been listed in **Table 7**, and the regime transition temperature from the regime II to the regime III was observed at 120 °C in their experiments. However, the regime I to regime II temperature did not present obviously. Theoretically, this regime transition temperature should be around 150 °C for the PLA materials according to other experimental data^[4], but not all the time, all three regimes could be observed. Therefore, it was decided to follow Tsuji's experimental procedure for the simulation work of this study and also to consider only one regime change from regime II to regime III.

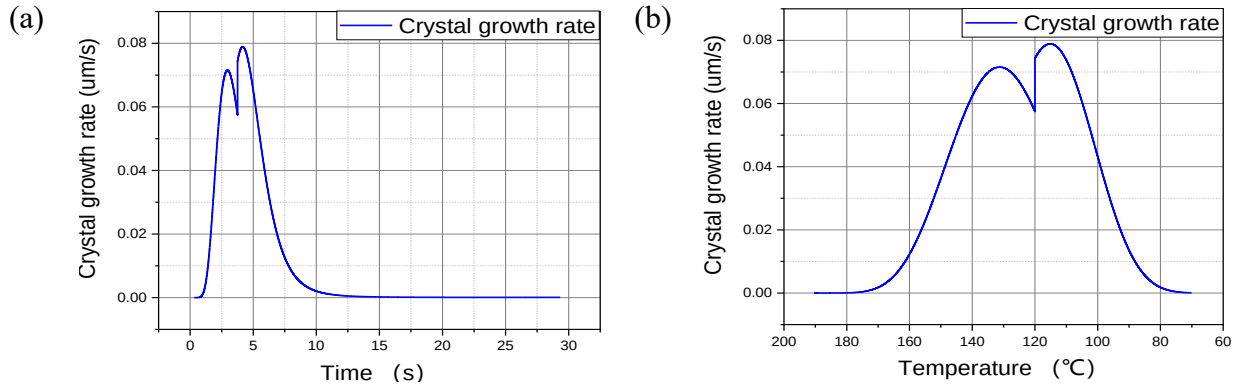


Fig. 39 The crystal growth rate as a function of (a) time and (b) temperature

Fig. 39 shows the crystal growth rate (G) as a function of time and temperature. As is apparent, the growth rate of crystal increased with time and temperature decreasing at first. After reaching the maximum value, it decreased with time and temperature. When the temperature dropped down to 120 °C, the regime II transferred to the regime III and this regime transformation is observable from the two peaks presented in **Fig. 39**. The first peak occurred at 2.96 s at a temperature of 131.3 °C which was the maximum growth rate in the regime II of 0.07154 $\mu\text{m/s}$. The next peak point occurs at 4.16 s and a temperature of 115.08 °C which was the maximum growth rate in the regime III of 0.07890 $\mu\text{m/s}$. The

maximum growth rate in the regime III was more significant than that in the regime II, which approved previous published work by Tsuji et al.^[23]

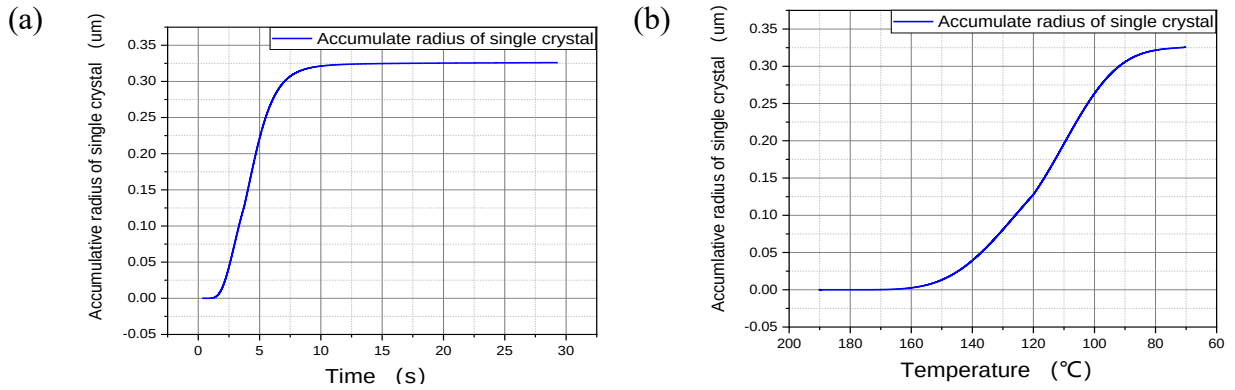


Fig. 40 The accumulative radius of the first crystal as a function of (a) time and (b) temperature

Since the growth rate values are known, the accumulative radius of the first crystal could be calculated from Eq. (41). As shown in **Fig. 40**, the accumulative radius of a single-crystal vary with the time and temperature. The radius of crystal growing began at around 1.5 s and at that time the temperature was 160 °C. And the end of radius growth was at 10 s and at that time the temperature was 80 °C. The maximum radius growth rate has occurred at 120 °C at around 3.7 s. The initial and final growth rate was relatively slow. Therefore, the whole curve has taken an S curve, and the final maximum radius of the crystal was 0.3259 um.

Next, the number of nucleation which was formed in the printed filaments would be considered. According to the theory of Hoffman et al.^[31], the nucleation density in an element could be calculated by equation (44) and (45). Based on the study of the PLA's isothermal crystallisation nucleation density by Sajjad et al.^[4], the PLA nucleation density would be as a function of temperature and the experiments data are collected from ^[23,90,113]. The nucleation density increases as the temperature decreases and the rate of increase become slower with the decreasing temperature. Finally, the nucleation density would be almost stable when the temperature is lower than 90 °C, and the density of nucleation is 5000 number / mm³. The volume of an element is by length × width × height as:

$$0.01 \times 0.01 \times 0.35 = 3.5 \times 10^{-5} \text{ mm}^3$$

Therefore, the number of nucleation forming in an element could be calculated by

multiplying the volume of the element with the reference density of nucleation:

$$3.5 \times 10^{-5} \times 5000 = 0.175$$

Which means every 6 elements will observe one formed crystal averagely. In this modelling work, the value $1/k$ in Eq. (45) was set as 5497.8^[24], then the Eq. (45) will become:

$$N = 0.175 \exp\left(-\frac{5497.8}{T \Delta T}\right) \quad T > T_g$$

The coefficients of A , B and C in Eq. (46) were polynomial fitted by the experiments data from presented by Zhou et al^[111]. Hence, the values of coefficients are 4.95, -1066.5 and 57600, respectively. Therefore, the formed nucleation rate in Eq. (47) can be expressed as:

$$n_s = \frac{0.175 \exp\left(-\frac{5497.8}{T \Delta T}\right)}{4.95T^2 - 1066.5T + 57600}$$

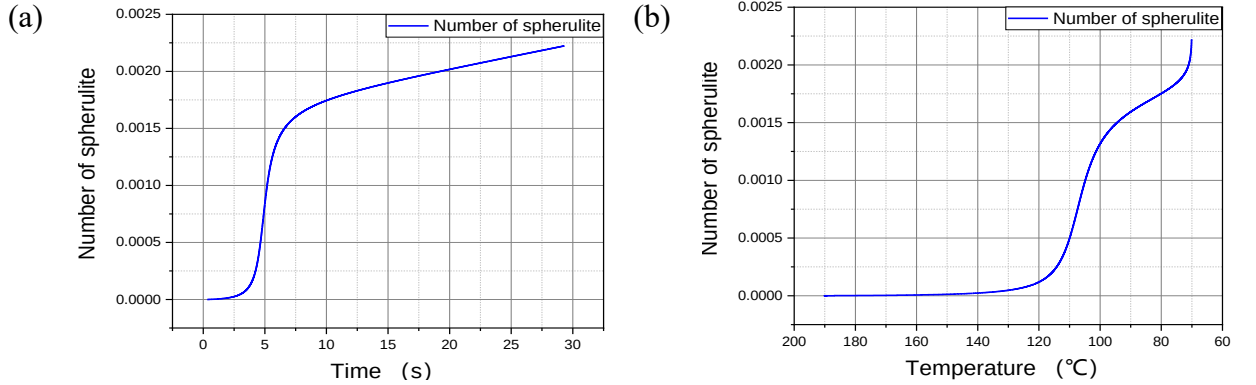


Fig. 41 Number of spherulites created as a function of (a) time and (b) temperature in an element

Fig. 41 showed the results of accumulative forming nucleation number varied with time and temperature as was calculated by Eq. (48). Under such temperature evolution circumstance, the number of spherulites growing began at around 2.5s and at that time the temperature was 150 °C and the maximum nucleation rate exhibited at the temperature of 107 °C at 5 s. Overall, the shape of the curve looked like “S” shape. Interestingly, at the end of these “S” curve, the value did not keep constant. The reason for this phenomenon was that the temperature was still higher than the glass transition temperature of PLA. Therefore, the forming nucleation process was still in progress, and the time, when the temperature was below 85 °C, had a low cooling rate compared with the previous periods, which resulted in

filament keeping long periods at this temperature range. Accordingly, the number of spherulites showed a dramatic increase at the end of the curve changed with temperature in **Fig. 41 (b)** but still kept the same low-forming speed as a function of time shown in **Fig. 41 (a)**.

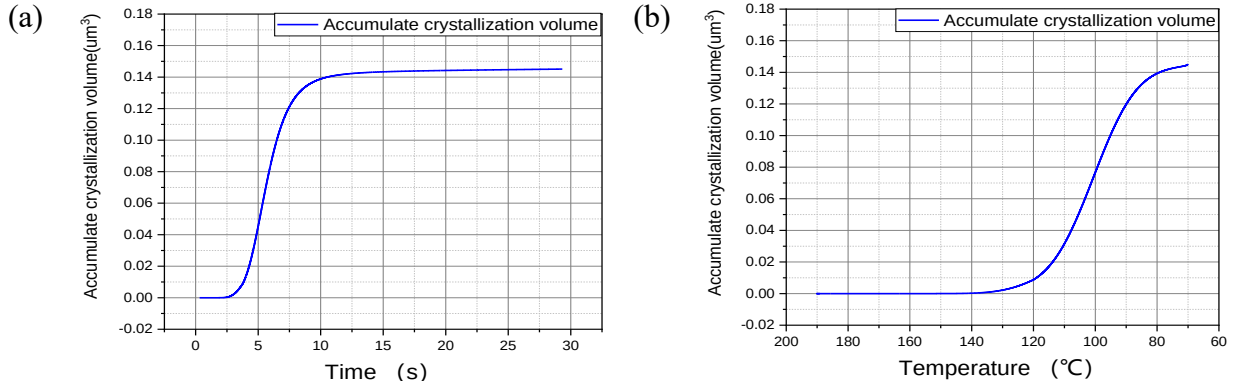


Fig. 42 The total crystallisation volume as a function of (a) time and (b) temperature in an element

Next, Eq. (49) can be used to determine the accumulation of the crystallisation volume. The crystal volume is the result of combining crystal growth rate and nucleation rate. **Fig. 42** showed that the total crystal volume in one element varied with time and temperature. The volume of crystal growing began at around 3 s and at that time the temperature was 140 °C. And the end of radius growth was at 12.5 s and at that time the temperature was 70 °C. The final volume of the total number of crystals was only 0.147 um^3 , and the volume of a single element was 35000 um^3 , the ratio of crystallisation was only a percentage of around 0.00042% compared to the total volume of the elements. Such a low crystallisation rate was due to the periods for crystallisation was less than 10 s. Compared with the time more than decades of minutes used in isothermal crystallisation to be 100% crystallisation ^[111], 10 s was only transient. Consequently, enlarging crystal times and decreasing cooling rate will assist the printed PLA to have more time in crystal growing periods, which will contribute to enhancing the mechanical properties of final products.

5.4 Computational modelling of multi-line

Since the proposed single line model has been well validated with two group of experimental works, it would be satisfied to extend from single line to the multi line work. As **Fig. 43** shown, the printing order and the length, width and height value of 2×2 multi-line has been presented. The time of the nozzle moving to the next line initial depositing position was set as 0.1 s.

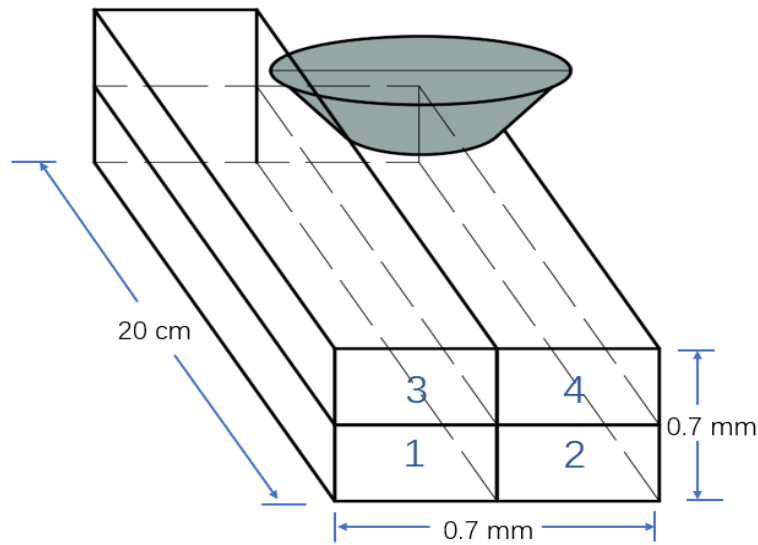


Fig. 43 2×2 multi-line dimensions and printing order

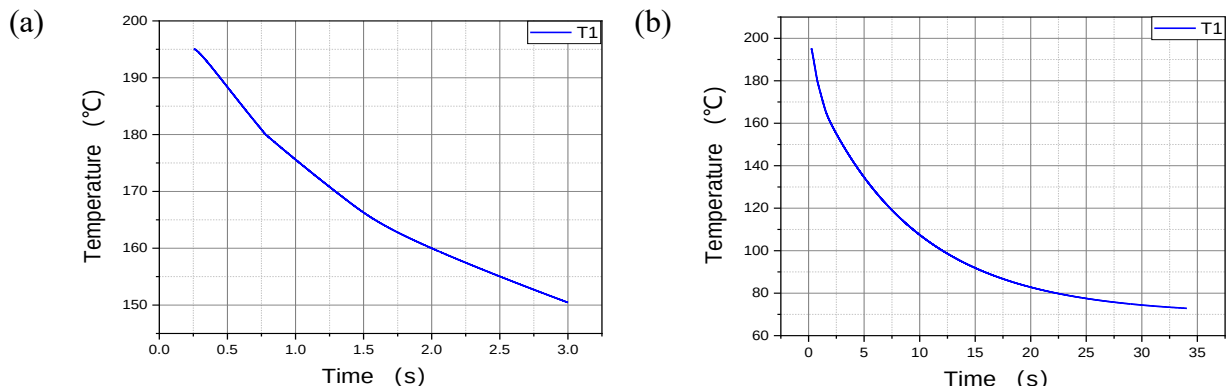


Fig. 44 the evolution of the first printed line (T1) bottom midpoint temperature from (a) 0 to 3 s and (b) 0 to 34 s

The evolution of the first printed line bottom midpoint temperature was shown below in **Fig. 44**. At first 3 seconds, the temperature changing curve was not as smooth as that in a single

line. At the time about 0.8 s and 1.6 s, the cooling rate decreased evidently due to the second line and the third line deposited next to and upper to the first line separately. The top and lateral surface, which should have convected and radiated with the air, had been replaced to contact with the high temperature printed filaments. The heat flow in the interface would be from the new filament to the elder filament rather than from the elder filament to the air. Therefore, the cooling rate become slower compared without reheating process by other filaments.

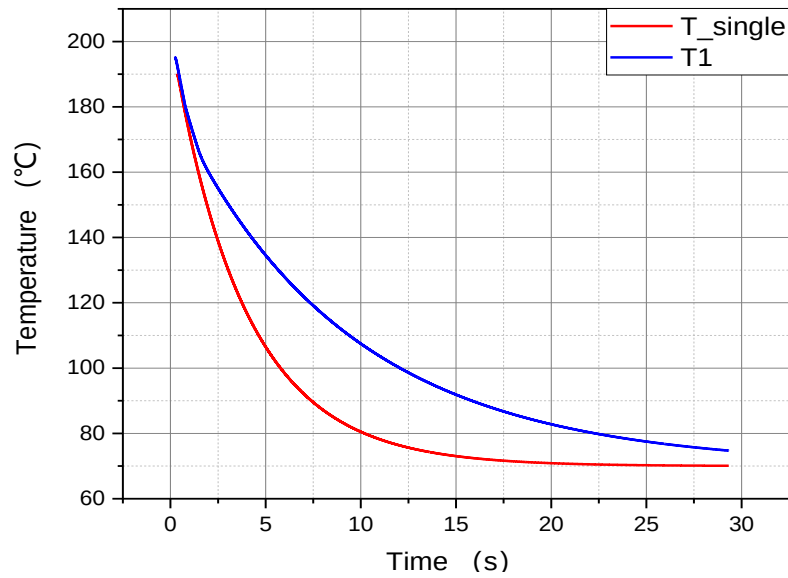


Fig. 45 Comparison between single line(red) with multi-line(blue) simulation results in the same detecting point with the same value for all critical parameters

In order to investigate the multi line work effects on the temperature evolution and crystallization behaviour of the printed filaments, the simulated results were compared by applied the proposed model. **Fig. 45** compared the single line(red) and multi-line(blue) simulation results in the same detecting point with same parameters' values. When the red line cooled down to the 100 °C, the temperature of the blue line was still in the 128 °C approximately. Moreover, the time for blue line cooling down to 100 °C was about 12 s, which delayed about 6 s and was doubled compared with the red line results. The red line reached 80 °C after being printed 10 s while the blue line reached 80 °C at 22.5 s. Consequently, the multi-line work helps filament keep in crystallising temperature longer by the reheating process, which will have the potential to get the better mechanical property

with large crystal.

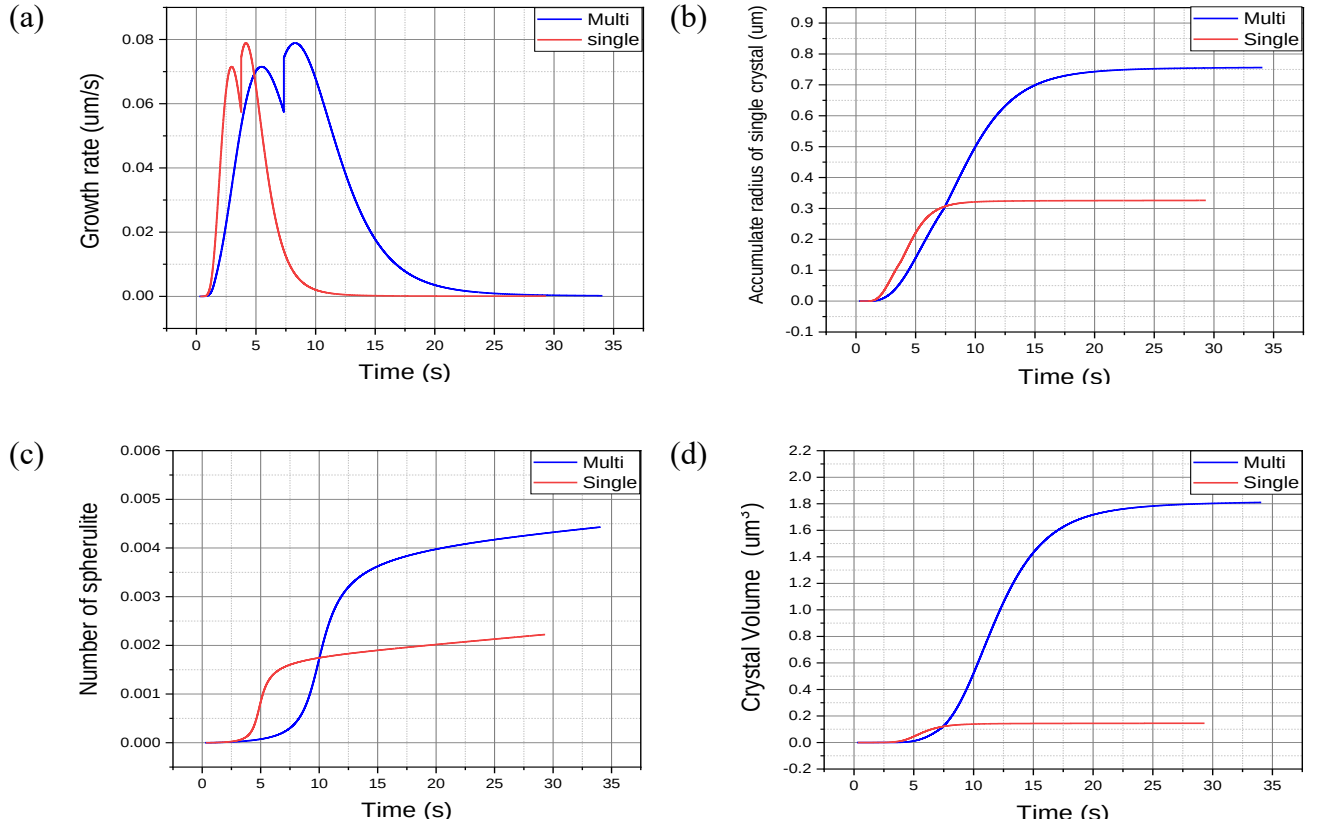


Fig. 46 Comparison of the single line (red) and multi-line (blue) crystallisation behaviour at the same position as a function of time by (a) Crystal growth rate; (b) Accumulate radius in the first crystal; (c) Number of formed spherulite; (d) The total crystal volume in single element.

In **Fig. 46**, the simulated results of the single line and multi-line crystallisation behaviour under the temperature profile mentioned above have been exhibited. In **Fig. 46 (a)**, the maximum crystal growth rate was the same in both situations. However, because of the cooling rate in the single line was more significant than the multiline work, the time reaching the maximum point in single-line work was faster than that in multi-line work. Besides, the area under the red curve was smaller than that in the blue curve, and the integrated results were shown in **Fig. 46 (b)**. The y-axis represented the first crystal radius grew with time. The maximum radius of the crystal was 0.33 μm and 0.76 μm , respectively. **Fig. 46 (c)** showed the number of spherulites formed vary with time and **Fig. 46 (d)** showed the total crystal volume in single element. The maximum volumes were 0.15 and 1.8 μm^3 ,

respectively. From the results, the volume grown in multi-line work was twelvefold as that in single-line work, which showed the significance of reheating effect and extending the crystallising time to the final results of crystal growth.

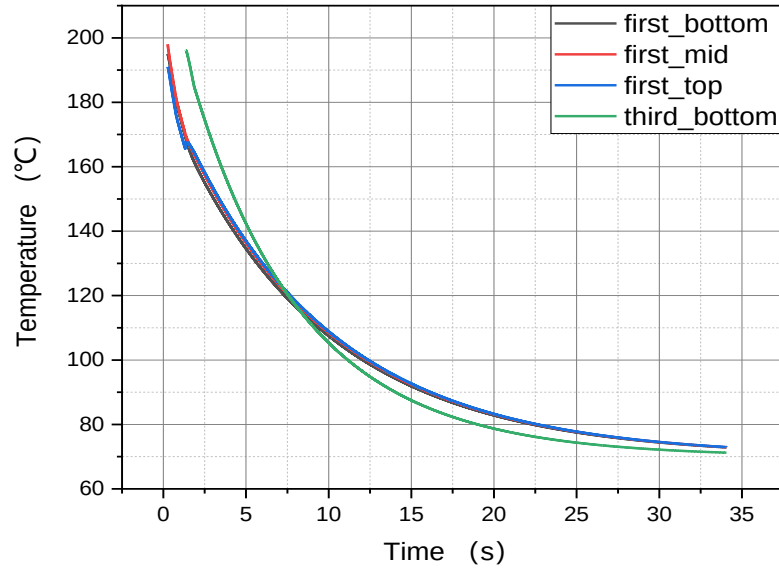


Fig. 47 Temperature distribution of the first line in the height direction

As similar to the situation in single line work, the internal temperature distribution of printed filaments will be different in the vertical direction. **Fig. 47** provided the temperature distribution in the same cross-section. In the beginning, the top surface detecting point had the lowest temperature due to it had the most significant area exposed to the air and generate heat exchange with surroundings. The bottom detecting point had the mid-value temperature of the three-detecting point. The heat loss convected with the platform and air was less than the top point. The centre point had the highest temperature because the thermal conductivity of PLA was low. Therefore, the heat loss of the centre point was the smallest. Then, when the third filament deposited on the first filament, the distribution of the temperature changed a lot. The top point in the first filament appeared a small temperature rising peak which began at 1.3 s. This phenomenon was due to the third filament deposited on the first filament as a heat source, and this heat source reheated the previous layer. Therefore, the rising peak showed at that time. Then, the temperature would decrease again as time went by.

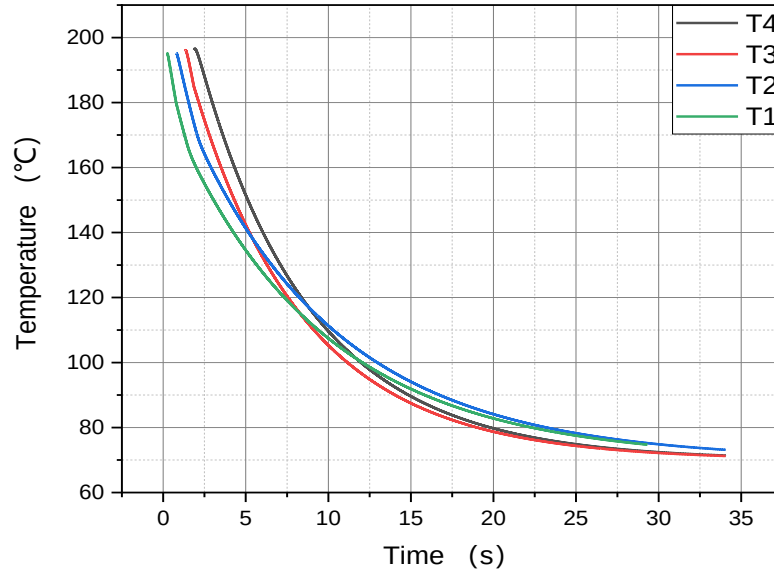


Fig. 48 4-line bottom midpoint simulation result from 0s to 34 s

After analysing the single filaments temperature distribution in the multi line work, all extruded filaments temperature evolution will be focused on. **Fig. 48** showed temperature changed at the same position in different filaments. T1 (green curve), T2(blue curve), T3(red curve) and T4 (black curve) were representing the first, second, third and fourth filaments, respectively. Each filament delays printing time between two neighbour order printed filament 0.6 s was the combination of previous filament printing time, nozzle moving time and depositing time from the nozzle to the platform or previous layer.

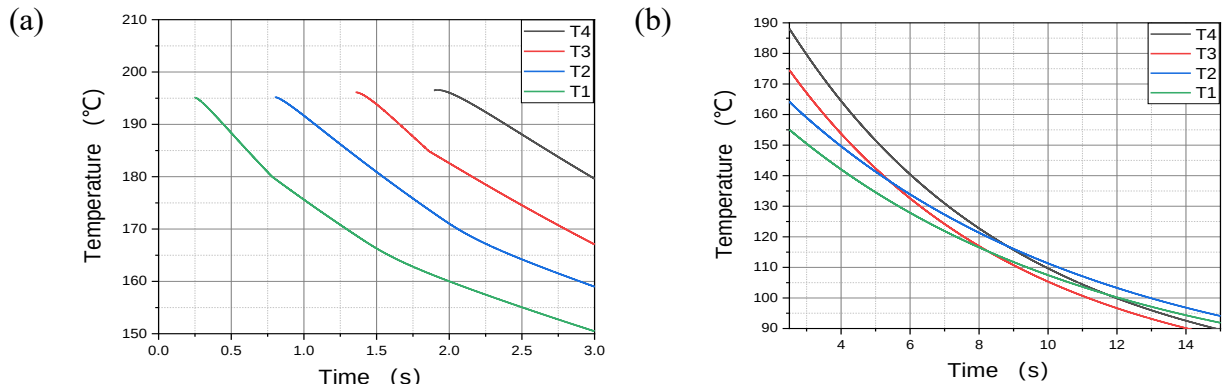


Fig. 49 4-line bottom midpoint simulation result from (a) 0s to 3s and (b) 2.5s to 15s

From the results, there are two periods arouse the notice, and the zoom-in results are shown in **Fig. 49**. The first one is from 0 s to 3 s in **Fig. 49 (a)**. Whenever the new filament started depositing, the previous filament temperature curve showed the cooling rate decreased,

which appeared in 0.8 s and 1.5 s on the green line, 2.05 s on the blue line and 1.85 s on the red line. The decreased cooling rate for the green line was due to the second line and the third line deposited next and upper to the first filament, respectively. Moreover, the decreased cooling rate for the blue line and red line was due to the fourth line deposition. The time delay between blue and red curve decreased cooling rate may due to the thermal conductivity is small. The influence of the new high-temperature filament needed time to transfer to the second line bottom, which performed as the time delay for the cooling rate decrease.

Fig. 49 (b) zooms in the time between 2.5 s to the 15 s, and there are four intersection points created by these four filaments dynamic temperature curve. The main reason for these curve intersections appearing is that the later filament covered the formerly deposited filament which reduced the convection and radiation area with the air. Therefore, the cooling rate for the later filament was more significant than the former filament. Also, the newly deposited filaments as a heat source reheated the previous filaments, which slowed down the rate of temperature falling. Besides, PLA is one of the low thermal conductivity materials, and the temperature difference distributed along the height direction. As time goes on, the temperature decreasing for the later filament was faster than the former filament, which is the reason why the intersection points would appear.

5.5 The influence of different parameters

In this section, a series of modelling results were implemented to investigate the influence of different parameters on temperature evolution of filaments and crystallinity, such as: printing speed, thermal convection coefficient, thermal contact conductance with platform, nozzle diameter of the FDM printer, latent heat of crystallisation and thermal contact conductance with neighbour filaments. When changing parameter in each section, other parameters were keeping as constant value as shown in **Table 7**.

5.5.1 The influence of printing speed

In order to investigate the influence of the printing speed on the cooling rate of filaments, the printing speed was set as 0.02, 0.05, 0.1 and 0.15 m/s , respectively. From **Fig. 50**, due to the extrusion speed was different, the time of filament contacted to the platform or previously printed filament was different. The slower the printing speed, the more time was spent in the semi-steady period and transition period. Therefore, when the filament contacted to the floor, the temperature cooled down much more compared to the high-speed extrusion way. However, this difference would not be too much. The maximum temperature gap between the black and green curve in **Fig. 50 (a)** is 4.3 $^{\circ}C$ approximately. Moreover, with time going by, this difference becomes smaller and smaller. Therefore, the printing speed has less effect on the single line work.

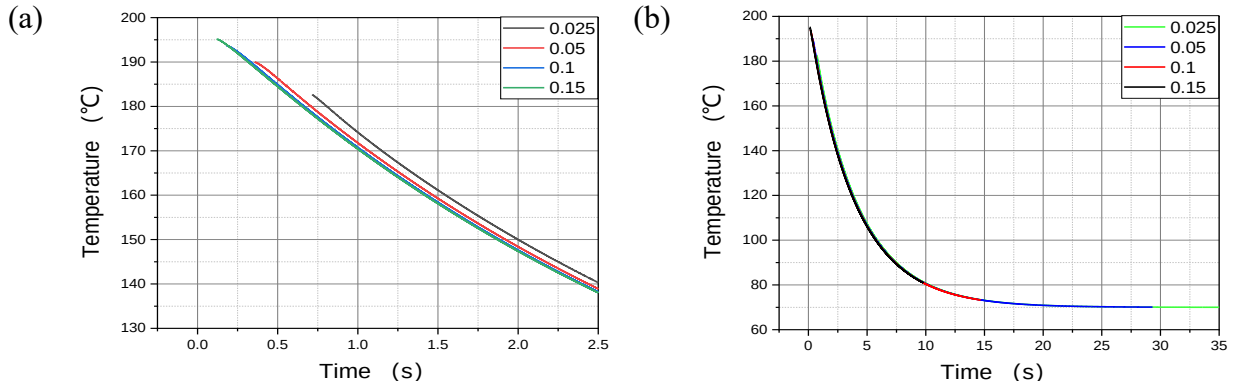


Fig. 50 single line temperature evolution with different printing speed from 0.025 m/s to 0.15 m/s : (a) 0 to 2.5 s; (b) 0 to 35 s

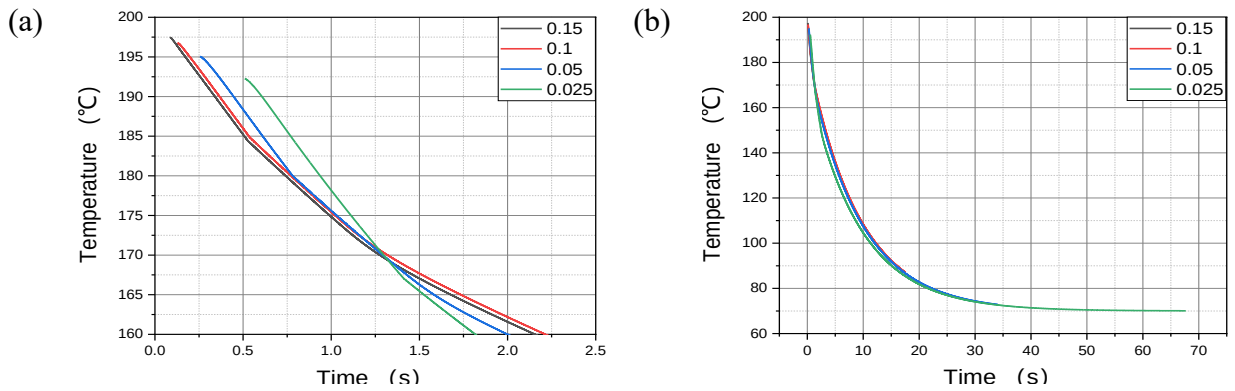


Fig. 51 multi-line temperature evolution with different printing speed from 0.025 m/s to 0.15 m/s : (a) 0 to 2.5 s; (b) 0 to 68 s

Fig. 51 presents the influence of printing speed on the multi-line works. Similarly, in the initial period, the temperatures were different when contacted to the platform. The reason for this difference has been explained in the single line work. Then, the cooling rate changed at a different time due to the different printing speed, which led to the upper layer deposited on the bottom layer with different time. Hence, with different cooling rate changing point, the later temperature showed different development. The faster the printing speed, the higher the temperature at the same time compared with other lower printing speeds. The longer the time remains at the crystal temperature, the more helpful it could be to assist polymer crystal forming and growing, which approve the experimental results of Tsouknidas et al. [64]

5.5.2 The influence of thermal convection coefficient

When the surrounding air flow is not violent and remains relatively calm, the thermal convection coefficient is at a relatively low value. On the contrary, the coefficient of thermal convection increases with the speed of air flowing across the surface of the filament. For the purpose of finding how the ambient would change the temperature evolution of printed filaments, the different value of thermal convection coefficient was investigated.

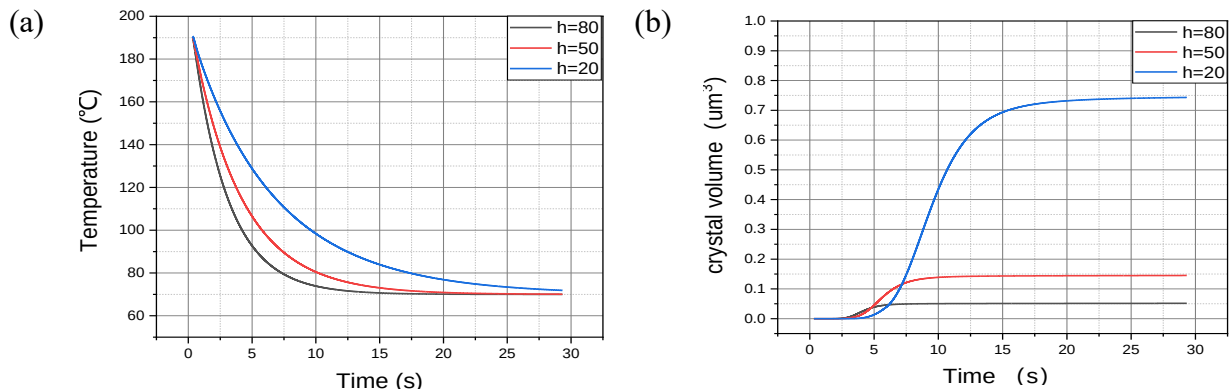


Fig. 52 single line (a) dynamic temperature changes and (b) total crystal volume in single element with different thermal convection coefficient: 20, 50 and 80 ($\text{W} / \text{m}^2 \cdot \text{K}$)

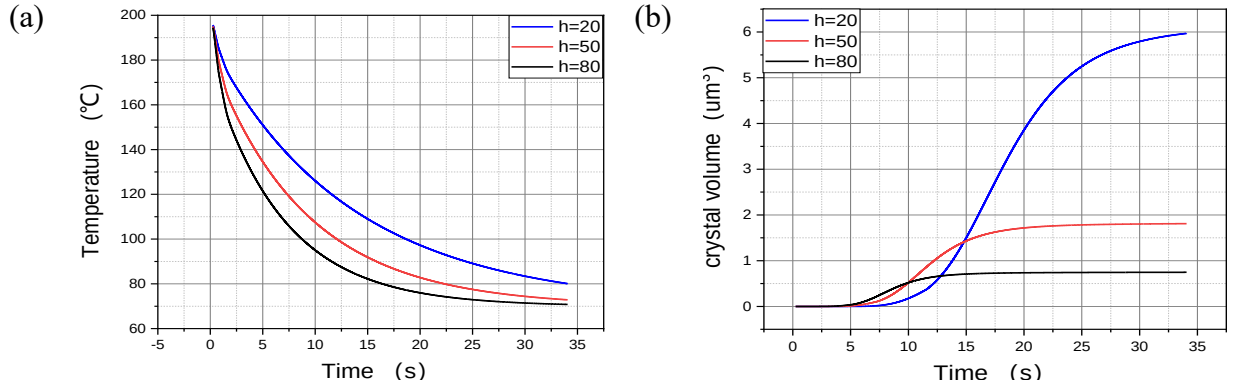


Fig. 53 multi-line (a) dynamic temperature changes and (b) total crystal volume in single element with different thermal convection coefficient: 20, 50 and 80 ($\text{W} / \text{m}^2 \cdot \text{K}$)

Fig. 52 and **Fig. 53** shows the temperature changes and crystal volume in a single element with different thermal convection coefficient by single and multi works separately. With the thermal convection coefficient decreasing, the cooling rate also decreased. Therefore, when the filament contacted to the platform, the initial temperature was different and started from 189.4, 190 and 190.6 $^{\circ}\text{C}$ for the three curves separately in **Fig. 52 (a)**. Then, the time extended in the rapid crystal growth rate under the low value of thermal convection coefficient condition. The maximum volume of final grown crystal reached 0.75 and 6 μm^3 with the h being equal to 20 ($\text{W} / \text{m}^2 \cdot \text{K}$), and when the thermal convection coefficient was 50 and 80, the maximum volume of crystal was only 0.15, 0.05 and 1.8, 0.75 μm^3 in single and multi works, respectively. Therefore, the forced convection will accelerate temperature decrease of the printed filaments, which results in the rate of crystallization decreased.

5.5.3 The influence of thermal contact conductance with the platform

Thermal contact conductance with the platform depends on the types of contacting materials with platform, surface roughness of the two materials interface and the interface pressure, etc. Compared with the thermal convection coefficient, the parameter thermal contact conductance with the platform had less effect on the cooling rate which was due to the area contact to the platform was less than the surface explored into the air directly.

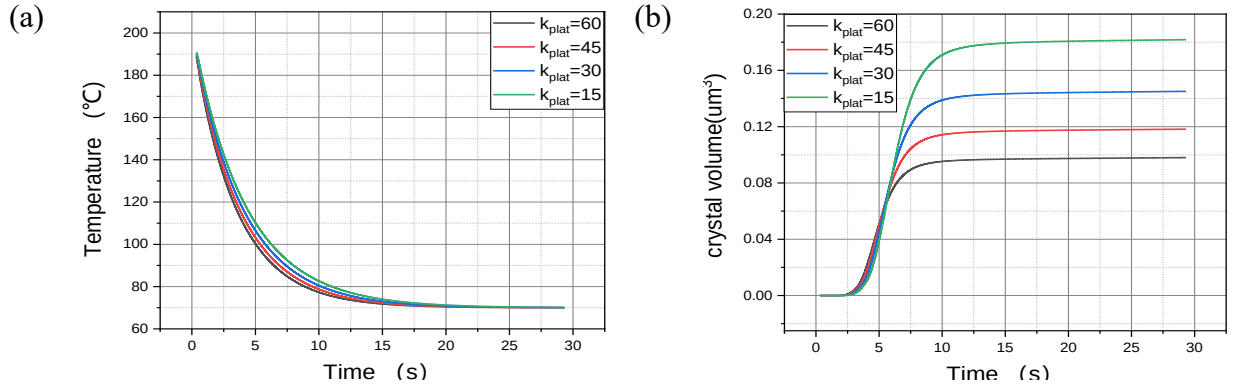


Fig. 54 single line (a) dynamic temperature changes and (b) total crystal volume in single element with different thermal contact conductance with the platform: 15, 30 45 and 60 ($W / m^2 \cdot K$)

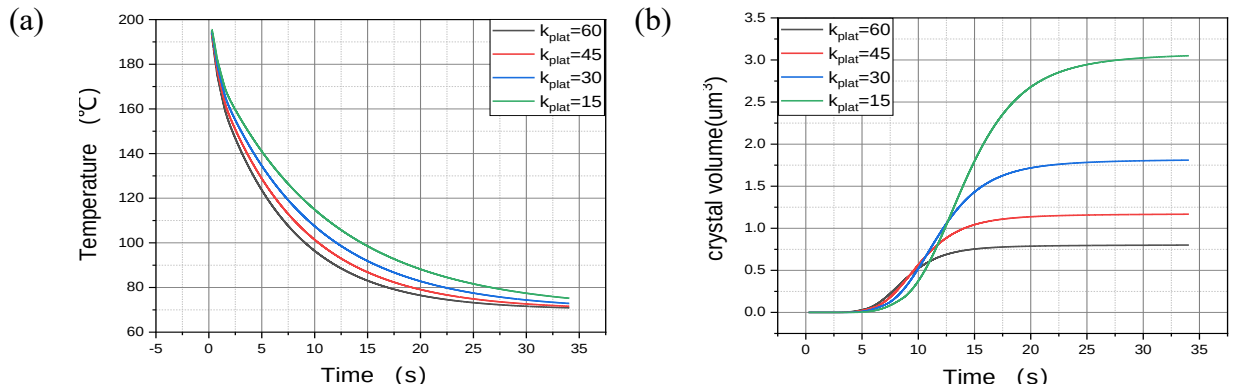


Fig. 55 multi line (a) dynamic temperature changes and (b) total crystal volume in single element with different thermal contact conductance with the platform: 15, 30 45 and 60 ($W / m^2 \cdot K$)

As shown in **Fig. 54 (a)**, the maximum temperature difference between $k_{plat}=15$ and $k_{plat}=60$ was 9.4 °C at about the 6 s, and the maximum final crystal volumes for these two situations were 0.18 and 0.10 μm^3 , respectively. In **Fig. 55 (a)**, the multi-line works further enhanced the results with the influence at the different thermal conductance with platform values. The temperature gap was increased compared with that in the single line work at the same time point. Moreover, the crystal volume disparity was also enlarged. The maximum value at $k_{plat}=15$ is 3.1 μm^3 compared with the minimum value at $k_{plat}=60$ is 0.8 μm^3 . Therefore, by changing the thermal convection coefficient via airflow speed and thermal contact conductance with the platform via contact level could effectively improve the crystal

volume in 3D printed filament both single and multi-line work.

5.5.4 The influence of the nozzle diameter

To understand the effect of nozzle diameter on the temperature evolution, the nozzle diameter was increased from 0.2 mm to 1.2 mm. There was a clear trend that the larger the volume extruded from the nozzle, the slower the temperature decreases in **Fig. 56** and **Fig. 57**. As a result, the time for crystal growth became more prolonged, and the volume of total crystal became more substantial compared with those extruded from a small diameter nozzle.

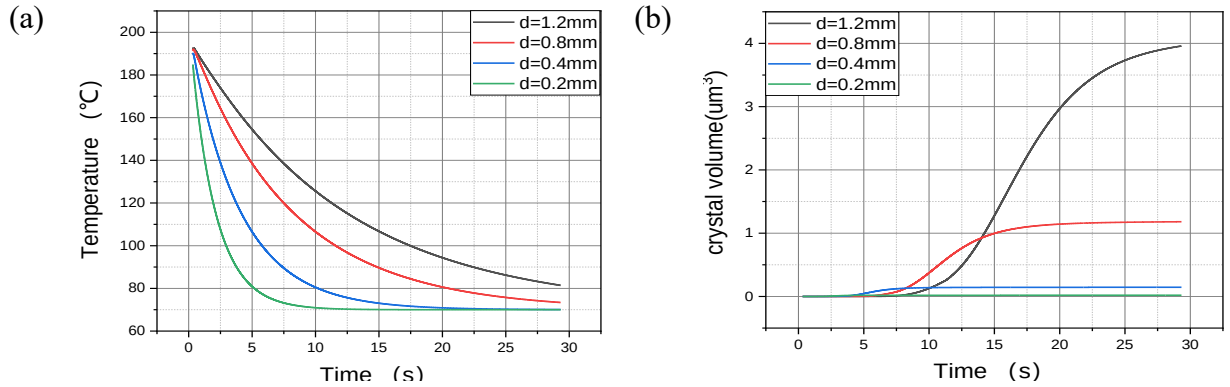


Fig. 56 single line (a) dynamic temperature changes and (b) total crystal volume in single element with different nozzle diameters: 0.2, 0.4, 0.8 and 1.2 mm

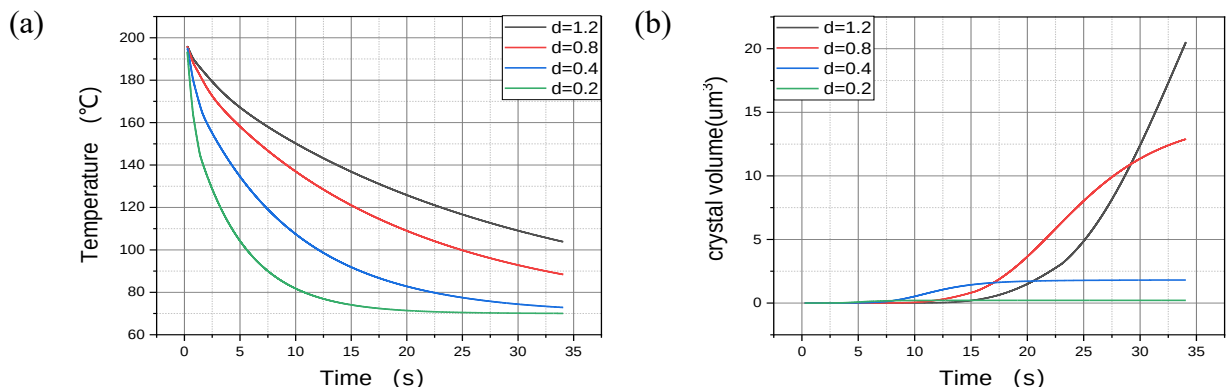


Fig. 57 multi line (a) dynamic temperature changes and (b) total crystal volume in single element with different nozzle diameters: 0.2, 0.4, 0.8 and 1.2 mm

Moreover, this phenomenon was also enhanced by multi works compared with single line work. Consequently, using a larger diameter nozzle is beneficial to decrease the cooling rate of filaments and increase the degree of crystallisation in printed filament. However, the resolution of the final product may be affected by the change of this process due to the change of the diameter.

5.5.5 The influence of latent heat of crystallisation

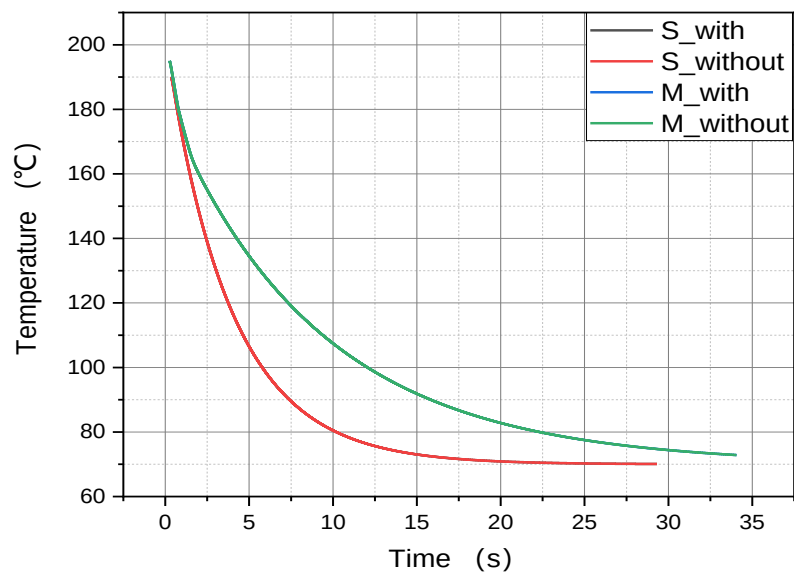


Fig. 58 result of a single (S_) and multi line (M_) work temperature evolution with and without latent heat of crystallisation

There were no significant differences found on temperature evolution results between with and without the influence by the latent heat of crystallisation both in single and multi line work, as shown in **Fig. 58**. Due to PLA is a semi-crystallised polymer and the time for extruded PLA filament crystallisation in suitable temperature was too short. Hence, the accumulated latent heat of crystallisation has little effect as a feedback on the filaments' temperature profile.

5.5.6 The influence of thermal contact conductance with filament

With the different setting in the FDM printing process, the deposited filament contact with the previous filament has different content. The better bonding between the two filaments, the better these two filaments proceeds heat transfer between them. If the bonding situation between two printed filaments performs well, the value k_{mat} would be tremendous, which means no or less thermal resistance would exist between the two filaments. In that situation, these two filaments could be seen as a combination of a giant volume filament. On the contrary, if these two filaments did not bond well, the thermal resistance between their interface would be enormous, and heat energy would hard transfer from one to another.

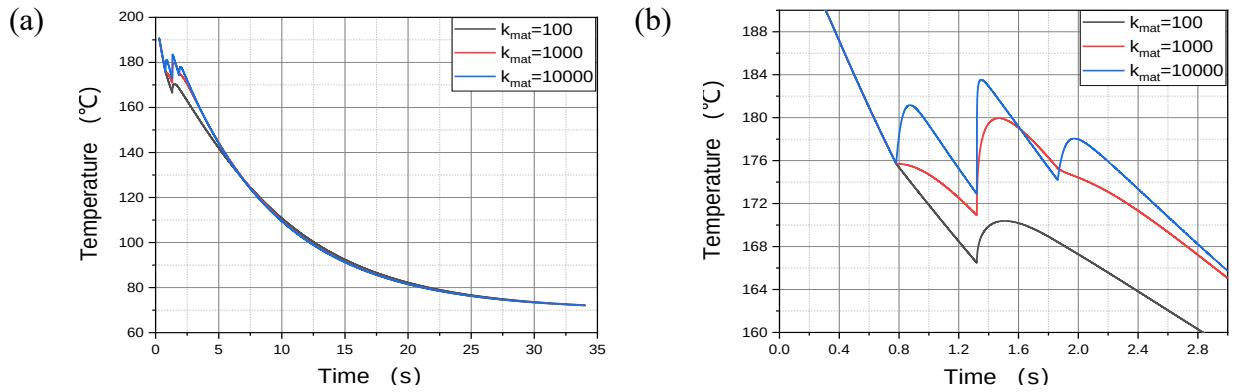


Fig. 59 multi line temperature evolution of the upper surface element with different values of thermal conductance with filament (k_{mat})

Fig. 59 (a) showed the temperature evolution with different value of thermal conductance with filament (k_{mat}). When three later filaments deposited in order, the presented number of reheating peaks, which was heated by the newly deposited filaments, were different in the different bonding situations as **Fig. 59 (b)** exhibits. The better the bonding, the more reheat peaks showing. Although the high value of k_{mat} helped the filament maintain at high temperature by reheat process, it also accelerated the cooling rate compared with the low value of k_{mat} for the formerly deposited filament, due to the small thermal resistance between bonded filament.

5.6 Summary

At first, the key parameters of experimental filaments used in the simulation have been measured by a series of experiments, which covered thermal conductivity, melting temperature, glass transition temperature and specific heat capacity. Then, two groups of temperature evolution experimental data from Varun and thermal camera detecting, were compared with the simulation results by applying same surrounding condition and materials parameters with experiments separately. Meanwhile, the crystallisation process was also modelled by the proposed model. After successfully applying single line model, the multi line works was carried out with temperature evolution and crystallisation process with re-melting process on the previous printed materials. The multi-line work helps printed materials keep in crystallising temperature longer by the reheating process, which will have the potential to get the better mechanical property with large crystal. Moreover, the influential of different parameters were investigated by the simulation with the proposed model. As was confirmed by the result, the thermal convection coefficient, the thermal contact conductance with the platform and the size of the nozzle diameter have a significant impact on the temperature evolution and degree of crystallization. On the contrary, the printing speed and latent heat of crystallization have little effect on temperature evolution.

Chapter 6 Conclusions

From the numerical modelling results presented in Chapter 5, it is possible to draw the following conclusions for the modelling of temperature evolution and crystal growth in additive manufacturing of polylactic acid (PLA):

- (1) A generic model was successfully developed to simulate the temperature evolution via a two-dimensional domain discretisation method incorporating with the Hoffman-Lauritzen theory to model the crystal growth during the FDM process, which could help to expect the crystal structure of real process.
- (2) The proposed model results showed good agreements with the experimental results of this study and also with the previously published data with maximum deviations of 4.4 and 4.3 °C, respectively.
- (3) The degree of crystallisation in a single printed line is meagre. This phenomenon could be due to two main reasons: First, crystallization process of PLA is really slow, and it needs tens of or even hundreds of minutes to reach maximum crystallinity, only 10 s provided might not be adequate to complete the crystallization process. Second, a single printed line cools down quite fast, and it is only a small period of the whole process that PLA is in an appropriate temperature range for the high-efficient crystallising. Therefore, a quite low rate of crystallisation is reasonable for a single printed filament.
- (4) Compared with a single line, the multi-lines could significantly improve the crystallisation rate by decreasing the cooling and reheating process. As a consequence, the adequate time for PLA crystallisation has been extended based on the temperature range. The final crystal rate is 10 times larger than that in the single printed line.
- (5) The influential factor level of various process parameters was investigated numerically on both single and multi line works. From the numerical simulation results, the thermal convection coefficient and nozzle diameter were the most crucial parameters which influenced the temperature evolutions. Also, the thermal contact conductance with the platform is influential. However, due to the area convecting with the platform is smaller

than compared the surface area explored to the ambient, the thermal contact conductance with platform has less impact on the temperature profile. Moreover, the parameters such as the printing speed, latent heat of crystallisation and thermal contact conductance has less impact on the whole temperature profile.

Chapter 7 Future work

Based on the current work and literature review, the following aspects are supposed to explore further in the future:

- (1) In the proposed model, we made a number of simplifying assumptions in order to simplify the model calculations, such as the platform is at a constant temperature and it will not be heated by the extruded filaments; the air temperature is also considered to be a constant; and the airflow bringing the heat energy away is also ignored; when the third line deposited on the first line, the radiation energy emitted from the second and third lines were not considered absorption between each other. Hence, improving of the model lifting these assumptions would be suitable.
- (2) The shape changes of the extruded filaments should be considered as the gravity effect and viscous force will act on the parts that have not solidified. In that case, the surface area exposed to the environment and the area contacted to the platform or previous filament would be different, which will lead to influence the accuracy of the simulation work.
- (3) In this work, the exact position of nucleation forms in an element was not set, and the average number of nucleation formed was considered to be function of time in each element. Also, the crystal morphology and growth behaviour were thought to be ideal. Any two independent nucleation would growth with no limits and will not contact to each other or any boundaries. In the future, the position of formed crystal could be simulated and independent nucleation contacting with other boundaries may also be considered.

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